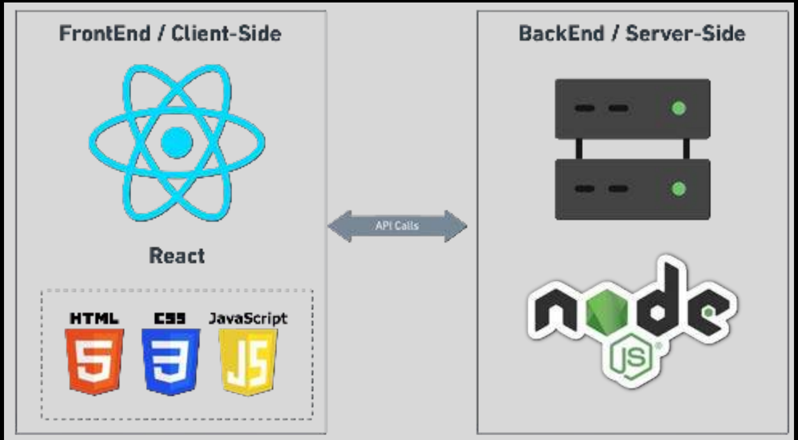
Playlist Link

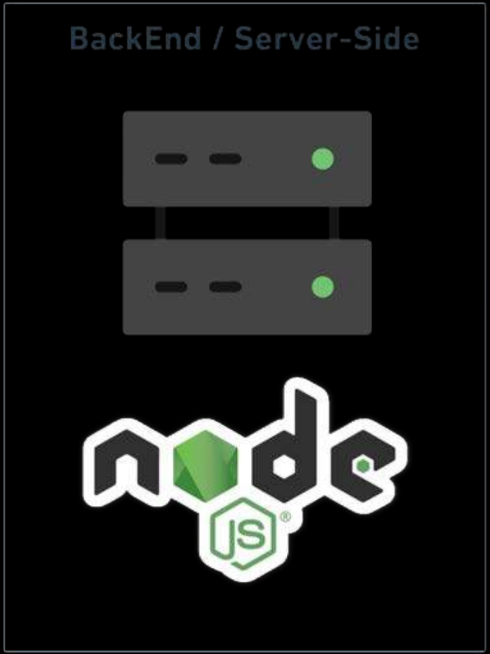
NodeJS

Complete Course



Introduction to NodeJS

1. Pre-requisites
2. What is NodeJS
3. NodeJs Features
4. JavaScript on Client
5. JavaScript on Server
6. Client Code vs Server Code
7. Other uses of NodeJs
8. Server architecture with NodeJs



JS is required for NodeJS

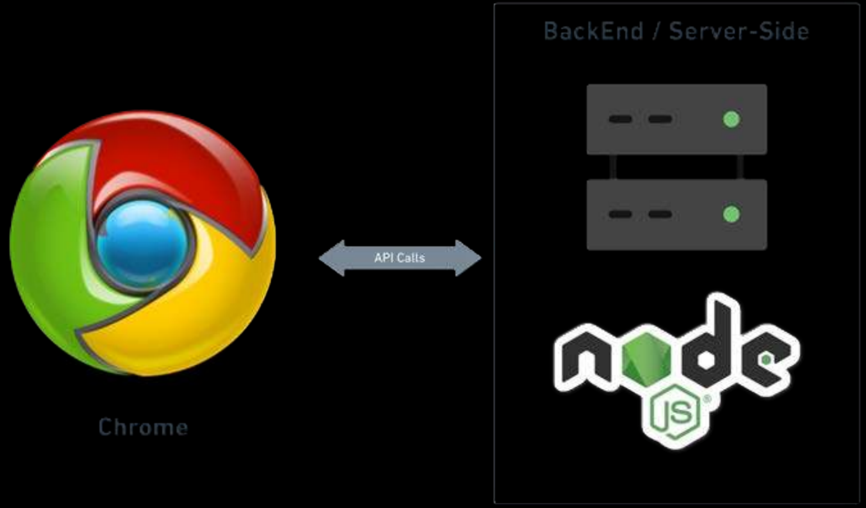
React is recommended before NodeJS

2.What is NodeJS

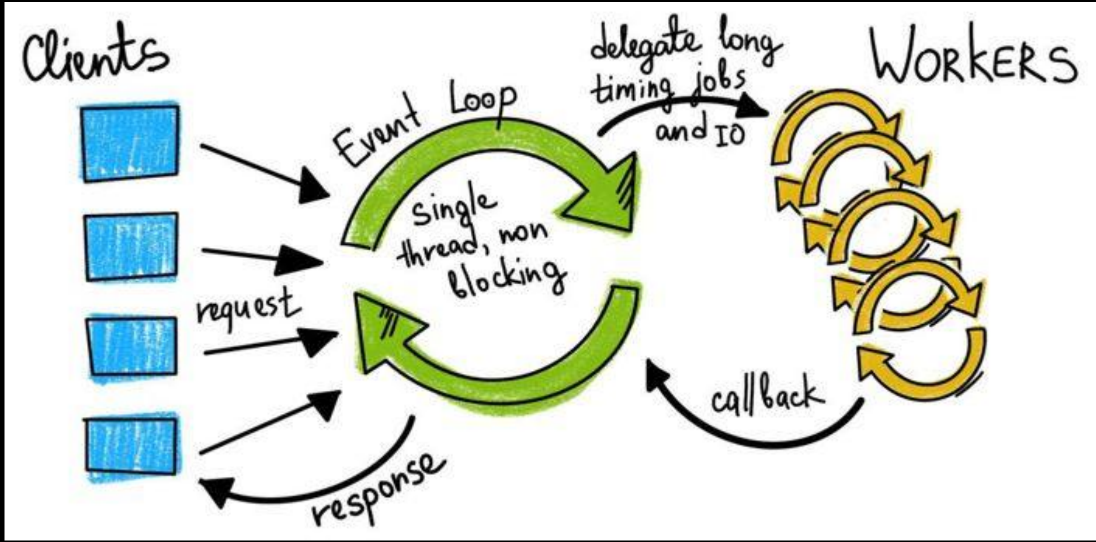
1. JavaScript Runtime: Node.js is an open-source, cross-platform runtime environment for executing JavaScript code outside of a browser.
2. NodeJs is a JavaScript in a different environment means Running JS on the server or any computer.
3. Built on Chrome's V8 Engine: It runs on the V8 engine, which compiles JavaScript directly to native machine code, enhancing performance.
4. V8 is written in C++ for speed.
5. V8 + Backend Features = NodeJs

2.What is NodeJS

1. Design: Features an event-driven, non-blocking I/O model for efficiency.
2. Full-Stack JavaScript: Allows using JavaScript on both server and client sides.
3. Scalability: Ideal for scalable network applications due to its architecture.
4. Versatility: Suitable for web, real-time chat, and REST API servers.



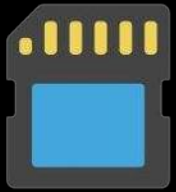
3. NodeJs Features
(Added)

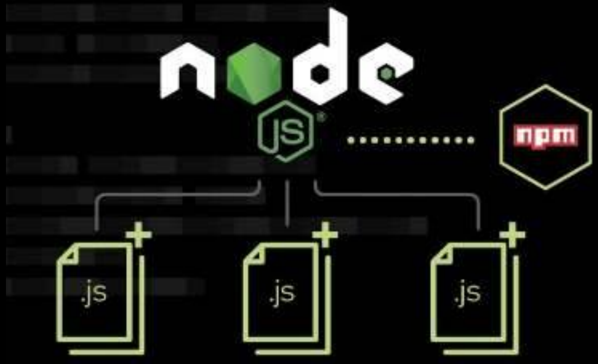


1. Non-blocking I/O: Designed to perform non-blocking operations by default, making it suitable for I/O-heavy operations.
2. Networking Support: Supports TCP/UDP sockets, which are crucial for building lower-level network applications that browsers can't handle.

3. NodeJs Features

(Added)

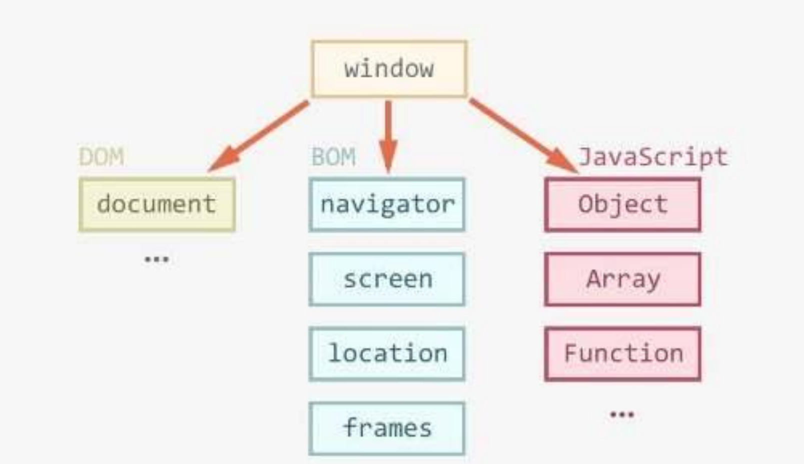


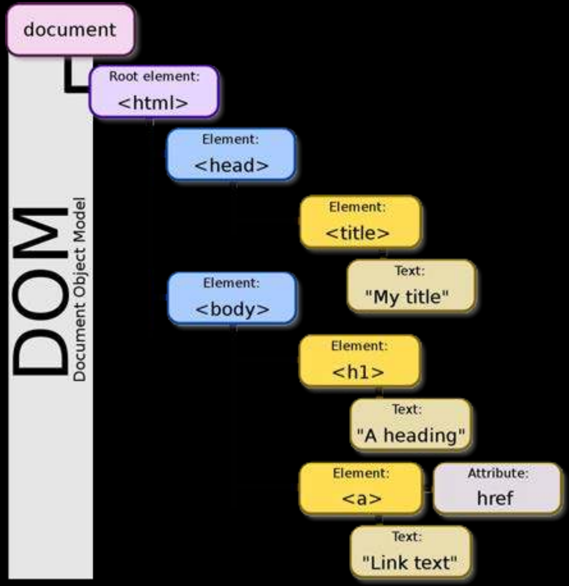


1. File System Access: Provides APIs to read and write files directly, which is not possible in browser environments for security reasons.
2. Server-Side Capabilities: Node.js enables JavaScript to run on the server, handling HTTP requests, file operations, and other server-side functionalities.
3. Modules: Organize code into reusable modules using require().

3. NodeJs Features

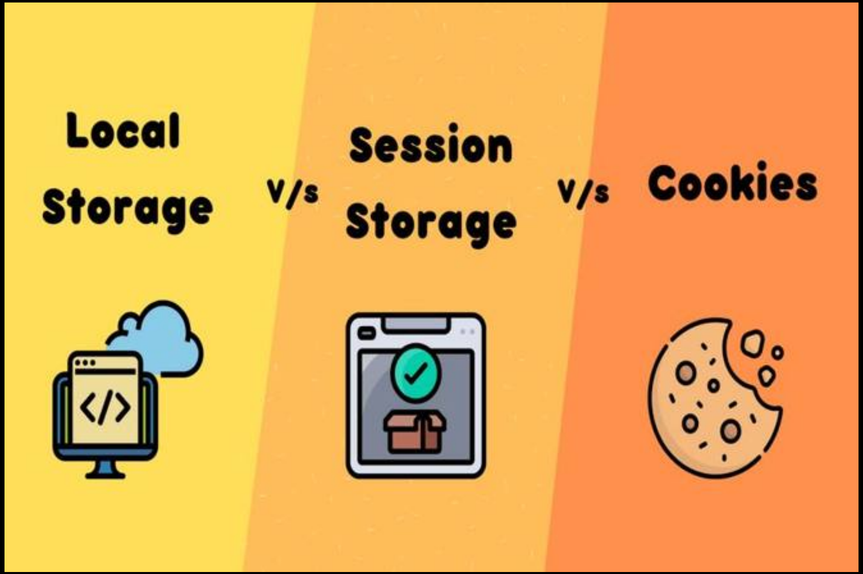
(Removed)

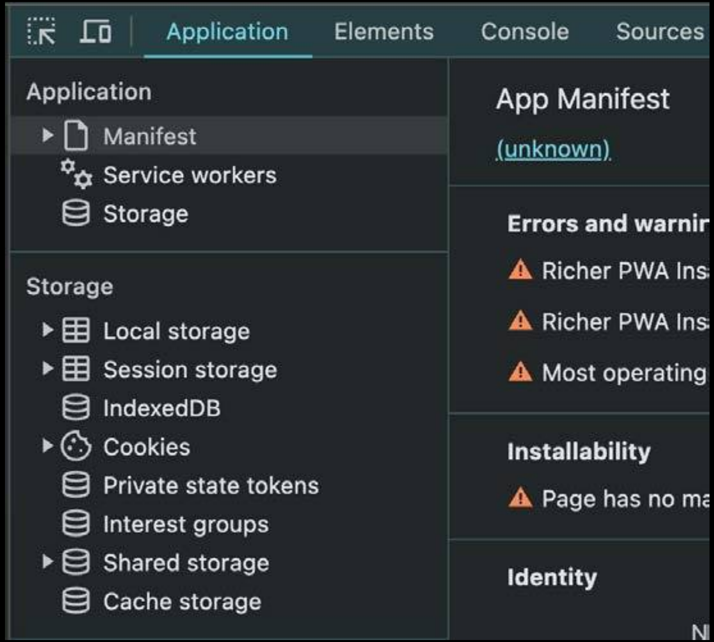




1. Window Object: The global window object, which is part of web browsers, is absent in Node.js.
2. DOM Manipulation: Node.js does not have a built-in Document Object Model (DOM), as it is not intended to interact with a webpage's content.
3. BOM (Browser Object Model): No direct interaction with things like navigator or screen which are part of BOM in browsers.

3. NodeJs Features
(Removed)





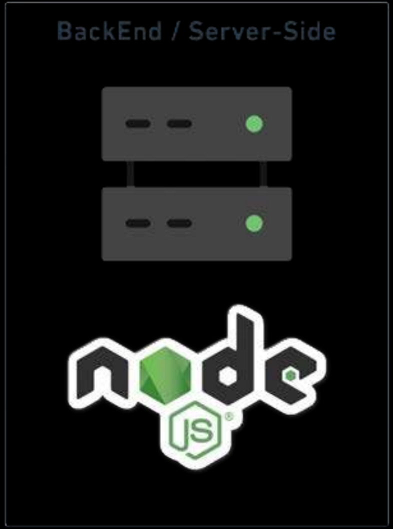
Web-Specific APIs: APIs like localStorage, sessionStorage, and browser-based fetch are not available in Node.js.

4. JavaScript on Client

1. Displays Web Page: Turns HTML code into what you see on screen.
2. User Clicks: Helps you interact with the web page.
3. Updates Content: Allows changes to the page using JavaScript.
4. Loads Files: Gets HTML, images, etc., from the server.

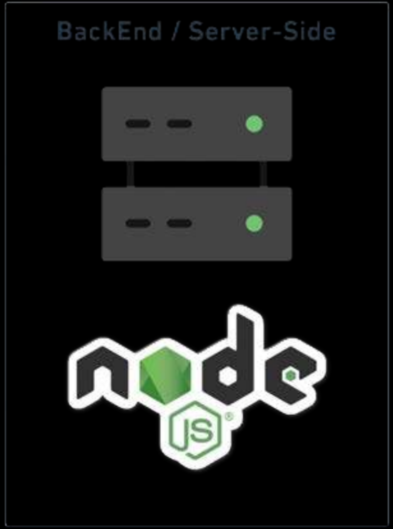
5. JavaScript on Server

1. Database Management: Stores, retrieves, and manages data efficiently through operations like CRUD (Create, Read, Update, Delete).
2. Authentication: Verifies user identities to control access to the system, ensuring that users are who they claim to be.
3. Authorization: Determines what authenticated users are allowed to do by managing permissions and access controls.
4. Input Validation: Checks incoming data for correctness, completeness, and security to prevent malicious data entry and errors.
5. Session Management: Tracks user activity across various requests to maintain state and manage user-specific settings.



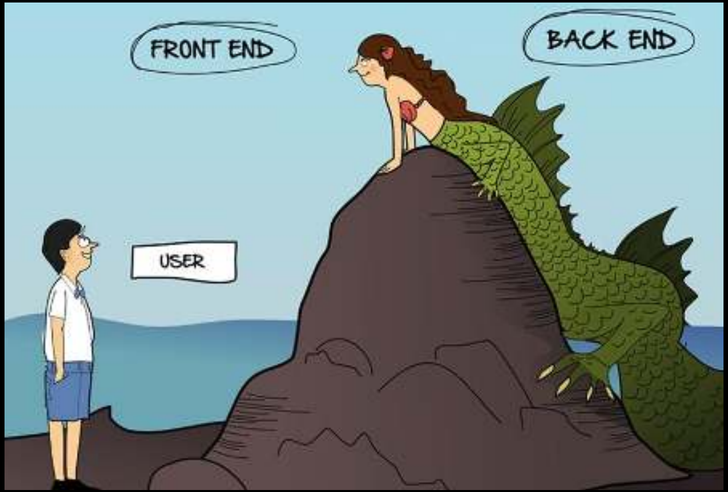
5. JavaScript on Server

6. API Management: Provides and handles interfaces for applications to interact, ensuring smooth data exchange and integration.
7. Error Handling: Manages and responds to errors effectively to maintain system stability and provide useful error messages.
8. Security Measures: Implements protocols to protect data from unauthorized access and attacks, such as SQL injection and cross-site scripting (XSS).
9. Data Encryption: Secures sensitive information by encrypting data stored in databases and during transmission.
10. Logging and Monitoring: Keeps records of system activity to diagnose issues and monitor system health and security.



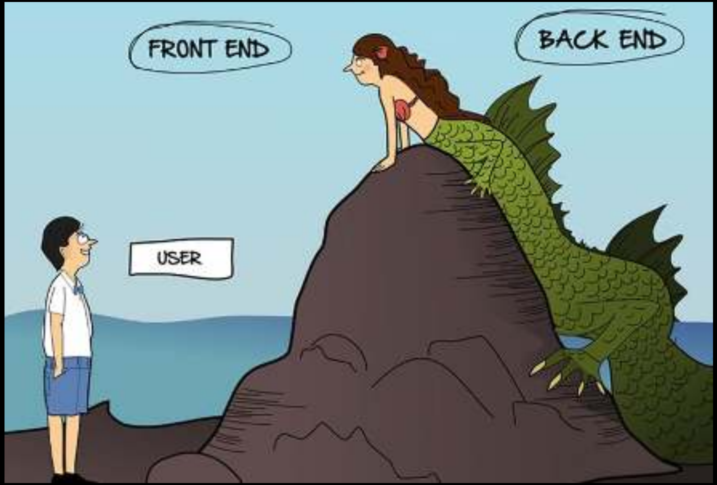
6. Client Code vs Server Code

1. User/client can't access server code directly.
2. Client must raise requests for particular APIs to access certain features or data.
3. Environment Access: Server-side JavaScript accesses server features like file systems and databases.
4. Security: Server-side code can handle sensitive operations securely, while client-side code is exposed and must manage security risks.
5. Performance: Heavy computations are better performed on the server to avoid slowing down the client.



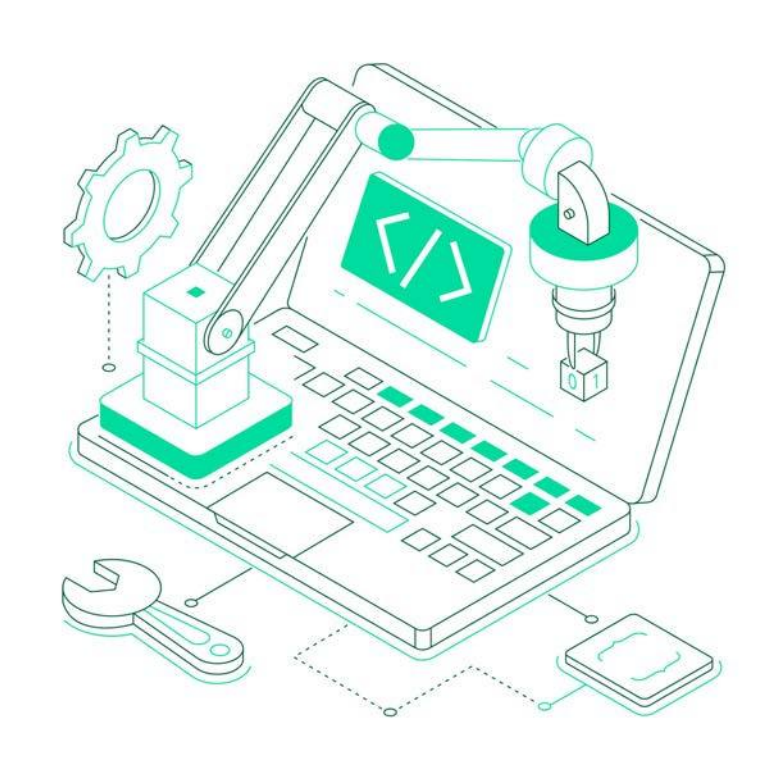
6. Client Code vs Server Code

6. Resource Utilization: Servers generally offer more powerful processing capabilities than client devices.
7. Data Handling: Server-side can directly manage large data sets and database interactions, unlike client-side JavaScript.
8. Asynchronous Operations: Server-side JavaScript is optimized for non-blocking I/O to efficiently manage multiple requests.
9. Session Management: Servers handle sessions and user states more comprehensively.
10. Scalability: Server-side code is designed to scale and handle requests from multiple clients simultaneously.

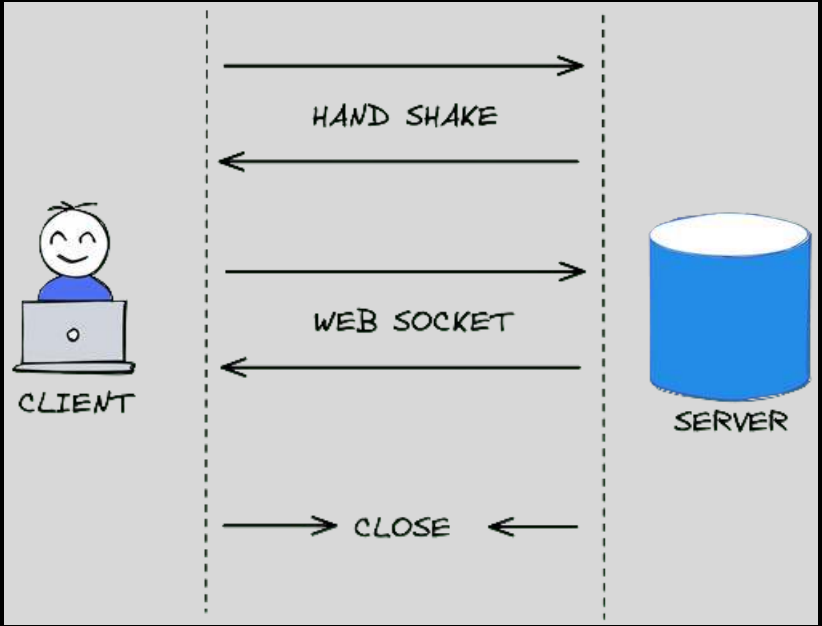


7. Other uses of NodeJs

1. Local Utility Scripts: Automates tasks and processes files locally, like using shell scripts but with JavaScript.
2. Internet of Things (IoT): Develops server-side applications for IoT devices, managing communications and data processing.
3. Scripting for Automation: Automates repetitive tasks in software development processes, such as testing and deployment.



7. Other uses of NodeJs



Real-Time Applications: Efficiently manages real-time data applications, such as chat apps and live updates, using WebSockets.

7. Other uses of NodeJs

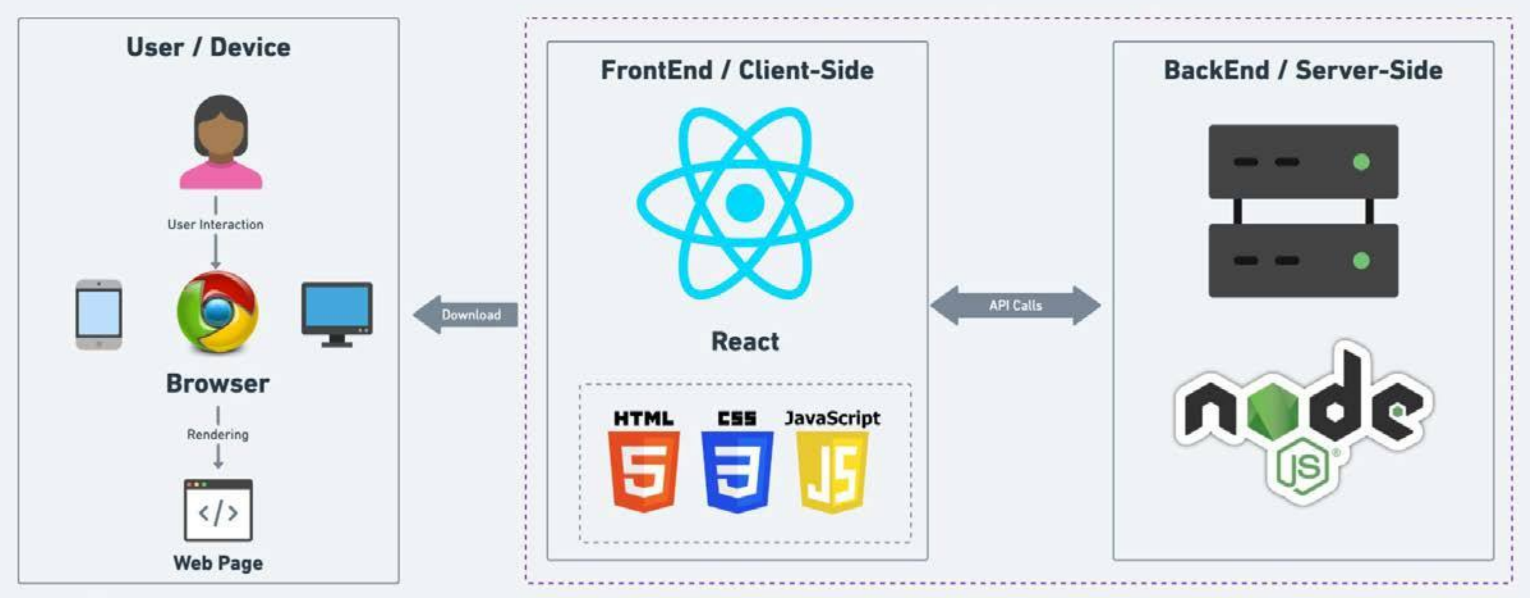
Desktop Applications: Creates cross-platform desktop applications using frameworks like Electron.

7. Other uses of NodeJs

Build Tools: Powers build processes for front-end technologies using tools like:

- Webpack
- Grunt
- Gulp
- Browserify
- Brunch
- Yeoman

8. Server architecture with NodeJs



Nodejs server will:

1. Create server and listen to incoming requests
2. Business logic: validation, connect to db, actual processing of data
3. Return response HTML, JSON, CSS, JS

Revision

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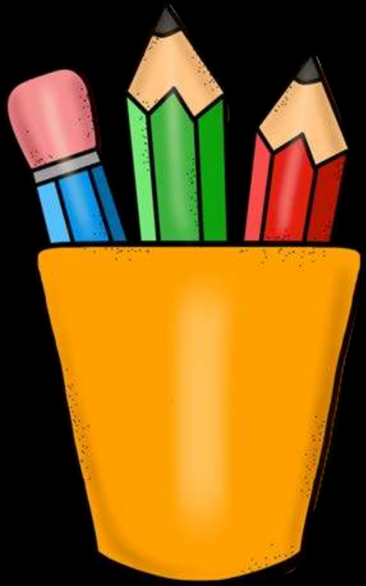


2. Installation of NodeJS

1. What is IDE
2. Need of IDE
3. MAC Setup
   - Install latest Node & VsCode
4. Windows Setup
   - Install latest Node & VsCode
5. Linux Setup
   - Install latest Node & VsCode
6. VsCode (Extensions and Settings)
7. Executing first .js file
8. What is REPL
9. Executing Code via REPL

2.1 What is IDE

1. IDE stands for Integrated Development Environment.
2. Software suite that consolidates basic tools required for software development.
3. Central hub for coding, finding problems, and testing.
4. Designed to improve developer efficiency.

2.2 Need of IDE

1. Streamlines development.
2. Increases productivity.
3. Simplifies complex tasks.
4. Offers a unified workspace.
5. IDE Features
    1. Code Autocomplete
    2. Syntax Highlighting
    3. Version Control
    4. Error Checking

MainActivity.kt

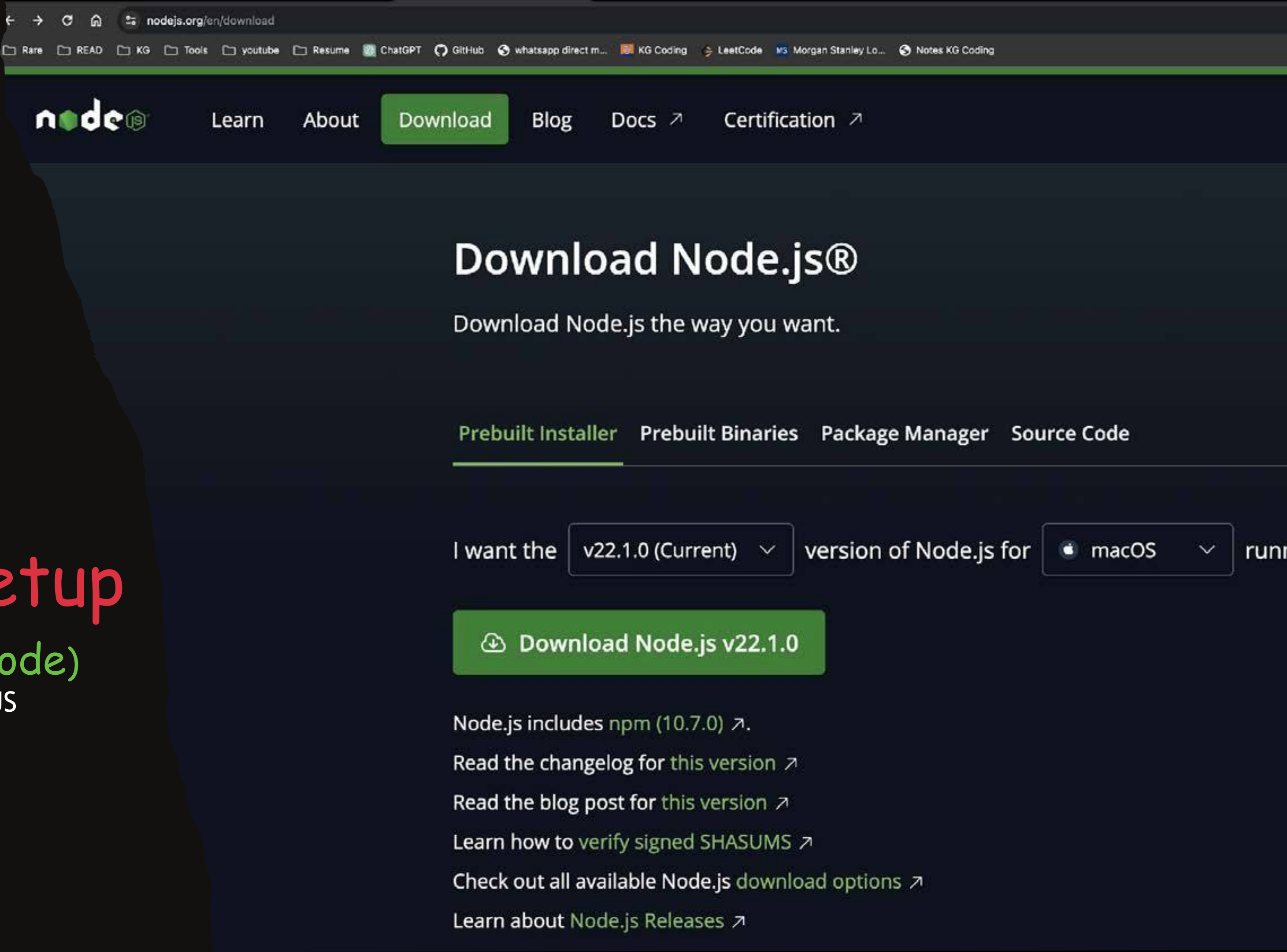
```kotlin
@Composable
fun MessageCard(msg: Message) {
    Row(modifier = Modifier.padding(all = 8.dp)) {
        Image(
            painter = painterResource(R.drawable.android_studio_logo),
            contentDescription = "Profile Picture",
            modifier = Modifier
                .size(45.dp)

        )


        Spacer(modifier = Modifier.width(8.dp))
        Column (Modifier
            .background(color = Color.White)) {
            Text(text = msg.author, color = Color.Black)
            Spacer(modifier = Modifier.height(1.dp))
            Text(text = msg.body, color = Color.Black)
        }
    }
}
```

2.3 MAC Setup

2.3 MAC Setup

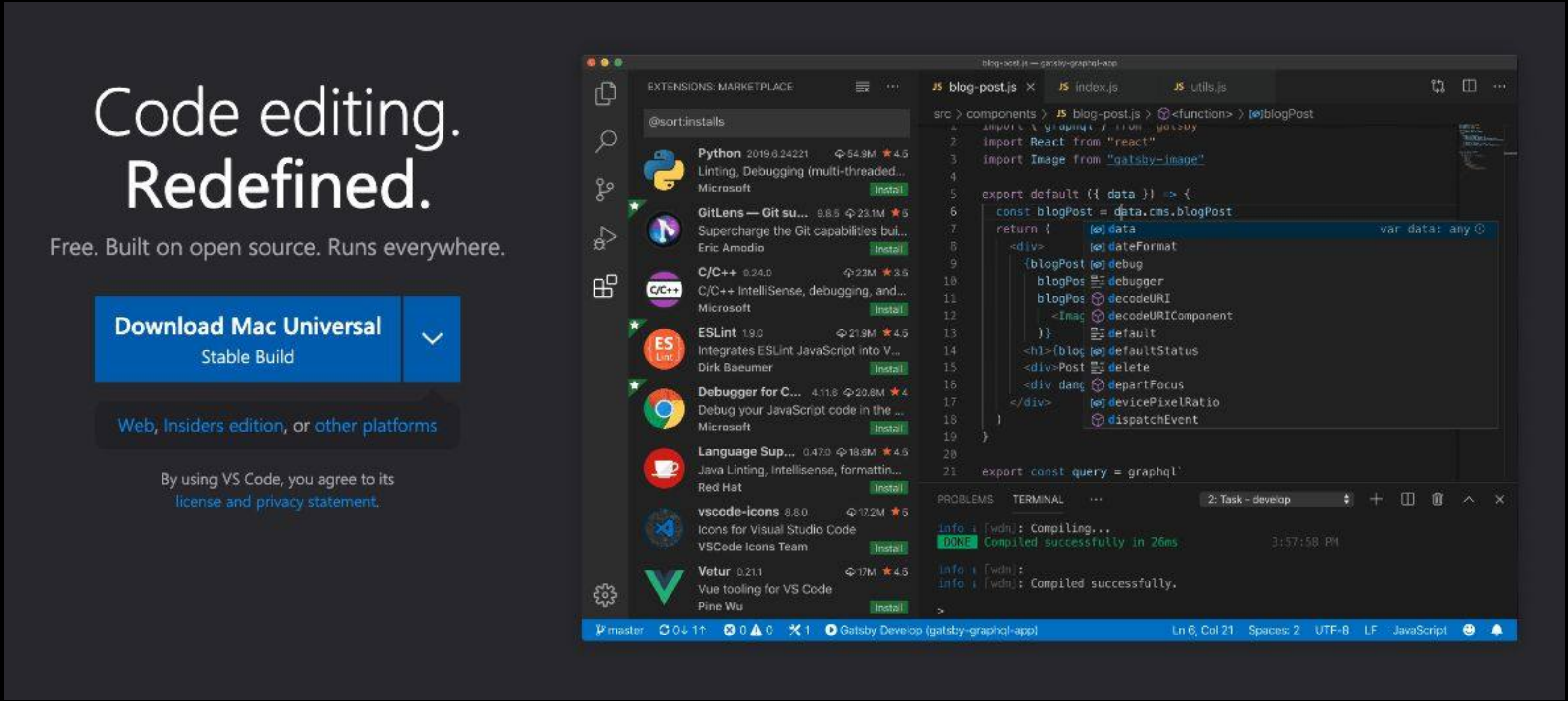
(Install latest Node)

Search Download NodeJS



2.3 MAC Setup
(Install VsCode)

Search VS Code on Google

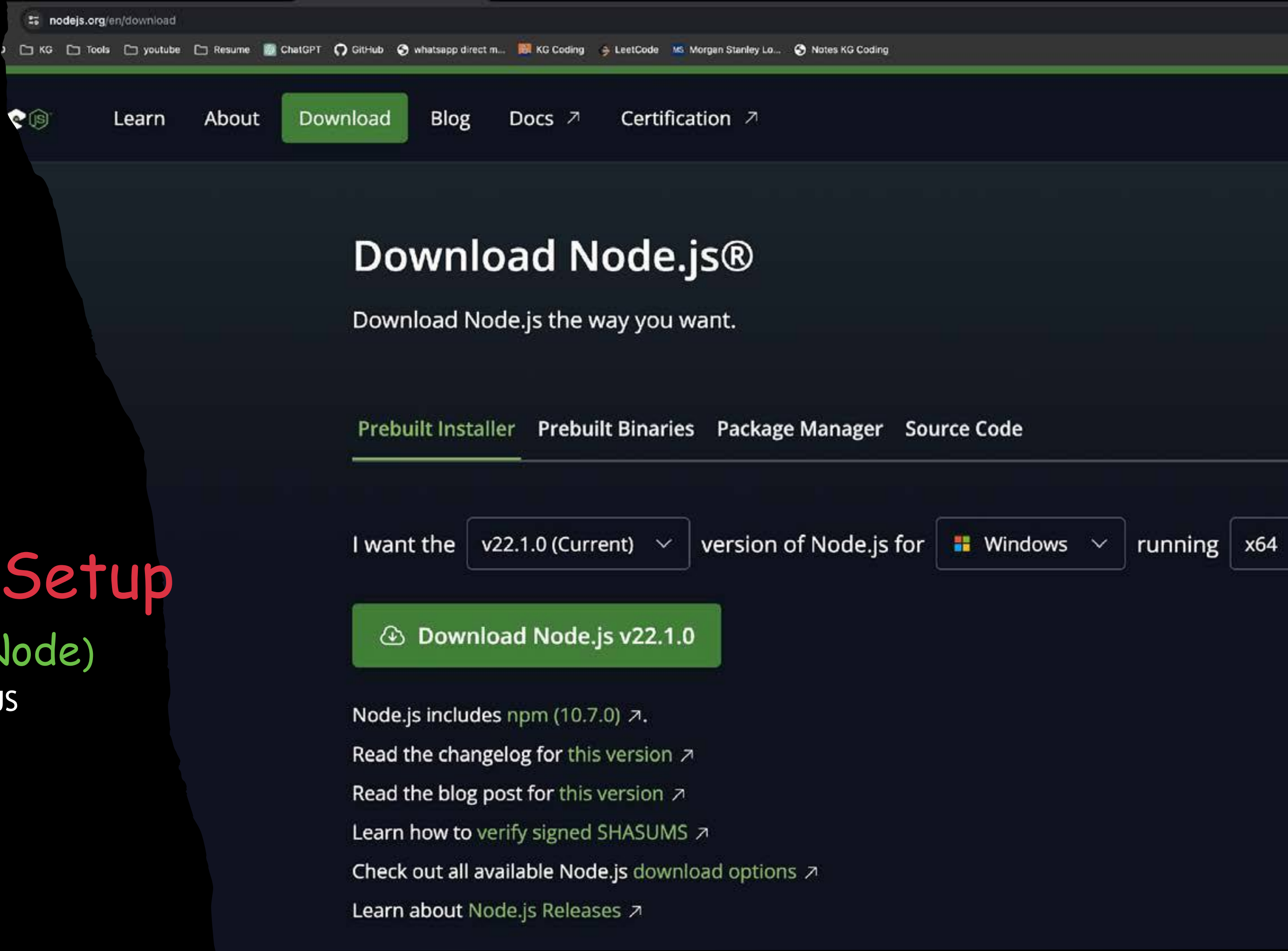


2.4 Windows Setup

2.4 Windows Setup

(Install latest Node)

Search Download NodeJS



2.4 Windows Setup

(Install VsCode)

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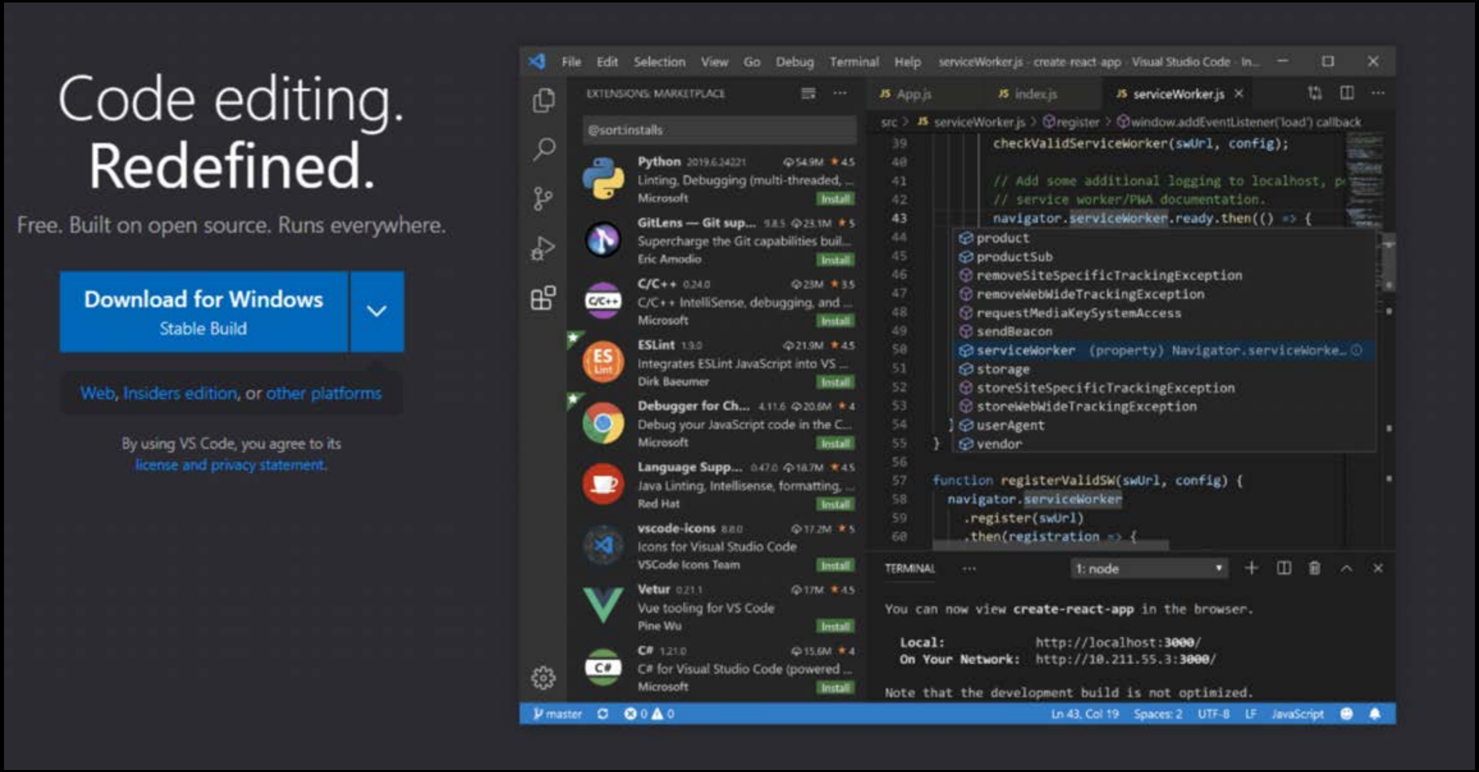
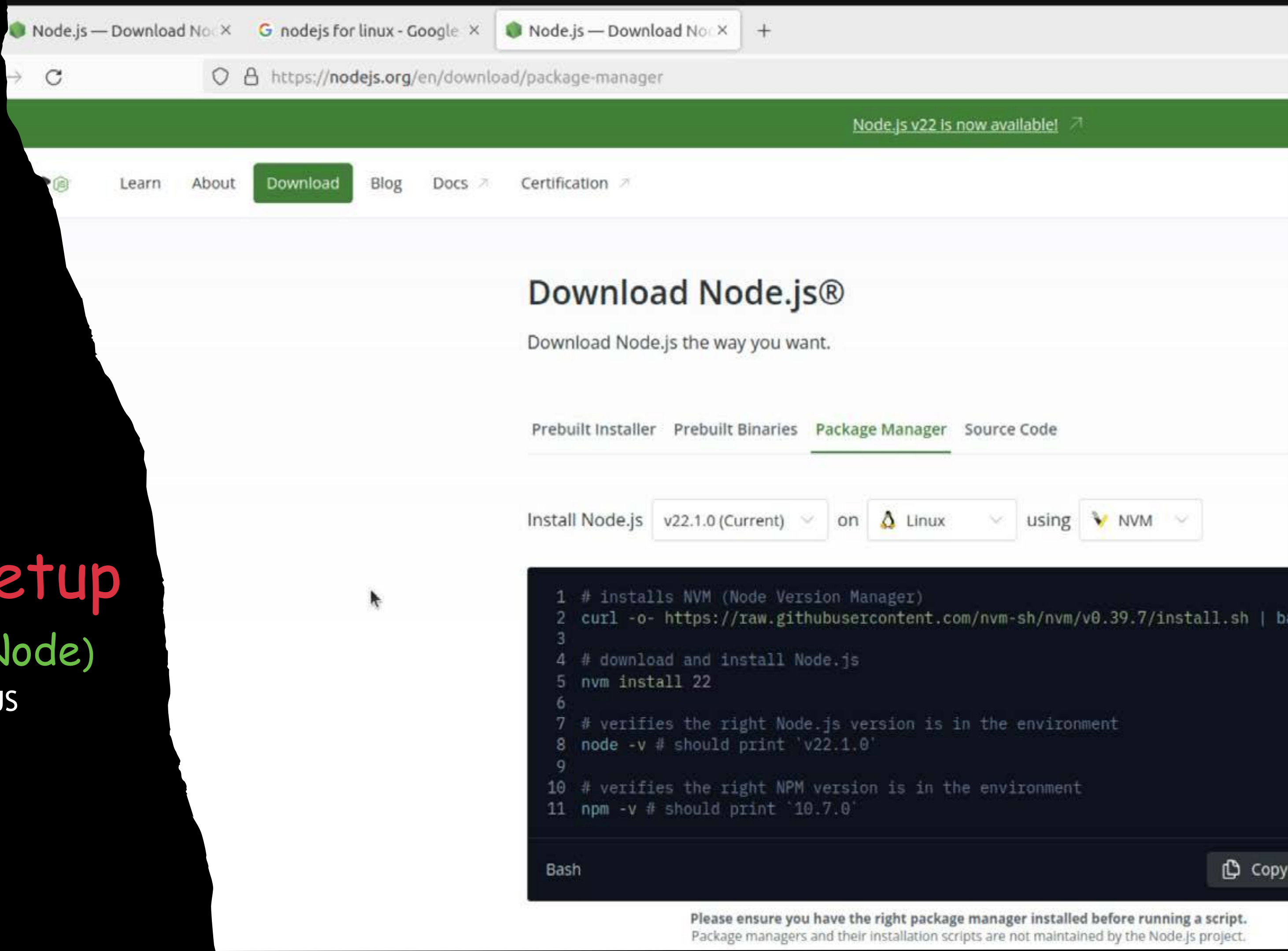


2.5 Linux Setup

2.5 Linux Setup

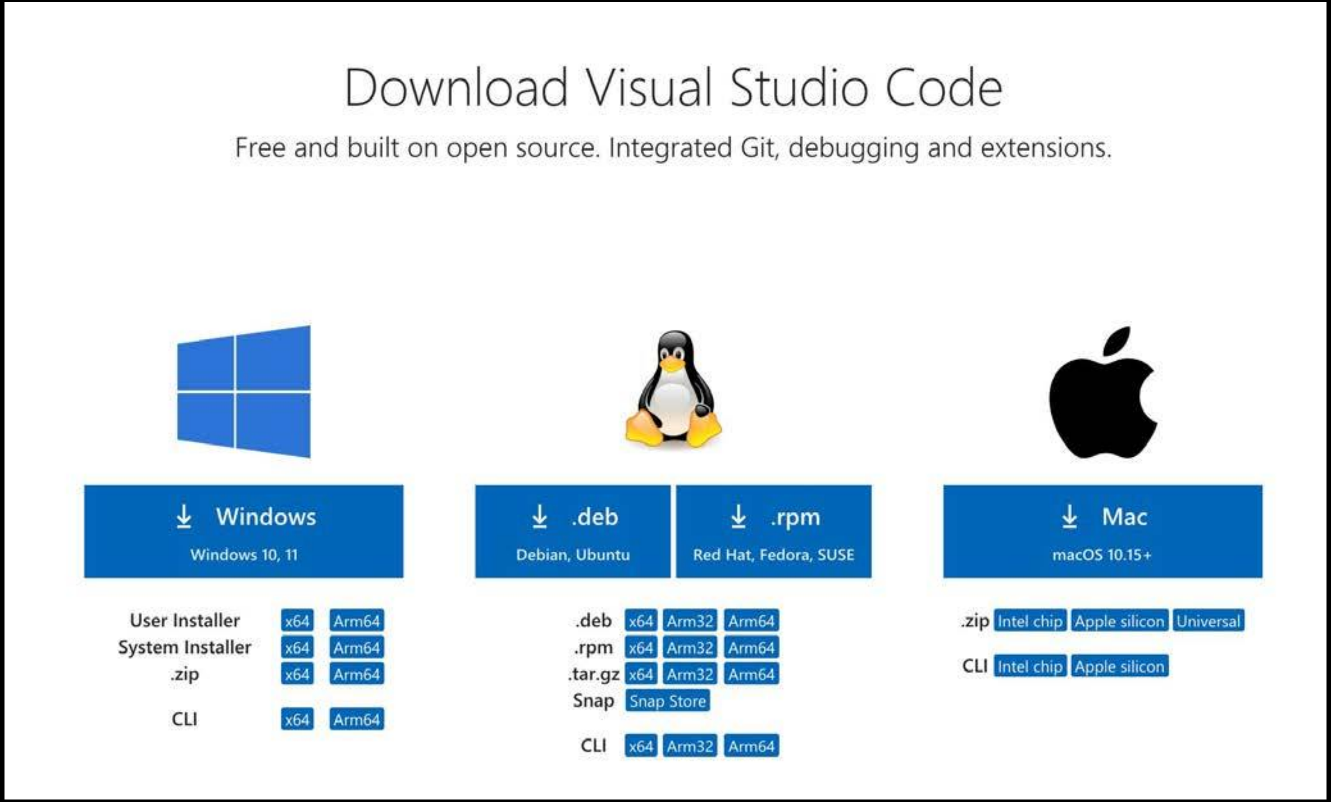
(Install latest Node)

Search Download NodeJS



2.5 Linux Setup
(Install VsCode)

Search VS Code on Google



2.6 VsCode

(Extensions and Settings)

1. Prettier (Format on Save)
2. Line Wrap
3. Tab Size from 4 to 2

Live Server

2.7 Executing first .js file

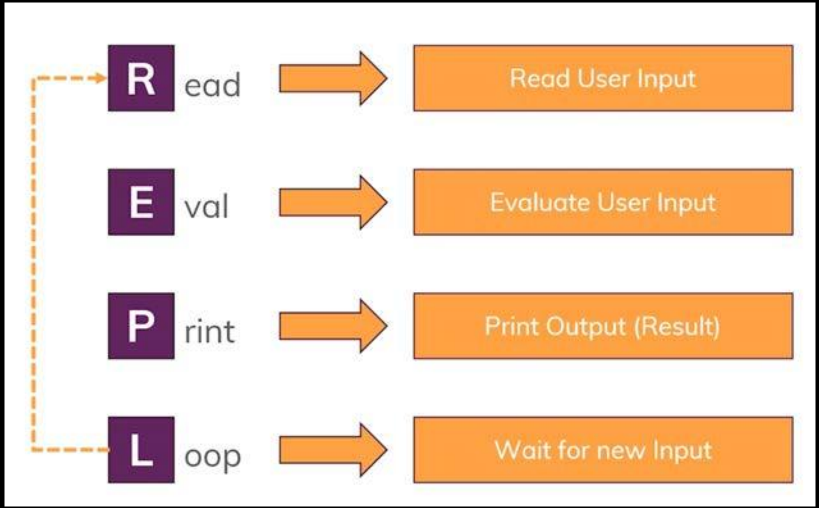
```javascript
const fs = require('fs');

// Define two variables
let a = 10;
let b = 5;

// Basic arithmetic operations
let sum = a + b;
let product = a * b;

// Prepare data to write
let data = `Sum: ${sum}\nProduct: ${product}`;
console.log(data);

// Write data to a local file
fs.writeFile('output.txt', data, (err) => {
    if (err) throw err;
    console.log('Data written to file');
});
```

```
prashantjain@Mac-mini Desktop % node test.js
Sum: 15
Product: 50
Data written to file
```

1. Streamlines Node Command: Use node filename.js to execute a JavaScript file in the Node.js environment.
2. Require Syntax: Use require('module') to include built-in or external modules, or other JavaScript files in your code.
3. Modular Code: require helps organize code into reusable modules, separating concerns and improving maintainability.
4. Caching: Modules loaded with require are cached, meaning the file is executed only once even if included multiple times.

2.8 What is REPL

1. Streamlines Interactive Shell: Executes JavaScript code interactively.
2. Quick Testing: Ideal for testing and debugging code snippets on the fly.
3. Built-in Help: Offers help commands via .help.
4. Session Management: Supports saving (.save) and loading (.load) code sessions.
5. Node.js API Access: Provides direct access to Node.js APIs for experimentation.
6. Customizable: Allows customization of prompt and behaviour settings.



2.9 Executing Code via REPL

```
prashantjain@Mac-mini Desktop % node
Welcome to Node.js v20.9.0.
Type ".help" for more information.
> 5 + 6
11
> console.log('KG Coding is the best');
KG Coding is the best
undefined
> fs.writeFile('output.txt', 'Writing to file', (err) => {
...     if (err) throw err;
...     console.log('Data written to file');
... });
undefined
> Data written to file
```

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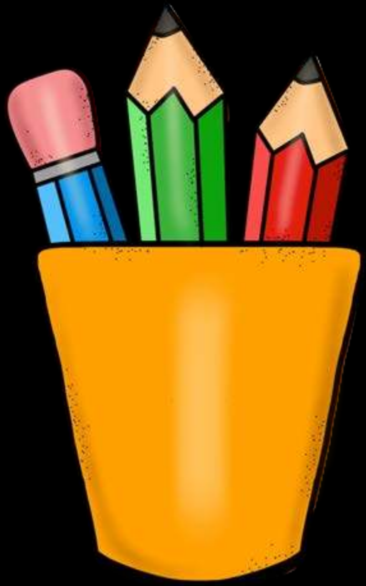
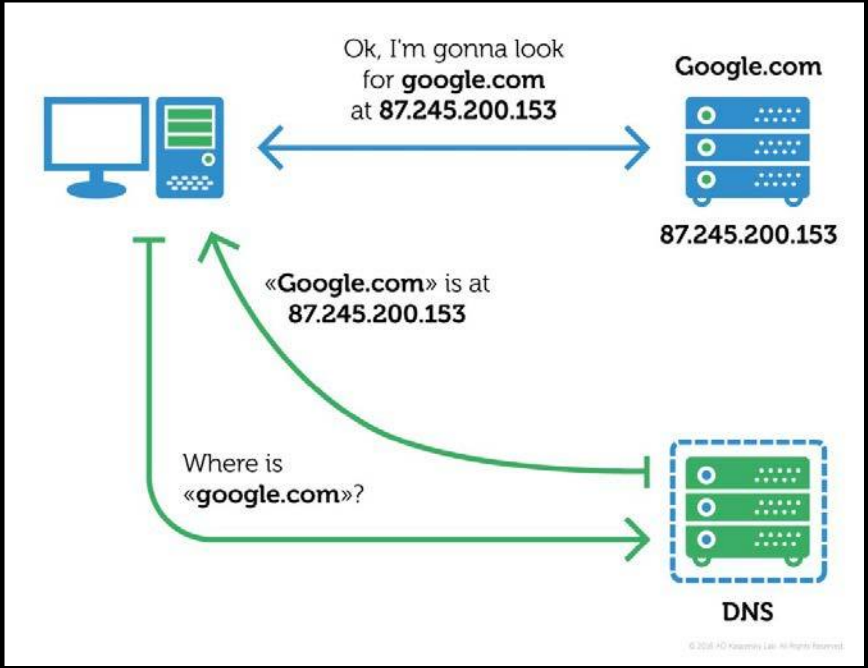


3. First Node Server

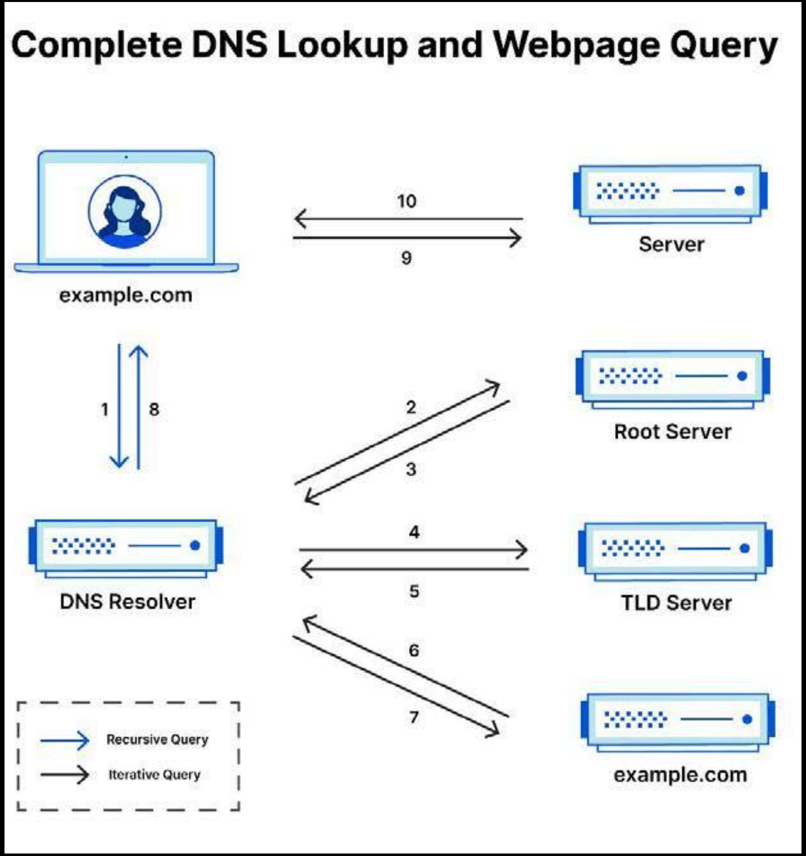
1. How DNS Works?
2. How Web Works?
3. What are Protocols?
4. Node Core Modules
5. Require Keyword
6. Creating first Node Server

3.1 How DNS Works?



1. Domain Name Entry: User types a domain (e.g., www.example.com) into the browser.
2. DNS Query: The browser sends a DNS query to resolve the domain into an IP address.
3. DNS Server: Provides the correct IP address for the domain.
4. Browser Connects: The browser uses the IP to connect to the web server and loads the website.

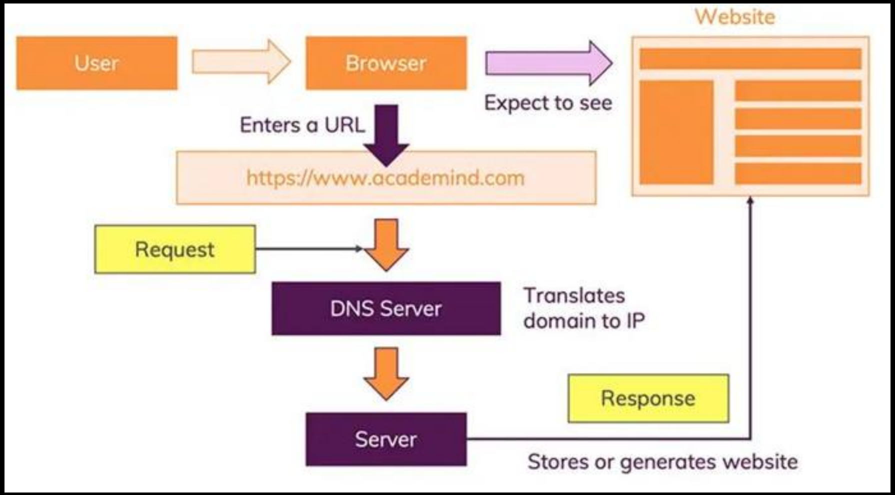
3.1 How DNS Actually Works?



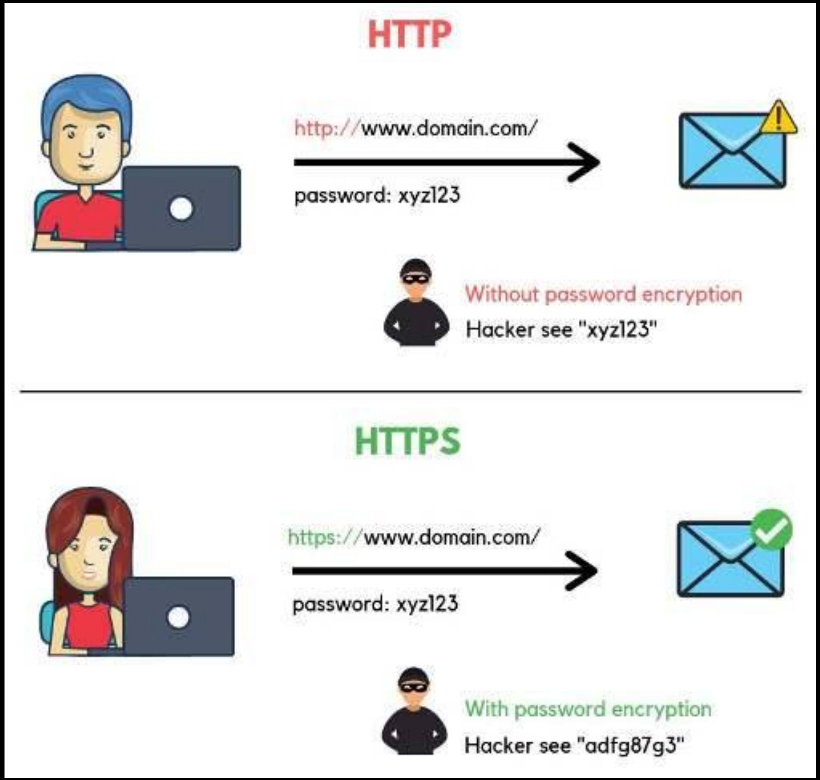
1. Root DNS: Acts as the starting point for DNS resolution. It directs queries to the correct TLD server (e.g., .com, .org).
2. TLD (Top-Level Domain) DNS: Handles queries for specific top-level domains (e.g., .com, .net) and directs them to the authoritative DNS server (e.g., Verisign for .com, PIR for .org)
3. Authoritative DNS: Contains the actual IP address of the domain and answers DNS queries with this information.(e.g., Cloudflare, Google DNS).

3.2 How Web Works?

1. Client Request Initiation: The client (browser) initiates a network call by entering a URL.
2. DNS Resolution: The browser contacts a DNS server to get the IP address of the domain.
3. TCP Connection: The browser establishes a TCP connection with the server's IP address.
4. HTTP Request: The browser sends an HTTP request to the server.
5. Server Processing: The server processes the request and prepares a response.
6. HTTP Response: The server sends an HTTP response back to the client.
7. Network Transmission: The response travels back to the client over the network.
8. Client Receives Response: The browser receives and interprets the response.
9. Rendering: The browser renders the content of the response and displays it to the user.



3.3 What are Protocols?



Http (HyperText Transfer Protocol):

- Facilitates communication between a web browser and a server to transfer web pages.
- Sends data in plain text (no encryption).
- Used for basic website browsing without security.

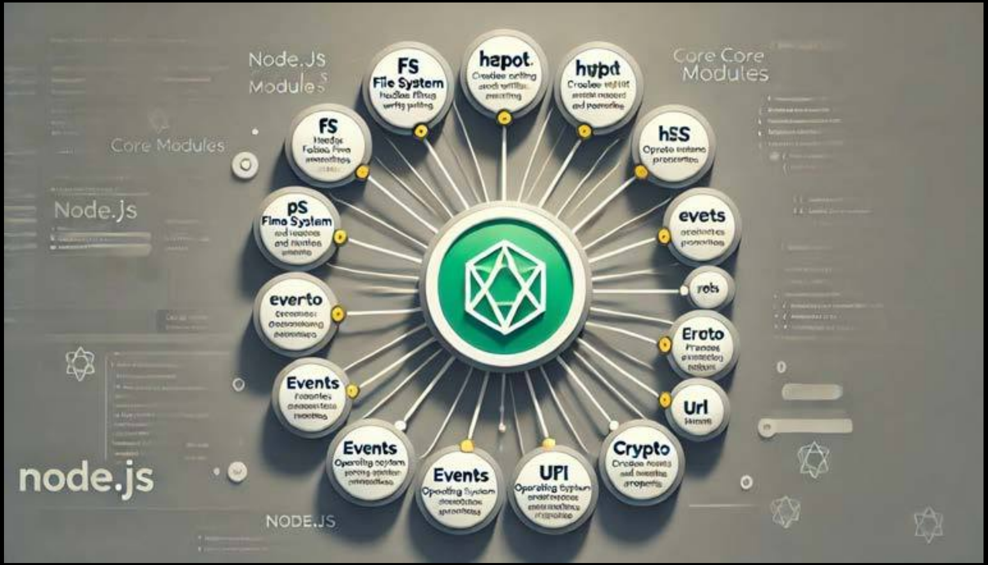
HTTPS (HyperText Transfer Protocol Secure):

- Secure version of HTTP, encrypts data for secure communication.
- Uses SSL/TLS to encrypt data.
- Used in online banking, e-commerce.

TCP (Transmission Control Protocol):

- Ensures reliable, ordered, and error-checked data delivery over the internet.
- Establishes a connection before data is transferred.

3.4 Node Core Modules



1. Built-in: Core modules are included with Node.js installation.
2. No Installation Needed: Directly available for use without npm install.
3. Performance: Highly optimized for performance.

3.4 Node Core Modules

1. fs (File System): Handles file operations like reading and writing files.
2. http: Creates HTTP servers and makes HTTP requests.
3. https: Launch a SSL Server.
4. path: Provides utilities for handling and transforming file
5. paths.os: Provides operating system-related utility methods and properties.
6. events: Handles events and event-driven programming.
7. crypto: Provides cryptographic functionalities like hashing and encryption.
8. url: Parses and formats URL strings.

3.5 Require Keyword

1. Purpose: Imports modules in Node.js.
2. Caching: Modules are cached after the first require call.
3. .js is added automatically and is not needed to at the end of module name.
4. Path Resolution: Node.js searches for modules in core, node_modules, and file paths.

Syntax:

const moduleName = require('module’);

```
// Load the built-in http module
const http = require('http');

// Load the third party express module
const express = require('express');

// Load the custom myModule module
const myModule = require('./myModule');
```

3.6 Creating first Node Server

```
// Simple Node.js server
const http = require('http');

function requestListener(req, res) {
    console.log(req);
}

http.createServer(requestListener);
```

3.6 Creating first Node Server

```javascript
// Simple Node.js server
const http = require('http');

http.createServer(function (req, res) {
  console.log(req);
});
```

3.6 Creating first Node Server

```
// Simple Node.js server
const http = require('http');

http.createServer((req, res) => {
  console.log(req);
});
```

Run the code with:

node app.js

3.6 Creating first Node Server

```javascript
// Simple Node.js server
const http = require('http');

const server = http.createServer((req, res) => {
  console.log(req);
});

server.listen(3000);
```

```
insecureHTTPParser: undefined,
requestTimeout: 300000,
headersTimeout: 60000,
keepAliveTimeout: 5000,
connectionsCheckingInterval: 30000,
requireHostHeader: true,
joinDuplicateHeaders: undefined,
rejectNonStandardBodyWrites: false,
_events: [Object: null prototype],
_eventsCount: 3,
_maxListeners: undefined,
_connections: 2,
_handle: [TCP],
_usingWorkers: false,
_workers: [],
_unref: false,
_listeningId: 2,
allowHalfOpen: true,
pauseOnConnect: false,
noDelay: true,
keepAlive: false,
keepAliveInitialDelay: 0,
highWaterMark: 65536,
httpAllowHalfOpen: false,
timeout: 0,
maxHeadersCount: null,
```

3.6 Creating first Node Server

```javascript
// Simple NodeJS server
const http = require('http');

const server = http.createServer((req, res) => {
  console.log(req);
});

const PORT = 3000;
server.listen(PORT, () => {
  console.log(`Server running at http://localhost:${PORT}/`);
});
```

Revision

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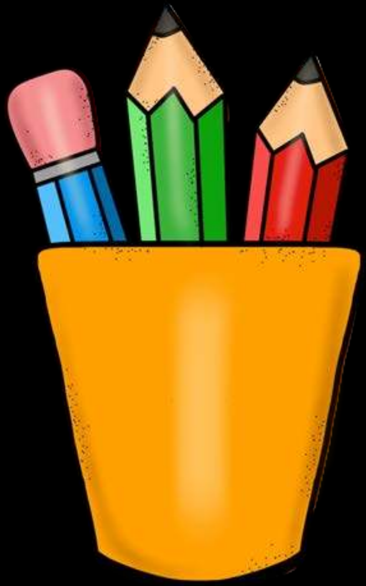
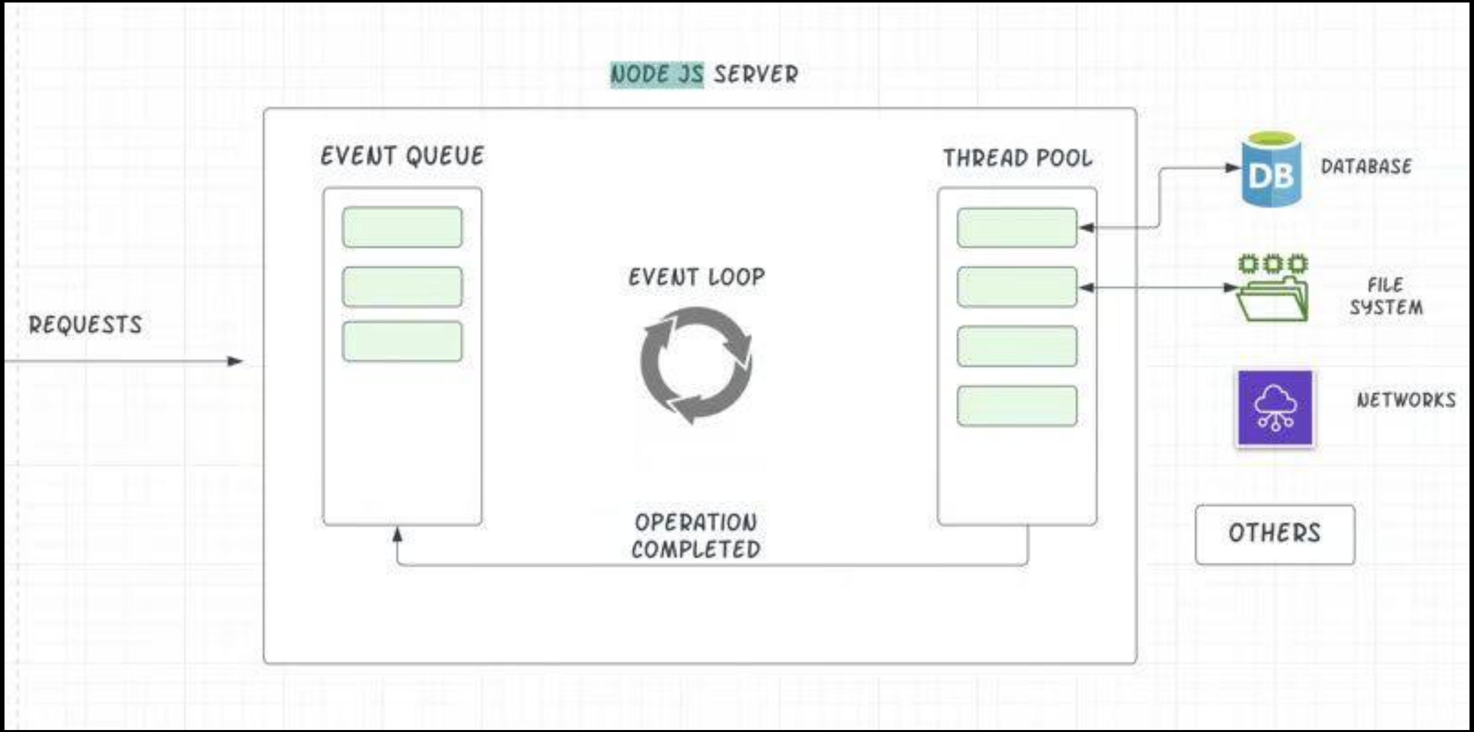


4. Request & Response

1. Node Lifecycle & Event Loop
2. How to exit Event Loop
3. Understand Request Object
4. Sending Response
5. Routing Requests
6. Taking User Input
7. Redirecting Requests

4.1 Node Lifecycle & Event Loop



4.2 How to exit Event Loop

```javascript
// Simple Node.js server
const http = require('http');

const server = http.createServer((req, res) => {
  console.log(req);
  process.exit(); // Stops event loop
});

const PORT = 3000;
server.listen(PORT, () => {
  console.log(`Server running at http://localhost:${PORT}/`);
});
```

4.3 Understand Request Object

```
  [Symbol(kHeaders)]: {
    host: 'localhost:3000',
    connection: 'keep-alive',
    'cache-control': 'max-age=0',
    'sec-ch-ua': '"Chromium";v="128", "Not;A=Brand";v="24", "Google Chrome";v="128"',
    'sec-ch-ua-mobile': '?0',
    'sec-ch-ua-platform': '"macOS"',
    'upgrade-insecure-requests': '1',
    'user-agent': 'Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (
36',
    accept: 'text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/web
nge;v=b3;q=0.7',
    'sec-fetch-site': 'none',
    'sec-fetch-mode': 'navigate',
    'sec-fetch-user': '?1',
    'sec-fetch-dest': 'document',
    'accept-encoding': 'gzip, deflate, br, zstd',
    'accept-language': 'en-US,en-IN;q=0.9,en;q=0.8,hi-IN;q=0.7,hi;q=0.6',
    cookie: 'token=eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJpZCI6ImM3M2Y0MzNjLTFlYzYtNDU
qvUSrMLqbP1uy5bTjnJEQHXc1c'
  },
  [Symbol(kHeadersCount)]: 32,
  [Symbol(kTrailers)]: null,
  [Symbol(kTrailersCount)]: 0
}
```

https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers

4.3 Understand Request Object

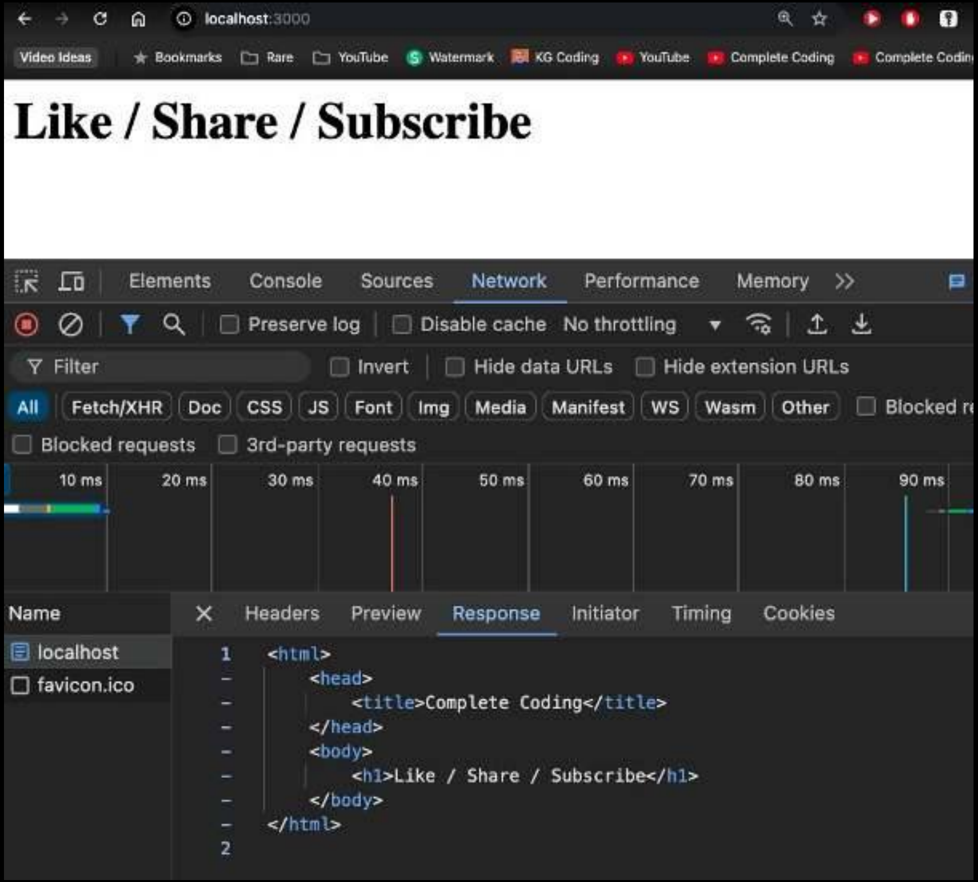
```
// Simple NodeJS server
const http = require('http');

const server = http.createServer((req, res) => {
  console.log(req.url, req.method, req.headers);
});

const PORT = 3000;
server.listen(PORT, () => {
  console.log(`Server running at http://localhost:${PORT}/`);
});
```

Use the browser to access:
http://localhost:3000/
http://localhost:3000/products

4.4 Sending Response

```javascript
// Simple NodeJS server
const http = require('http');

const server = http.createServer((req, res) => {
  //res.setHeader('Content-Type', 'json');
  res.setHeader('Content-Type', 'text/html');
  res.write('<html>');
  res.write('<head><title>Complete Coding</title></head>');
  res.write('<body><h1>Like / Share / Subscribe</h1></body>');
  res.write('</html>');
  res.end();
});

const PORT = 3000;
server.listen(PORT, () => {
  console.log(`Server running at http://localhost:${PORT}/`);
});
```

4.4 Sending Response



4.5 Routing Requests

```
const server = http.createServer((req, res) => {
  res.setHeader('Content-Type', 'text/html');
  res.write('<html>');
  res.write('<head><title>Complete Coding</title></head>');

  if (req.url === '/') {
    res.write('<h1>Welcome to Home page</h1>');
    res.end();
  } else if (req.url.toLowerCase() === '/products') {
    res.write('<h1>Products</h1>');
    res.end();
  }

  res.write('<body><h1>Like / Share / Subscribe</h1></body>');
  res.write('</html>');
  res.end();
});
```

4.5 Routing Requests

```
prashantjain@Prashants-Mac-mini node % node app.js
Server running at http://localhost:3000/
node:_http_outgoing:699
    throw new ERR_HTTP_HEADERS_SENT('set');
    ^

Error [ERR_HTTP_HEADERS_SENT]: Cannot set headers after they are sent to the client
    at ServerResponse.setHeader (node:_http_outgoing:699:11)
    at Server.<anonymous> (/Users/prashantjain/workspace/Test Project/node/app.js:14:7)
    at Server.emit (node:events:520:28)
    at parserOnIncoming (node:_http_server:1146:12)
    at HTTPParser.parserOnHeadersComplete (node:_http_common:118:17) {
  code: 'ERR_HTTP_HEADERS_SENT'
}

Node.js v22.5.1
```

4.5 Routing Requests

```
const server = http.createServer((req, res) => {
  res.setHeader('Content-Type', 'text/html');
  res.write('<html>');
  res.write('<head><title>Complete Coding</title></head>');

  if (req.url === '/') {
    res.write('<h1>Welcome to Home page</h1>');
    return res.end();
  } else if (req.url.toLowerCase() === '/products') {
    res.write('<h1>Products</h1>');
    return res.end();
  }

  res.write('<body><h1>Like / Share / Subscribe</h1></body>');
  res.write('</html>');
  return res.end();
});
```

4.6 Taking User Input

```
if (req.url === '/') {
  res.write('<h1>Welcome to Home page</h1>');
  res.write('<form action="/submit-details" method="POST">');
  res.write('<input type="text" id="name" name="name" placeholder="Enter your
name"><br><br>');
  res.write('<label for="gender">Gender:</label>');
  res.write('<input type="radio" id="male" name="gender" value="male">');
  res.write('<label for="male">Male</label>');
  res.write('<input type="radio" id="female" name="gender" value="female">');
  res.write('<label for="female">Female</label><br><br>');
  res.write('<button type="submit">Submit</button>');
  res.write('</form>');
  return res.end();
}
```

4.7 Redirecting Requests

```
} else if (req.method == 'POST' &&
           req.url.toLowerCase() === '/submit-details') {
  fs.writeFileSync('user-details.txt', 'Prashant Jain');
  res.statusCode = 302;
  res.setHeader('Location', '/');
  return res.end();
}
```

```
1   Prashant Jain
```

Practise Set

Create a page that shows a navigation bar of Myntra with the following links:

A. Home
B. Men
C. Women
D. Kids
E. Cart

Clicking on each link page should navigate to that page and a welcome to section text is shown there.

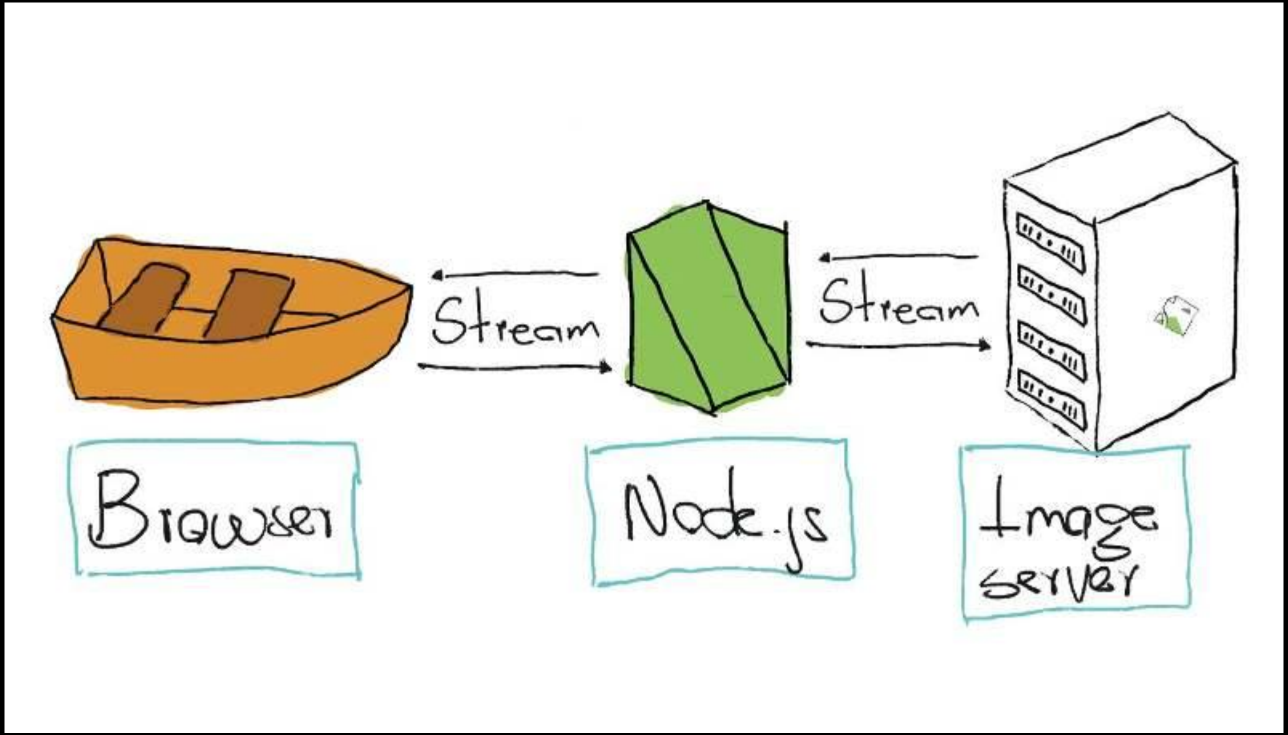
Revision

1. Node Lifecycle & Event Loop
2. How to exit Event Loop
3. Understand Request Object
4. Sending Response
5. Routing Requests
6. Taking User Input
7. Redirecting Requests

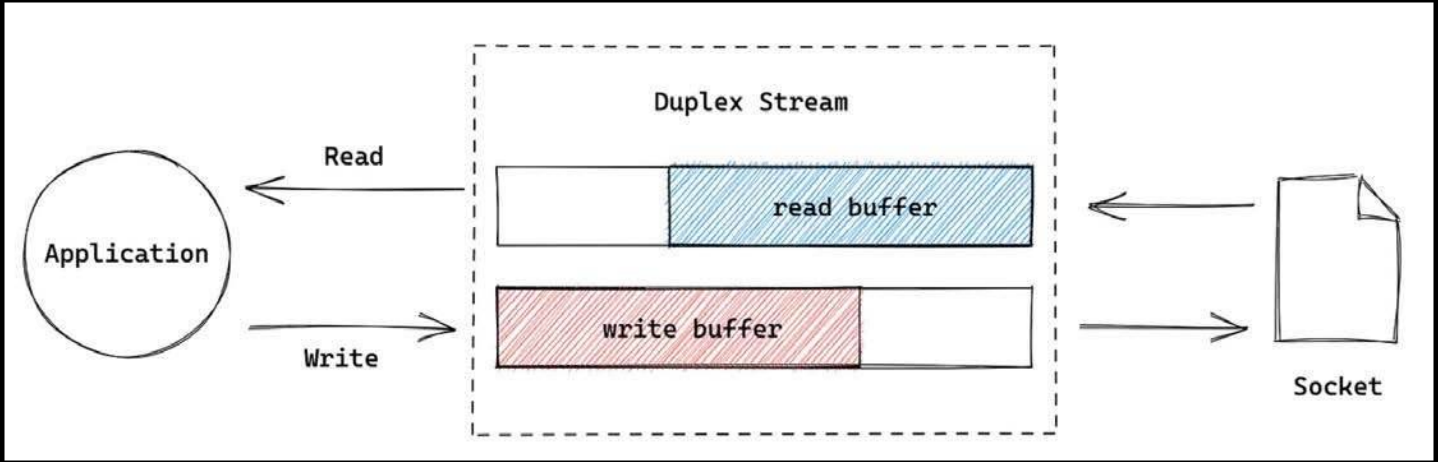
5. Parsing Request

1. Streams
2. Chunks
3. Buffers
4. Reading Chunk
5. Buffering Chunks
6. Parsing Request
7. Using Modules

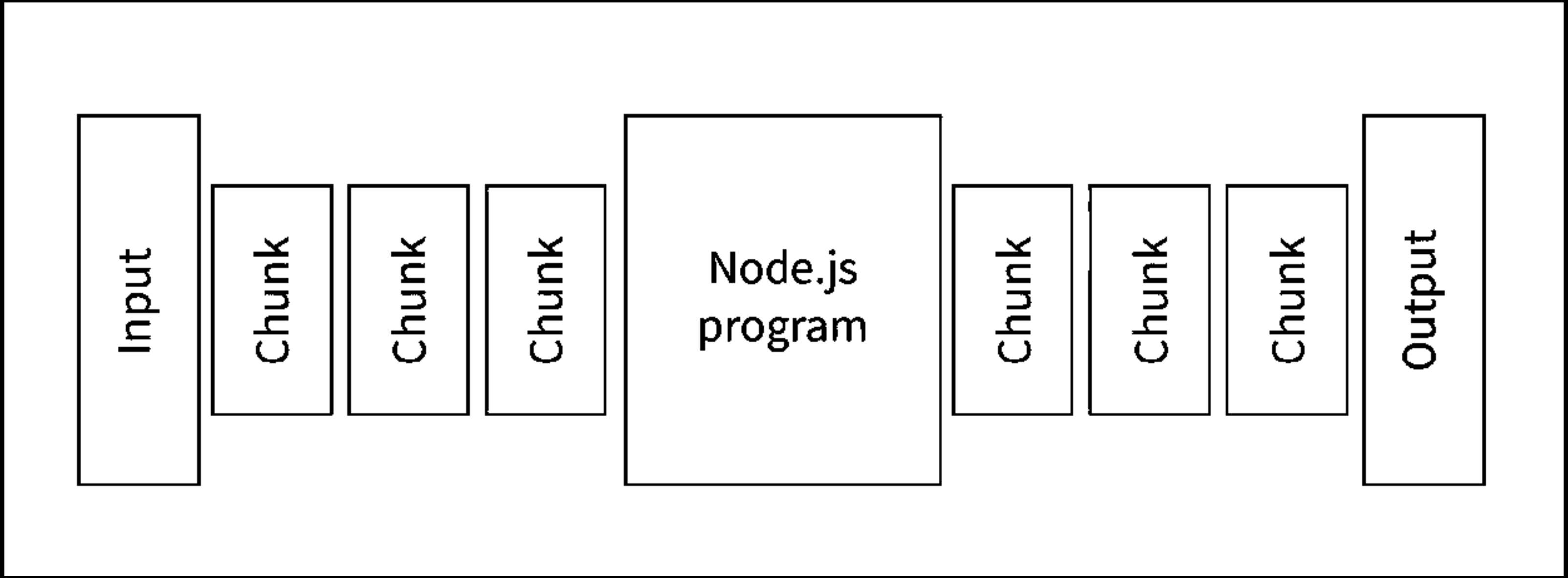
5.1 Streams



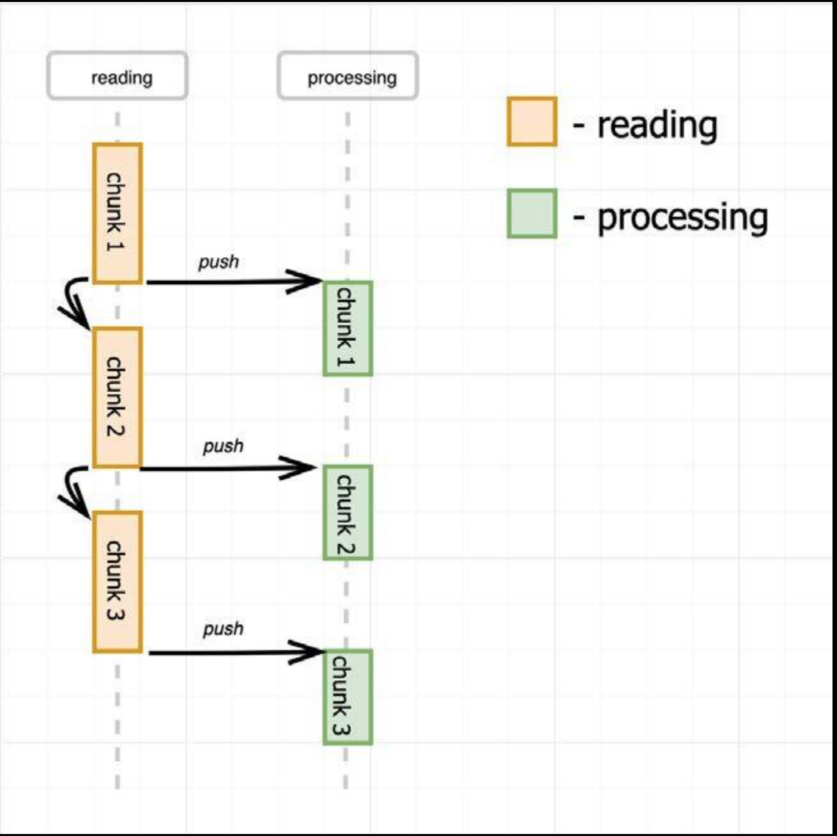
5.1 Streams


Duplex Stream
Read
read buffer
Application
write buffer
Write
Socket

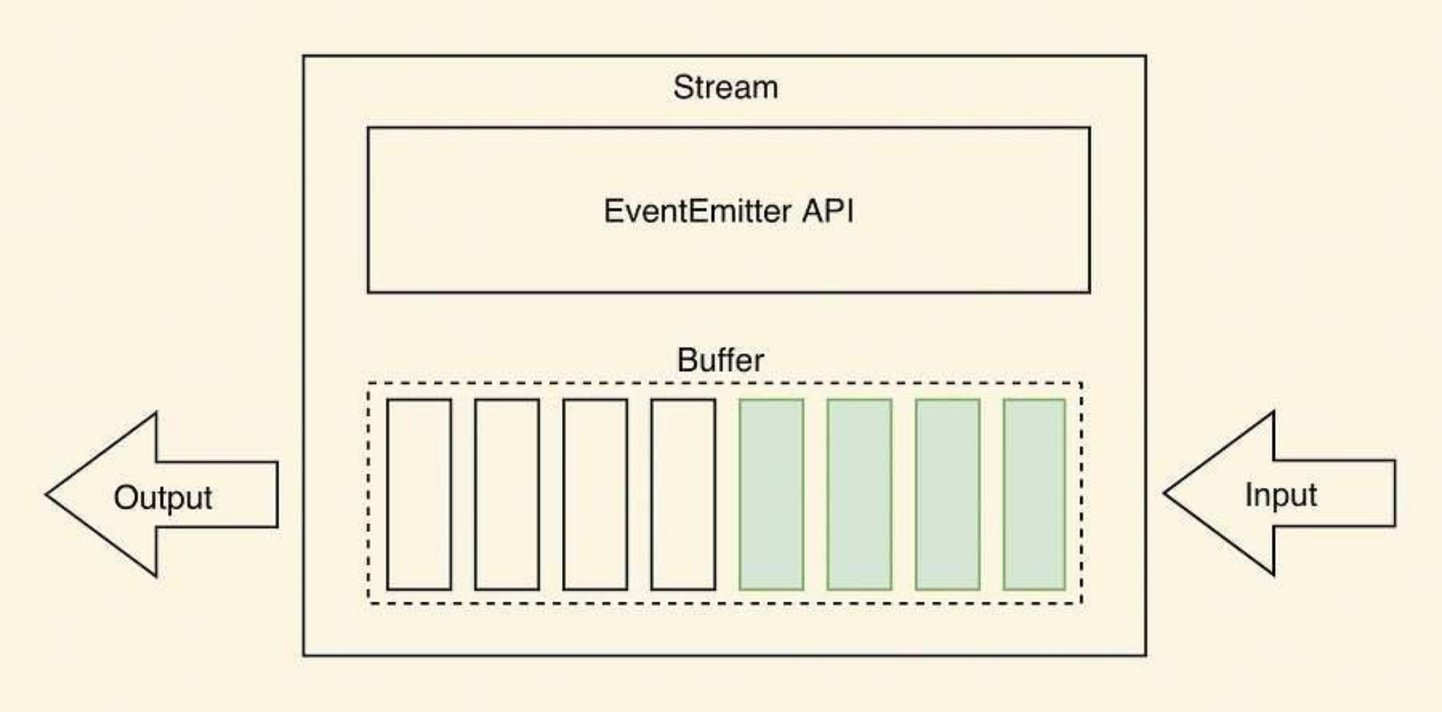
5.2 Chunks


Input
Chunk
Chunk
Chunk
Node.js program
Chunk
Chunk
Chunk
Output

5.2 Chunks


reading
processing
- reading
- processing
chunk 1
push
chunk 1
chunk 2
push
chunk 2
chunk 3
push
chunk 3

5.3 Buffers



5.4 Reading Chunk

```
} else if (
  req.method == "POST" &&
  req.url.toLowerCase() === "/submit-details"
) {
  req.on("data", (chunk) => {
    console.log(chunk);
  });
  fs.writeFileSync("user-details.txt", "Prashant Jain");
  res.setHeader("Location", "/");
  res.statusCode = 302;
  return res.end();
}
```

```
prashantjain@Prashants-Mac-mini node % node app.js
Server running at http://localhost:3000/
<Buffer 6e 61 6d 65 3d 50 72 61 73 68 61 6e 74 26 67 65 6e 64 65 72 3d 6d 61 6c 65>
```

5.5 Buffering Chunks

```
const body = [];
req.on("data", (chunk) => {
  console.log(chunk);
  body.push(chunk);
});
req.on("end", () => {
  const parsedBody = Buffer.concat(body).toString();
  console.log(parsedBody);
});
fs.writeFileSync("user-details.txt", "Prashant Jain");
```

```
prashantjain@Prashants-Mac-mini node % node app.js
Server running at http://localhost:3000/
<Buffer 6e 61 6d 65 3d 50 72 61 73 68 61 6e 74 26 67 65 6e 64 65 72 3d 6d 61 6c 65>
name=Prashant&gender=male
```

5.6 Parsing Request

```
req.on("end", () => {
  const parsedBody = Buffer.concat(body).toString();
  console.log(parsedBody);
  const params = new URLSearchParams(parsedBody);
  const jsonObject = {};
  for (const [key, value] of params.entries()) {
    jsonObject[key] = value;
  }
  console.log(jsonObject);
  // Output: { name: 'Prashant', gender: 'male' }
});
fs.writeFileSync("user-details.txt", "Prashant Jain");
res.setHeader("Location", "/");
res.statusCode = 302;
return res.end();
```

5.6 Parsing Request

```
req.on("end", () => {
  const parsedBody = Buffer.concat(body).toString();
  console.log(parsedBody);
  const params = new URLSearchParams(parsedBody);
  const jsonObject = {};
  for (const [key, value] of params.entries()) {
    jsonObject[key] = value;
  }
  const jsonString = JSON.stringify(jsonObject);
  console.log(jsonString);
  fs.writeFileSync("user-details.txt", jsonString);
});
res.setHeader("Location", "/");
res.statusCode = 302;
return res.end();
```

```
1   {"name":"Prashant","gender":"male"}
```

5.7 Using Modules

JS app.js
JS handler.js

JS handler.js > ...

```
const fs = require("fs");

const requestHandler = (req, res) => {
  if (req.url === "/") {
    res.setHeader("Content-Type", "text/html");


module.exports = requestHandler
```

JS app.js > ...

```
// Simple NodeJS server
const http = require('http');
const requestHandler = require('./handler');

const server = http.createServer(requestHandler);
```

5.7 Using Modules

```
// Method 1: Multiple exports using object
module.exports = {
  handler: requestHandler,
  extra: "Extra"
};

// Method 2: Setting multiple properties
module.exports.handler = requestHandler;
module.exports.extra = "Extra";

// Method 3: Shortcut using exports
exports.handler = requestHandler;
exports.extra = "Extra";
```

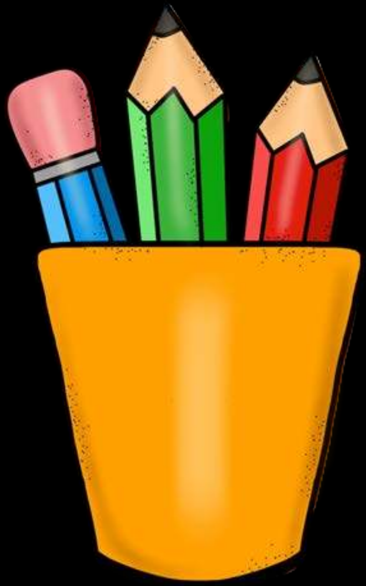
Practise Set

Create a Calculator

1. Create a new Node.js project named "Calculator".
2. On the home page (route "/"), show a welcome message and a link to the calculator page.
3. On the "/calculator" page, display a form with two input fields and a "Sum" button.
4. When the user clicks the "Sum" button, they should be taken to the "/calculate-result" page, which shows the sum of the two numbers.
    - Make sure the request goes to the server.
    - Create a separate module for the addition function.
    - Create another module to handle incoming requests.
    - On the "/calculate-result" page, parse the user input, use the addition module to calculate the sum, and display the result on a new HTML page.

Revision

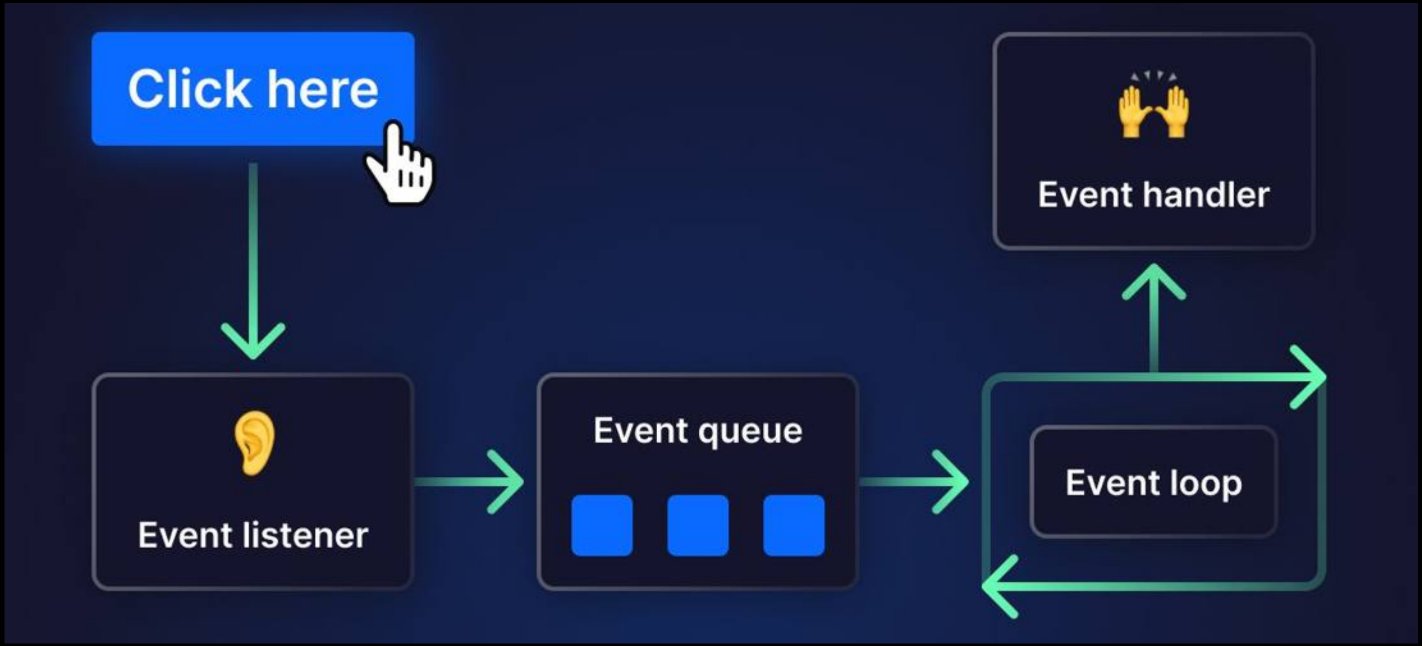
1. Streams
2. Chunks
3. Buffers
4. Reading Chunk
5. Buffering Chunks
6. Parsing Request
7. Using Modules



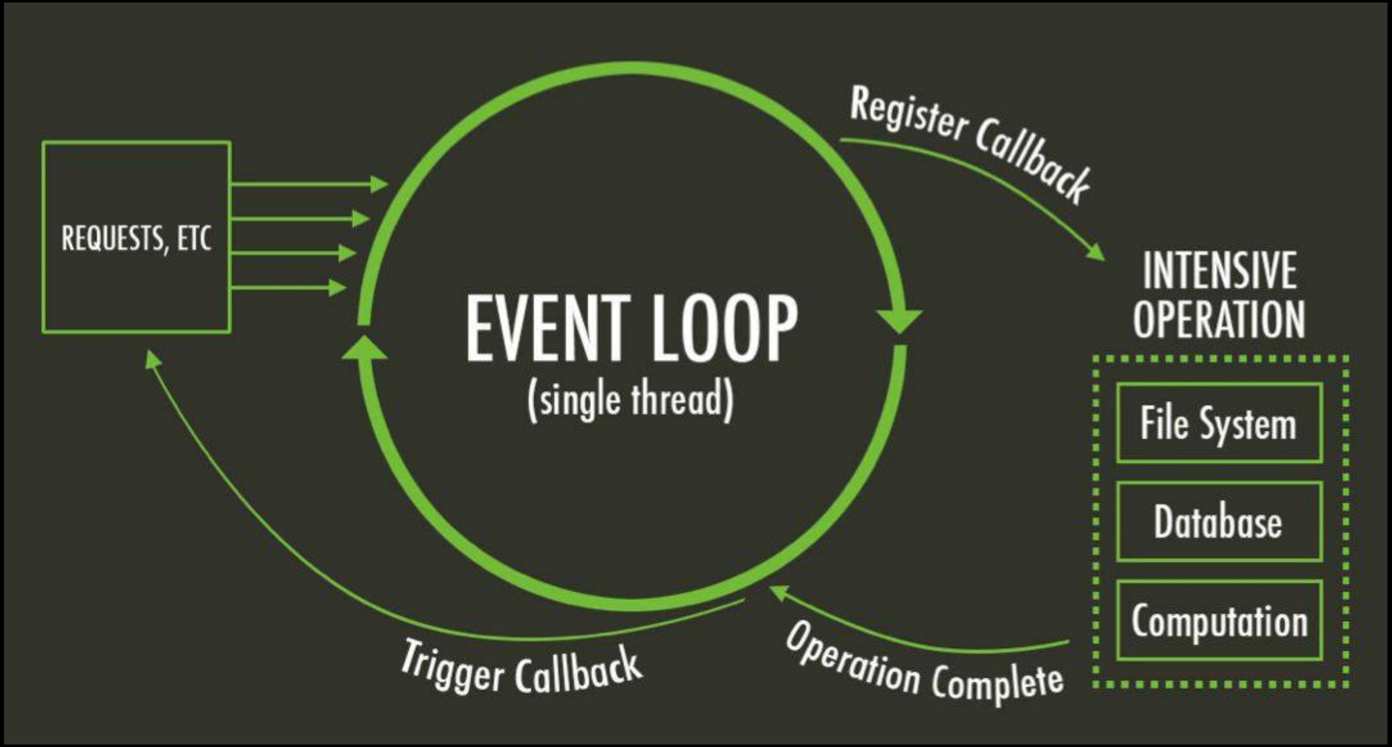
6. Event Loop

1. Event Driven
2. Single Threaded
3. V8 vs libuv
4. Node Runtime
5. Event Loop
6. Async Code
7. Blocking Code

6.1 Event Driven



6.2 Single Threaded



6.3 V8 vs libuv

V8:

1. Open-source JavaScript engine by Google.
2. Used in Chrome and Node.js.
3. Compiles JavaScript to native machine code.
4. Ensures high-performance JavaScript execution.

libuv:

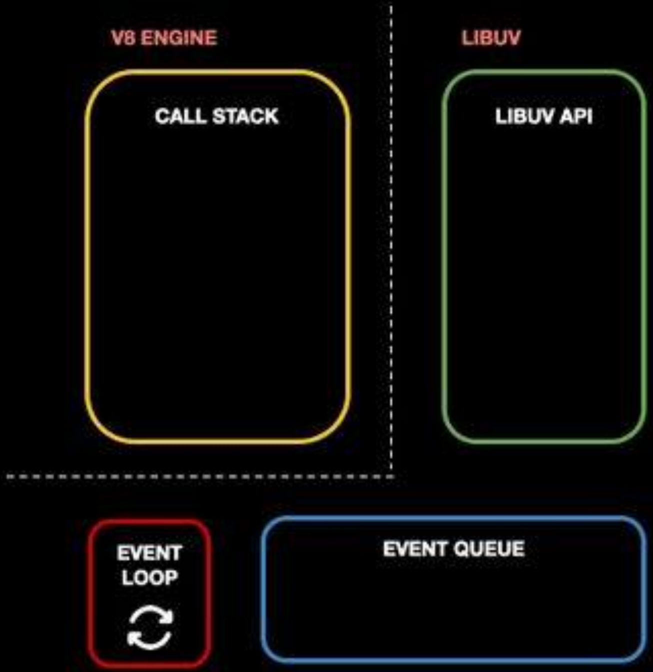
1. Multi-platform support library for Node.js.
2. Handles asynchronous I/O operations.
3. Provides event-driven architecture.
4. Manages file system, networking, and timers non-blockingly across platforms.

6.4 Node Runtime

An invoked function is added to the call stack. Once it returns a value, it is popped off.

```
console.log("Starting Node.js");

db.query("SELECT * FROM public.cars", function (err, res)  {
  console.log("Query executed");
});

console.log("Before query result");
```

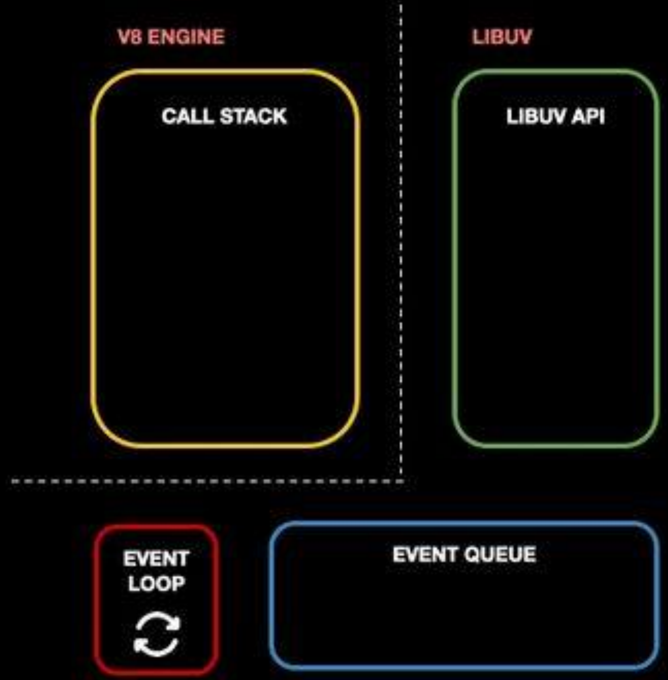
OUTPUT



6.4 Node Runtime

Database queries or other I/O ops do not block Node.js single thread because Libuv API handles them.

```
console.log("Starting Node.js");

db.query("SELECT * FROM public.cars", function (err, res)  {
  console.log("Query executed");
});

console.log("Before query result");
```



OUTPUT

```
Starting Node.js
```

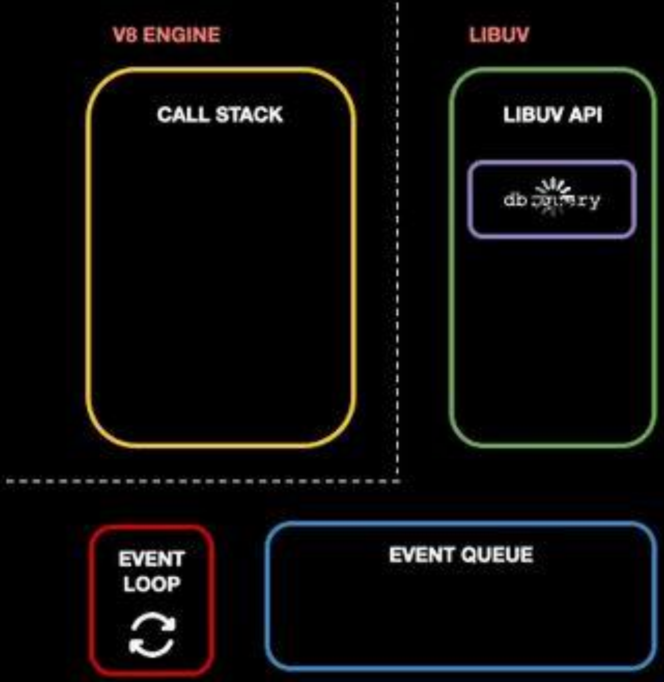
6.4 Node Runtime

While Libuv asynchronously handles I/O operations, Node.js single thread keeps running code.

```
console.log("Starting Node.js");

db.query("SELECT * FROM public.cars", function (err, res)  {
  console.log("Query executed");
});

console.log("Before query result");
```

OUTPUT

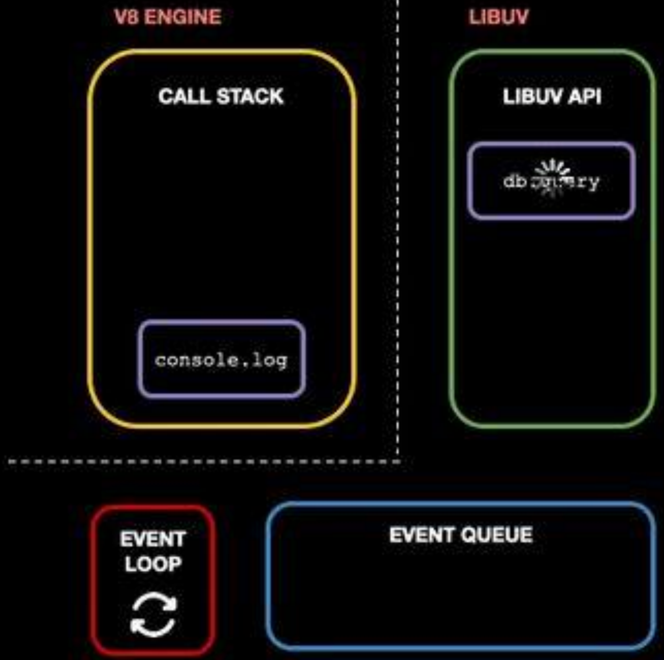
```
Starting Node.js
```



6.4 Node Runtime

Callbacks of completed queries are moved to the event queue. If the call stack is empty, the event loop checks for callbacks and transfers the first.

```
console.log("Starting Node.js");

db.query("SELECT * FROM public.cars", function (err, res) {
  console.log("Query executed");
});

console.log("Before query result");
```

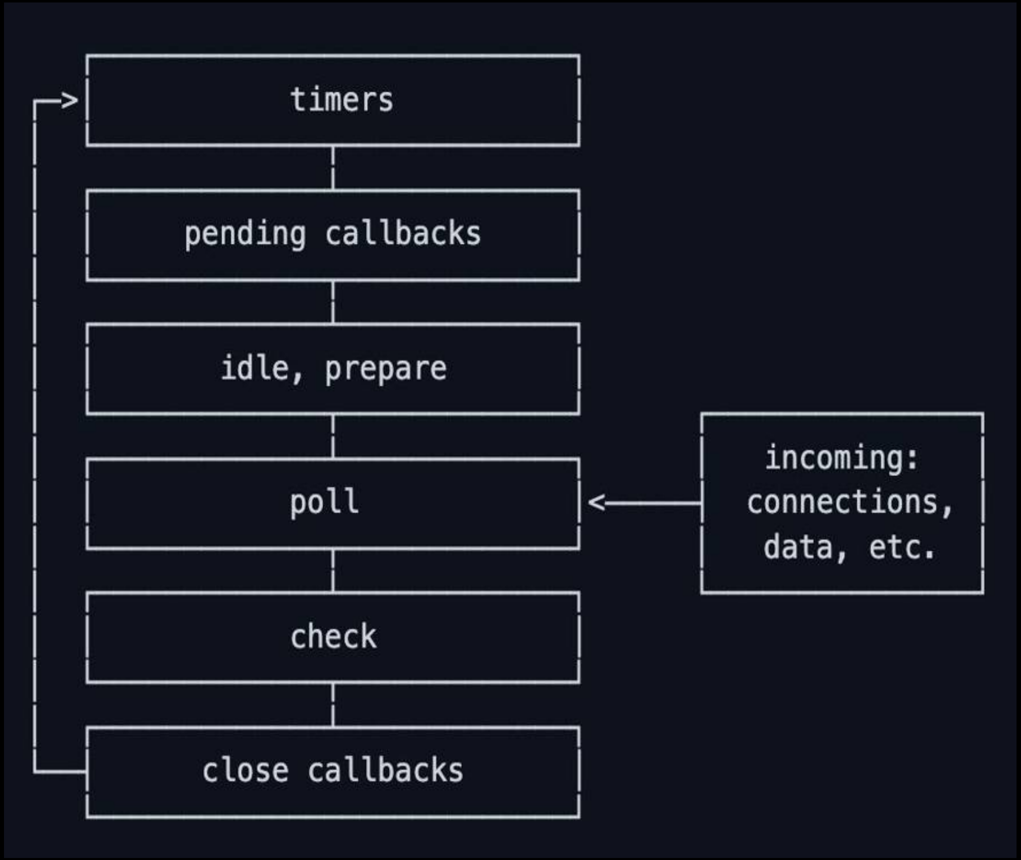


OUTPUT

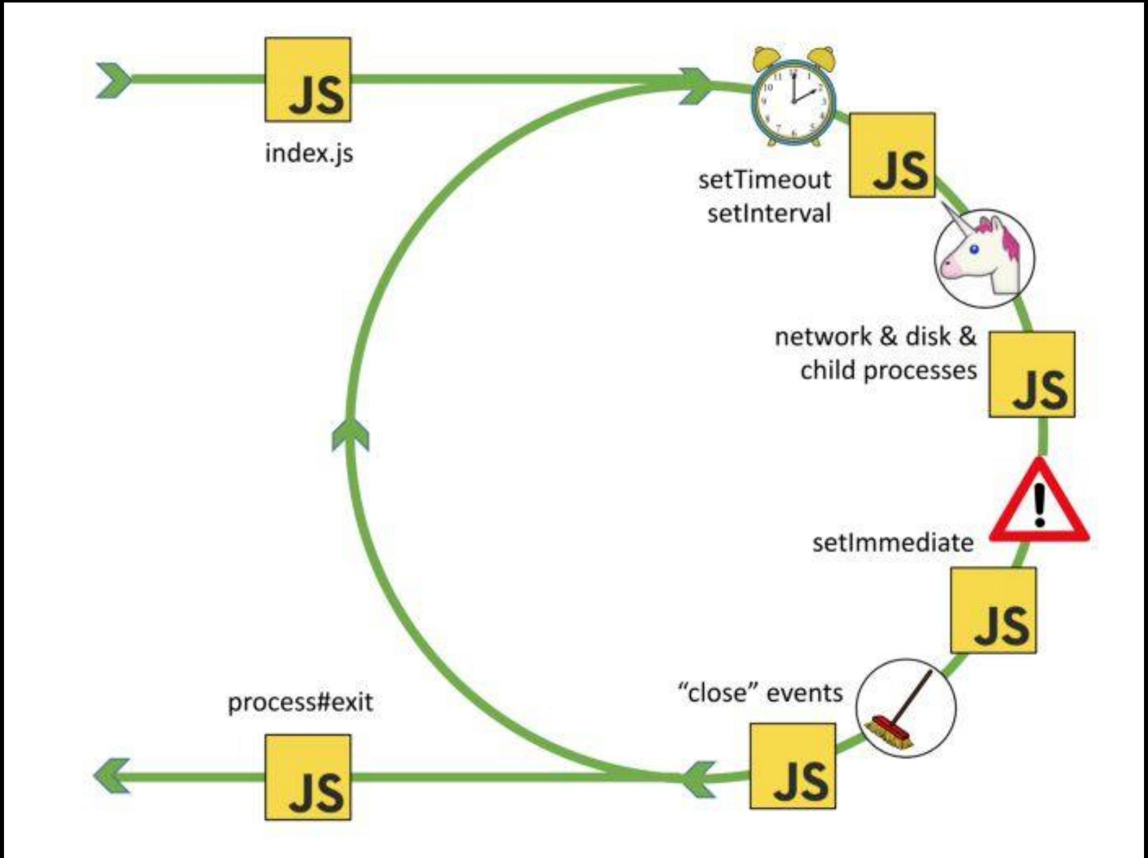
```
Starting Node.js

Before query result
```

6.5 Event Loop

- timers: this phase executes callbacks scheduled by `setTimeout()` and `setInterval()`.
- pending callbacks: executes I/O callbacks deferred to the next loop iteration.
- idle, prepare: only used internally.
- poll: retrieve new I/O events; execute I/O related callbacks (almost all with the exception of close callbacks, the ones scheduled by timers, and `setImmediate()`); node will block here when appropriate.
- check: `setImmediate()` callbacks are invoked here.
- close callbacks: some close callbacks, e.g. `socket.on('close', ...)`.



6.5 Event Loop



6.6 Async Code

```
        const jsonString = JSON.stringify(jsonObject);
        console.log(jsonString);
        fs.writeFileSync("user-details.txt", jsonString);
        res.setHeader("Location", "/");
        res.statusCode = 302;
        res.end();
      });
    }

  res.write('<body><h1>Like / Share / Subscribe</h1></
  body>');
  res.write('</html>');
  return res.end();
```

```
Error [ERR_HTTP_HEADERS_SENT]: Cannot set headers after they are sent to the client
    at ServerResponse.setHeader (node:_http_outgoing:699:11)
    at IncomingMessage.<anonymous> (/Users/prashantjain/workspace/Test Project/node/app.js:44:11)
    at IncomingMessage.emit (node:events:532:35)
    at endReadableNT (node:internal/streams/readable:1696:12)
    at process.processTicksAndRejections (node:internal/process/task_queues:82:21) {
  code: 'ERR_HTTP_HEADERS_SENT'
}
```

6.6 Async Code

```
req.on("end", () => {
  const parsedBody = Buffer.concat(body).toString();
  console.log(parsedBody);
  const params = new URLSearchParams(parsedBody);
  const jsonObject = {};
  for (const [key, value] of params.entries()) {
    jsonObject[key] = value;
  }
  const jsonString = JSON.stringify(jsonObject);
  console.log(jsonString);
  fs.writeFileSync("user-details.txt", jsonString);
  res.setHeader("Location", "/");
  res.statusCode = 302;
  return res.end();
});
```

6.7 Blocking Code

```
const jsonString = JSON.stringify(jsonObject);
console.log(jsonString);
// BLOCKING EVERTHING
fs.writeFileSync("user-details.txt", jsonString);
res.setHeader("Location", "/");
```

6.7 Blocking Code

```
console.log(jsonString);
// Async Operation
fs.writeFile("user-details.txt", jsonString, error => {
  res.setHeader("Location", "/");
  res.statusCode = 302;
  return res.end();
});
```

Run & Observe

Blocking vs Async

```
const fs = require('fs');

console.log('1. Start of script');

// Synchronous (blocking) operation
console.log('2. Reading file synchronously');
const dataSync = fs.readFileSync('user-details.txt', 'utf8');
console.log('3. Synchronous read complete');

// Asynchronous (non-blocking) operation
console.log('4. Reading file asynchronously');
fs.readFile('user-details.txt', 'utf8', (err, dataAsync) => {
 if (err) throw err;
 console.log('6. Asynchronous read complete');
});

console.log('5. End of script');
```

```
1. Start of script
2. Reading file synchronously
3. Synchronous read complete
4. Reading file asynchronously
5. End of script
6. Asynchronous read complete
```

Run & Observe

Event Loop Sequence

```
console.log('1. Start of script');

// Microtask queue (Promise)
Promise.resolve().then(() => console.log('2. Microtask 1'));

// Timer queue
setTimeout(() => console.log('3. Timer 1'), 0);

// I/O queue
const fs = require('fs');
fs.readFile('user-details.txt', () => console.log('4. I/O operation'));

// Check queue
setImmediate(() => console.log('5. Immediate 1'));

// Close queue
process.on('exit', (code) => {
 console.log('6. Exit event');
});

console.log('7. End of script');
```

```
1. Start of script
7. End of script
2. Microtask 1
3. Timer 1
5. Immediate 1
4. I/O operation
6. Exit event
```

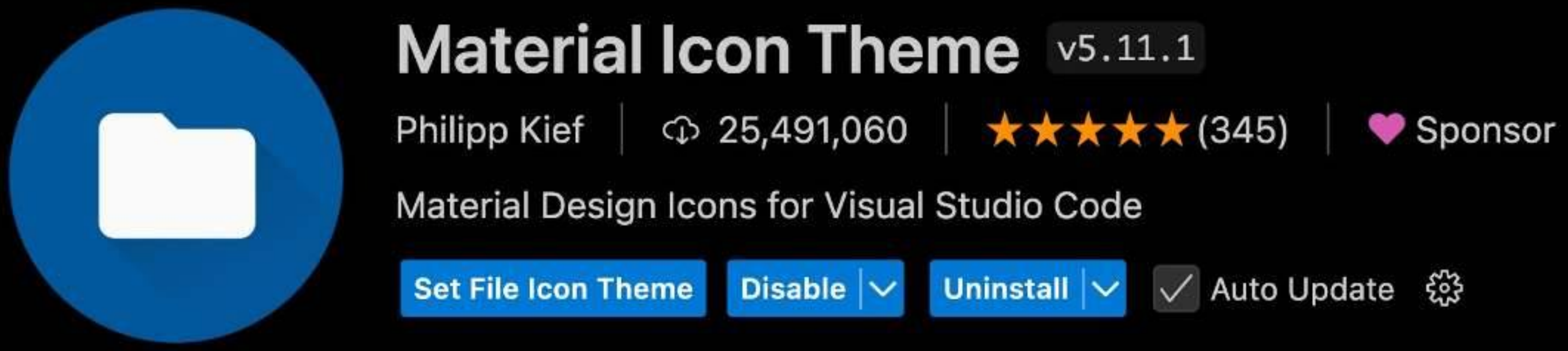
Revision

1. Event Driven
2. Single Threaded
3. V8 vs libuv
4. Node Runtime
5. Event Loop
6. Async Code
7. Blocking Code

7. NPM & Tools

1. Install Material Icons
2. npm init
3. npm Scripts
4. npm Packages
5. Installing Packages
6. Installing nodemon
7. Using nodemon

7.1 Install Material Icons



7.2 npm init

npm init

```
prashantjain@Prashants-Mac-mini user % npm init
This utility will walk you through creating a package.json file.
It only covers the most common items, and tries to guess sensible defaults.

See `npm help init` for definitive documentation on these fields
and exactly what they do.

Use `npm install <pkg>` afterwards to install a package and
save it as a dependency in the package.json file.

Press ^C at any time to quit.
package name: (user) User Backend
Sorry, name can only contain URL-friendly characters and name can no longer contain capital letters.
package name: (user) user-backend
version: (1.0.0)
description: This project will have the backend code of our User project.
entry point: (user.js) app.js
test command:
git repository:
keywords:
author: Complete Coding
license: (ISC)
About to write to /Users/prashantjain/workspace/NodeJS_Complete_YouTube/Chapter 6 - Event Loop/user/package.json:
```

7.3 npm Scripts

```
npm start
npm run khul-ja-sim-sim
```

```json
{
  "name": "user-backend",
  "version": "1.0.0",
  "description": "This project will have the backend code of our User project.",
  "main": "app.js",
  ▷ Debug
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1",
    "start": "node app.js",
    "khul-ja-sim-sim": "node app.js"
  },
  "author": "Complete Coding",
  "license": "ISC"
}
```

7.4 npm Packages

1. npm: Node.js package manager for code sharing.
2. Package: Reusable code or library.
3. package.json: Defines package metadata and dependencies.
4. Versioning: Manages different package versions.
5. Local/Global: Install packages locally or globally.
6. Registry: Public storage for open-source packages.
7. Examples: Express, React, Lodash.

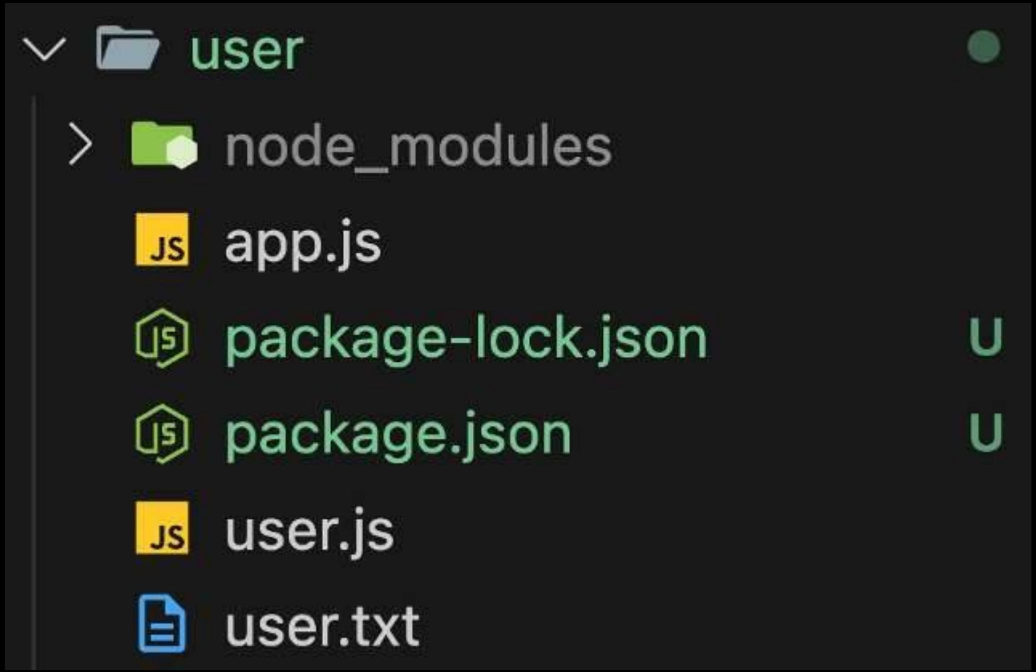
7.5 Installing Packages

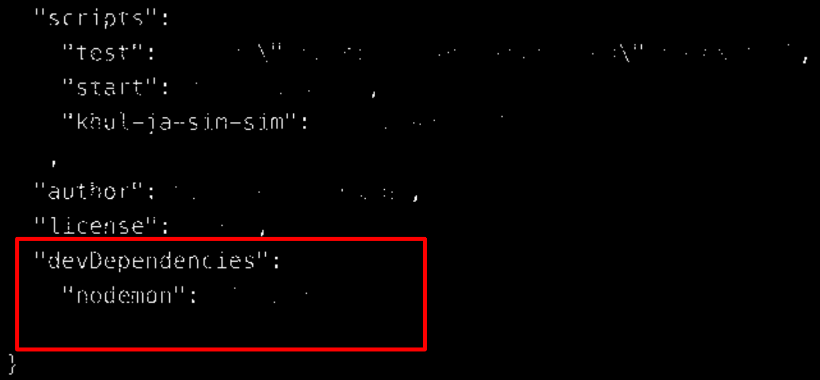
npm install <package-name>

1. -save: Adds the package to the project's dependencies in package.json.
2. -save-dev: Adds the package to the project's devDependencies (used only in development) in package.json.
3. -g: Installs the package globally, making it available system-wide, not just in a specific project.
4. -save-exact: Installs the exact version specified without updating for newer versions.
5. -force: Forces npm to fetch and install packages even if they are already installed.

7.6 Installing nodemon

npm install nodemon --save-dev





npm install

Recreates node_modules

7.7 Using nodemon

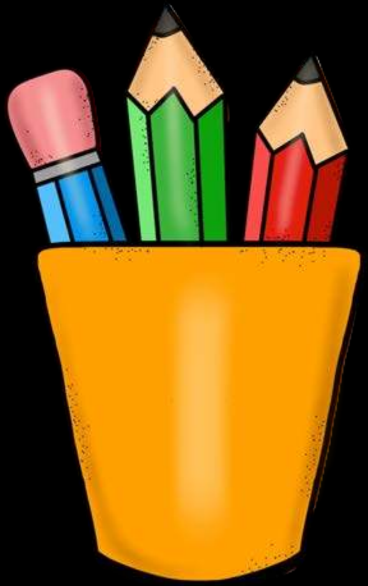
```
"scripts": {
  "test": "echo \"Error: no test specified\" && exit 1",
  "start": "nodemon app.js",
  "khul-ja-sim-sim": "node app.js"
},
```

```
prashantjain@Prashants-Mac-mini user % nodemon app.js
zsh: command not found: nodemon
prashantjain@Prashants-Mac-mini user % npm start

> user-backend@1.0.0 start
> nodemon app.js

[nodemon] 3.1.7
[nodemon] to restart at any time, enter `rs`
[nodemon] watching path(s): *.*
[nodemon] watching extensions: js,mjs,cjs,json
[nodemon] starting `node app.js`
Server running on address http://localhost:3001
```

npm install nodemon -g

Revision

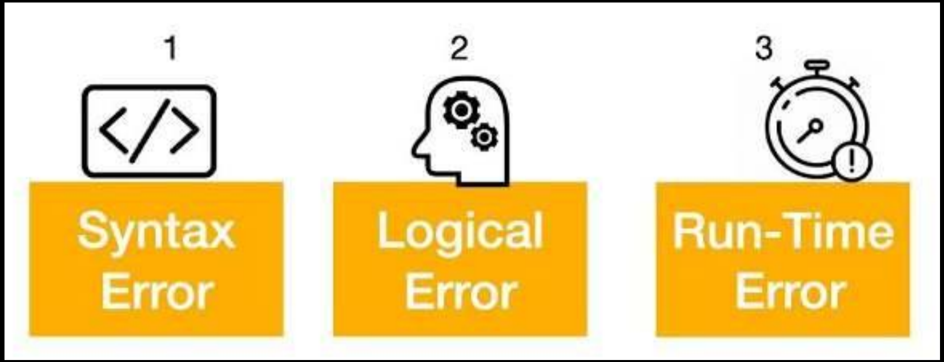
1. Install Material Icons
2. npm init
3. npm Scripts
4. npm Packages
5. Installing Packages
6. Installing nodemon
7. Using nodemon



8. Errors & Debugging

1. Types of Errors
2. Syntax Errors
3. Runtime Errors
4. Logical Errors
5. Using the Debugger
6. Debugger with Async Code
7. Restart Debug with nodemon

8.1 Types of Errors



1. Syntax Error: An error in the code's structure, causing it to not compile or run (e.g., missing semicolon).
2. Logical Error: The code runs but produces incorrect results due to faulty logic (e.g., wrong formula).
3. Runtime Error: An error that occurs while the program is running, often due to invalid operations

8.2 Syntax Errors

```
// Missing parenthesis in function call
console.log("Hello, world"

// Unclosed string literal
let message = "Welcome to Node.js;

// Improper use of reserved keywords
let new = 5;

// Incorrect variable declaration (const needs an initial value)
const myVar;
```

8.3 Runtime Errors

```
Error [ERR_HTTP_HEADERS_SENT]: Cannot set headers after they are sent to the client
    at ServerResponse.setHeader (node:_http_outgoing:699:11)
    at IncomingMessage.<anonymous> (/Users/prashantjain/workspace/Test Project/node/app.js:44:11)
    at IncomingMessage.emit (node:events:532:35)
    at endReadableNT (node:internal/streams/readable:1696:12)
    at process.processTicksAndRejections (node:internal/process/task_queues:82:21) {
  code: 'ERR_HTTP_HEADERS_SENT'
}
```

```
// Reference Error (x is not defined)
console.log(x);

// Type Error (num is not a function)
let num = 10;
num();

// Invalid JSON parse (SyntaxError)
let jsonString = "{ name: 'John' }";  // Invalid JSON (single quotes)
JSON.parse(jsonString);

// File not found error (fs module)
const fs = require('fs');
fs.readFileSync('nonexistentFile.txt');  // Throws Error: ENOENT (file not found)
```

8.4 Logical Errors

```
let x = 5;
if (x = 10) {  // Assignment instead of comparison
  console.log("x is 10");  // Incorrectly prints this
}

let arr = [1, 2, 3, 4, 5];
for (let i = 0; i <= arr.length; i++) {
  console.log(arr[i]);  // Prints undefined at the end of the loop
}

let num = "10";
console.log(num + 5);  // Expected result: 15, prints 105
```

8.5 Using the Debugger

Step 1

Go Run Terminal Window Help

Start Debugging

Run Without Debugging

Step 2

Select debugger

Node.js

VS Code Extension Development

Web App (Chrome)

Web App (Edge)

Install an extension for JavaScript...

Step 3: Put a breakpoint

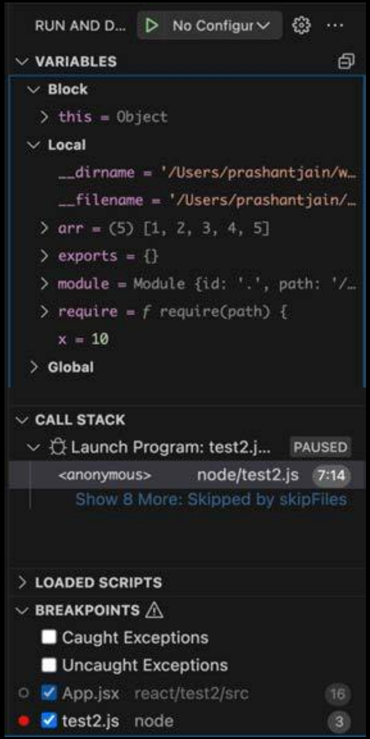
```
let x = 5;
if (x = 10) {  // Assignment instead of comparison
    console.log("x is 10");  // Incorrectly prints this
}
```

8.5 Using the Debugger

Step 4: Use the tools

Step 5: Hover

Step 6: Debug Panel



8.5 Using the Debugger

Step 7: Using Debug Console

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
/opt/homebrew/bin/node ./node/test2.js
→ x
10
→ x++
10
→ x === 15
false
>
```

8.6 Debugger with Async Code

```
  console.log("Here");
  req.on("end", () => {
    const parsedBody = Buffer.concat(body).toString();
    console.log(parsedBody);
    const params = new URLSearchParams(parsedBody);
    const jsonObject = {};
    for (const [key, value] of params.entries()) {
      jsonObject[key] = value;
    }
    const jsonString = JSON.stringify(jsonObject);
    console.log(jsonString);
    // Async Operation
    fs.writeFile("user-details.txt", jsonString, error => {
      res.setHeader("Location", "/");
      res.statusCode = 302;
      return res.end();
    });
  });
}
```

8.7 Restart Debug with nodemon

Run Terminal Window

Start Debugging

Run Without Debugging

Open Configurations

Add Configuration...

```
"version": "0.2.0",
"configurations": [
  {
    "type": "node",
    "request": "launch",
    "name": "Launch Program",
    "skipFiles": [
      "<node_internals>/**"
    ],
    "program": "${workspaceFolder}/node/test2.js",
    "restart": true,
    "runtimeExecutable": "nodemon",
    "console": "integratedTerminal"
  }
]
}
```

Practise Set

Debug and fix Syntax, Runtime and Logical Errors

```
function calculateArea(width, height {
 return width + height;
}

let width = 10 height = 5;

if (area > 100) {
 console.log("The area is large.");
} else {
 console.log("The area is small.");
}

if (width + height > 100) {
 console.log("Area is greater than or equal to 100");
}
```

Revision

1. Types of Errors
2. Syntax Errors
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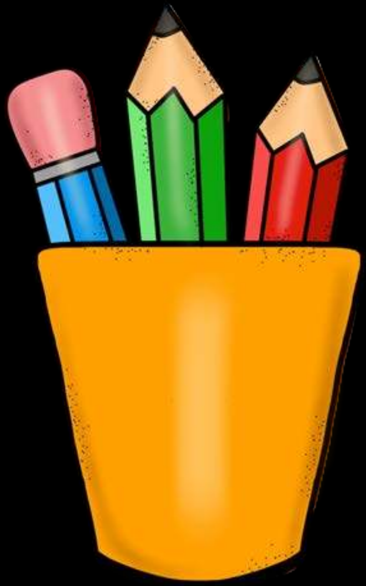
4. Logical Errors
5. Using the Debugger
6. Debugger with Async Code
7. Restart Debug with nodemon

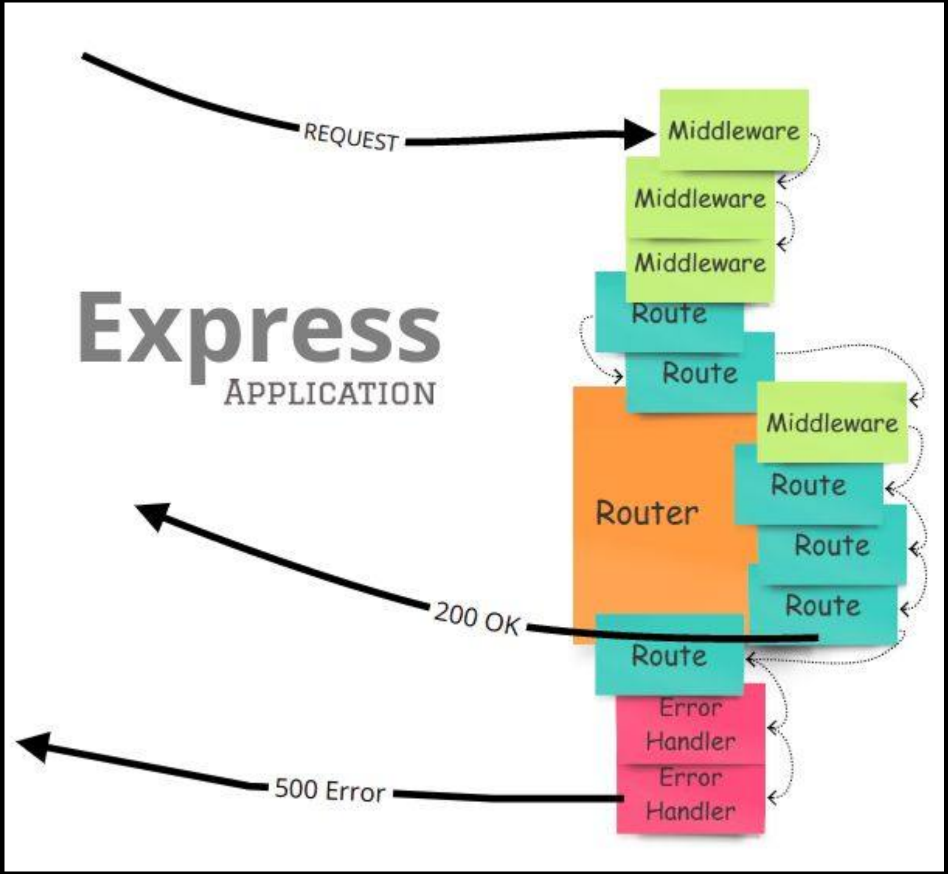


9. Starting with Express.js

1. What is Express.js
2. Need of Express.js
3. Installing Express.js
4. Adding Middleware
5. Sending Response
6. Express DeepDive
7. Handling Routes

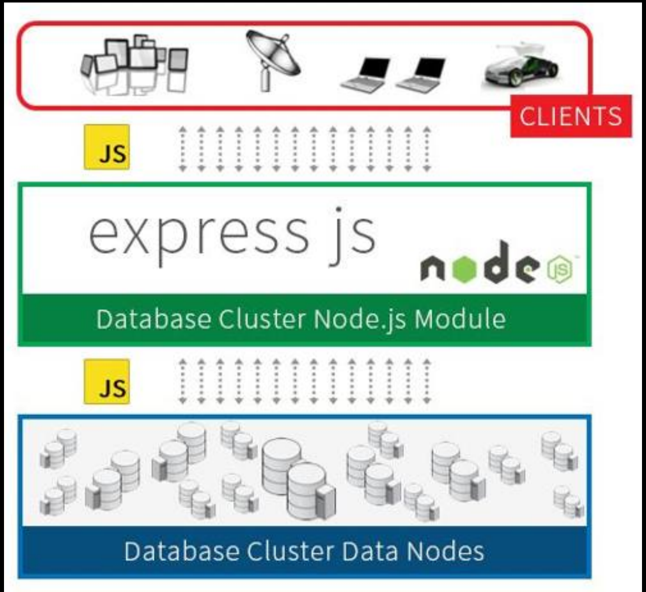
9.1 What is Express.js

1. Express.js is a minimal and flexible web application framework for Node.js.
2. It provides a robust set of features for building single-page, multi-page, and hybrid web applications.
3. Express.js simplifies server-side coding by providing a layer of fundamental web application features.



9.2 Need of Express.js

1. Express.js Simplifies Server Creation: Helps in quickly setting up and running a web server without the need for complex coding.
2. Routing Management: Provides a powerful routing mechanism to handle URLs and HTTP methods effectively.
3. Middleware Support: Allows the use of middleware to handle requests, responses, and any middle operations, making code modular and maintainable.
4. API Development: Facilitates easy and efficient creation of RESTful APIs.
5. Community and Plugins: Has a large ecosystem with numerous plugins and extensions, accelerating development time.



9.3 Installing Express.js

EXPRESS Js

```
$ npm install express
```

To install Express temporarily and not add it to the dependencies list:

```
$ npm install express --no-save
```

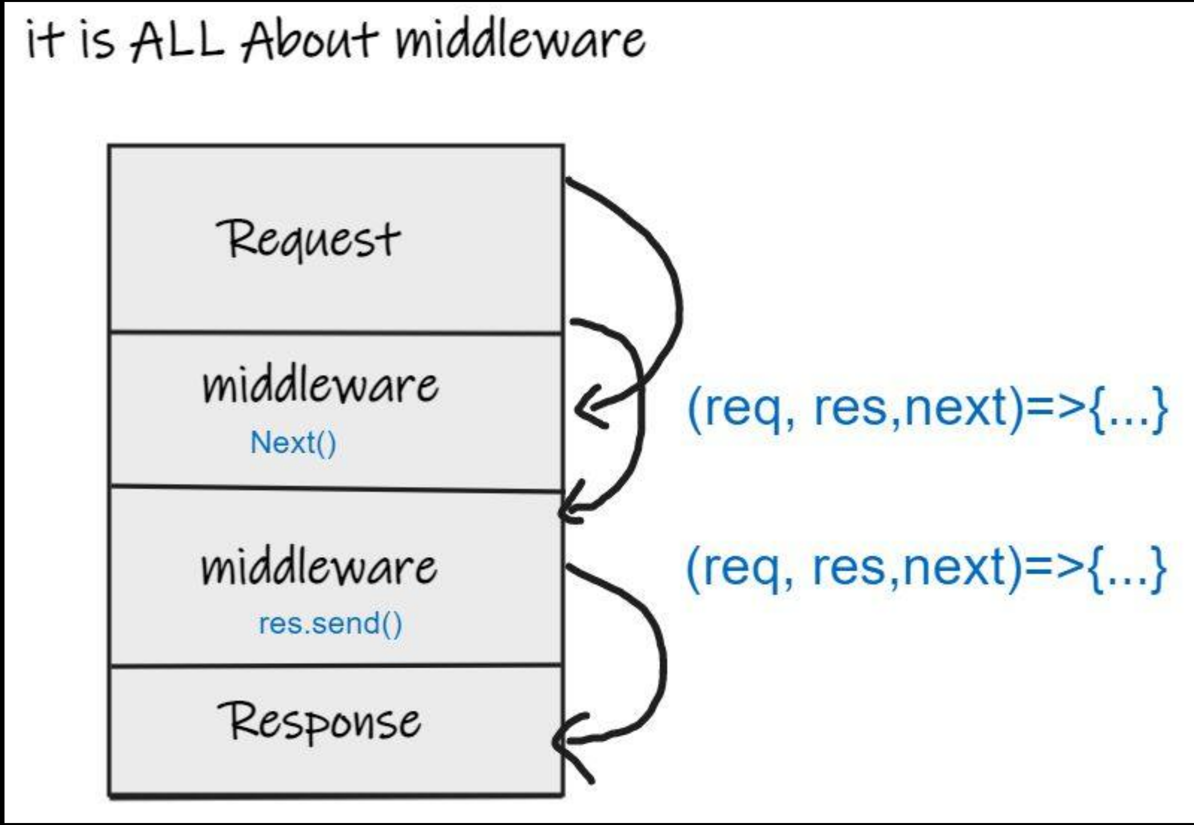
By default with version npm 5.0+, `npm install` adds the module to the `dependencies` list in the `package.json` file; with earlier versions of npm, you must specify the `--save` option explicitly. Then, afterwards, running `npm install` in the app directory will automatically install modules in the dependencies list.

9.3 Installing Express.js

```javascript
// Core Modules
const http = require('http');
// External Modules
const express = require('express');

const app = express();

const server = http.createServer(app);

const PORT = 3000;
server.listen(PORT, () => {
  console.log(`Server running at http://localhost:${PORT}/`);
});
```

```
prashantjain@Prashants-Mac-mini node % npm start

> node@1.0.0 start
> nodemon app.js

[nodemon] 3.1.7
[nodemon] to restart at any time, enter `rs`
[nodemon] watching path(s): *.*
[nodemon] watching extensions: js,mjs,cjs,json
[nodemon] starting `node app.js`
Server running at http://localhost:3000/
```

9.4 Adding Middleware

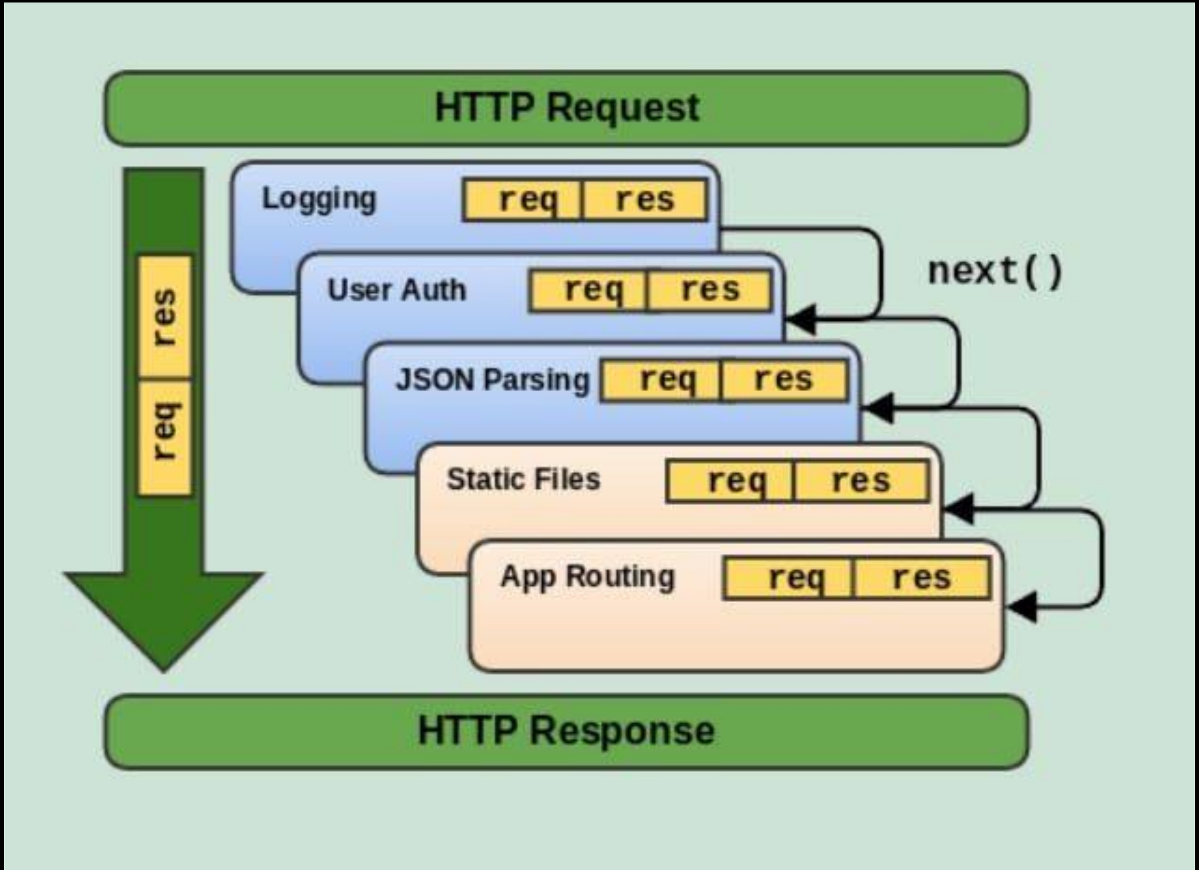


9.4 Adding Middleware

```
// Adding Middleware
app.use((req, res, next) => {
  console.log("First Middleware", req.url, req.method);
  next();
});

app.use((req, res, next) => {
  console.log("Second Middleware", req.url, req.method);
});

const server = http.createServer(app);
```

```
[nodemon] restarting due to changes...
[nodemon] starting `node app.js`
Server running at http://localhost:3000/
First Middleware / GET
Second Middleware / GET
```

9.4 Adding Middleware



9.5 Sending Response

```
app.use((req, res, next) => {
  console.log("Second Middleware", req.url, req.method);
  res.send('<p>Welcome to Complete Coding</p>');
});
```

localhost:3000

localhost:3000

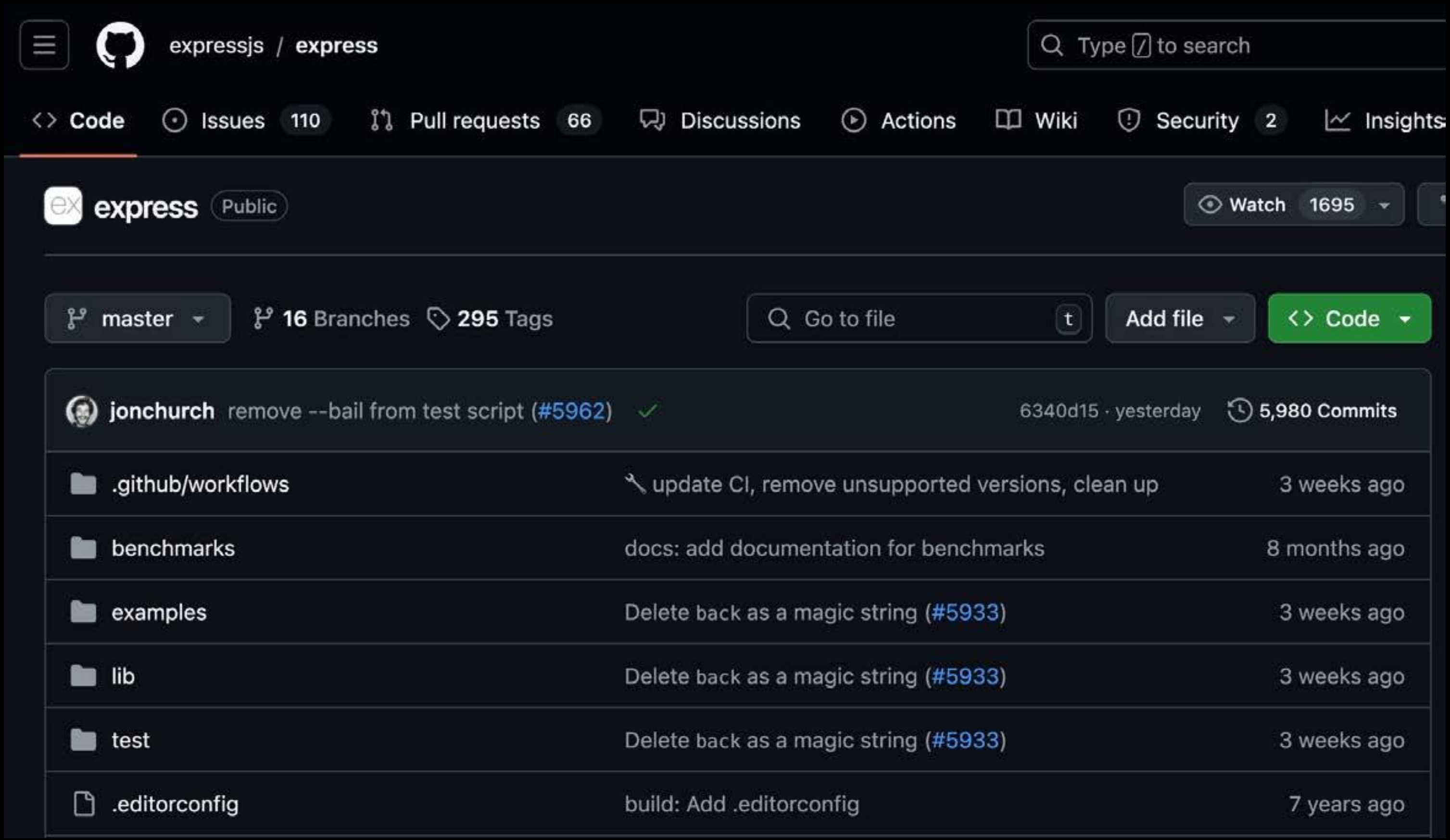
Welcome to Complete Coding

▾Response Headers

Raw

Connection:	keep-alive
Content-Length:	33
Content-Type:	text/html; charset=utf-8
Date:	Wed, 02 Oct 2024 07:45:48 GMT
Etag:	W/"21-FoFd61RXqayGnfodqIQO62RgMzY"
Keep-Alive:	timeout=5
X-Powered-By:	Express

9.6 Express DeepDive



9.6 Express DeepDive

EXPRESS Js

```
/**
 * Send a response.
 *
 * Examples:
 *
 *     res.send(Buffer.from('wahoo'));
 *     res.send({ some: 'json' });
 *     res.send('<p>some html</p>');
 *
 * @param {string|number|boolean|object|Buffer} body
 * @public
 */

res.send = function send(body) {
  var chunk = body;
  var encoding;
  var req = this.req;
  var type;

  // settings
  var app = this.app;

```

```
// string defaulting to html
case 'string':
  if (!this.get('Content-Type')) {
    this.type('html');
  }
  break;
case 'boolean':
```

9.6 Express DeepDive

```
// Core Modules
const http = require('http');
// External Modules
const express = require('express');

const app = express();

// Adding Middleware
app.use((req, res, next) => {
  console.log("First Middleware", req.url, req.method);
  next();
});

app.use((req, res, next) => {
  console.log("Second Middleware", req.url, req.method);
  res.send('<p>Welcome to Complete Coding</p>');
});

const PORT = 3000;
app.listen(PORT, () => {
  console.log(`Server running at http://localhost:${PORT}/`);
});
```

9.7 Handling Routes

EXPRESS Js

app.use([path,] callback [, callback...])

Mounts the specified middleware function or functions at the specified path: the middleware function is executed when the base of the requested path matches `path`.

Arguments

Argument	Description	Default
`path`	The path for which the middleware function is invoked; can be any of: • A string representing a path. • A path pattern. • A regular expression pattern to match paths. • An array of combinations of any of the above. For examples, see Path examples.	'/' (root path)

9.7 Handling Routes

```
const app = express();
app.use('/', (req, res, next) => {
  console.log("Hello World");
  //res.send('<p>Welcome to Submit Details Page</p>');
  next();
});

app.use('/submit-details', (req, res, next) => {
  console.log("Second Middleware", req.url, req.method);
  res.send('<p>Welcome to Submit Details Page</p>');
});

app.use('/', (req, res, next) => {
  console.log("Second Middleware", req.url, req.method);
  res.send('<p>Welcome to Complete Coding</p>');
});
```

1. Order matters
2. Can not call next() after send()
3. “/” matches everything
4. Calling res.send implicitly calls res.end()

9.7 Handling Routes

```
var express = require('express');
var app = express();

app.get('/', function(req, res, next) {
  next();
})

app.listen(3000);
```

HTTP method for which the middleware function applies.

Path (route) for which the middleware function applies.

The middleware function.

Callback argument to the middleware function, called "next" by convention.

HTTP response argument to the middleware function, called "res" by convention.

HTTP request argument to the middleware function, called "req" by convention.

9.7 Handling Routes

```
const app = express();
app.post('/', (req, res, next) => {
  console.log("Hello World");
  //res.send('<p>Welcome to Submit Details Page</p>');
  next();
});

app.use('/submit-details', (req, res, next) => {
  console.log("Second Middleware", req.url, req.method);
  res.send('<p>Welcome to Submit Details Page</p>');
});

app.use('/', (req, res, next) => {
  console.log("Second Middleware", req.url, req.method);
  res.send('<p>Welcome to Complete Coding</p>');
});
```

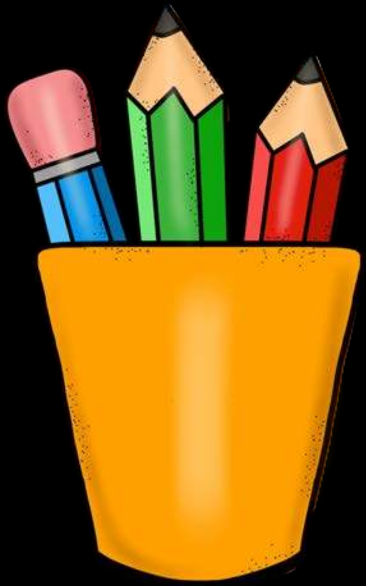
Practise Set

Create a new project.

1. Install nodemon and express.
2. Add two dummy middleware that logs request path and request method respectively.
3. Add a third middleware that returns a response.
4. Now add handling using two more middleware that handle path /, a request to /contact-us page.
5. Contact us should return a form with name and email as input fields that submits to /contact-us page also.
6. Also handle POST incoming request to /contact-us path using a separate middleware.

Revision

1. What is Express.js
2. Need of Express.js
3. Installing Express.js
4. Adding Middleware
5. Sending Response
6. Express DeepDive
7. Handling Routes



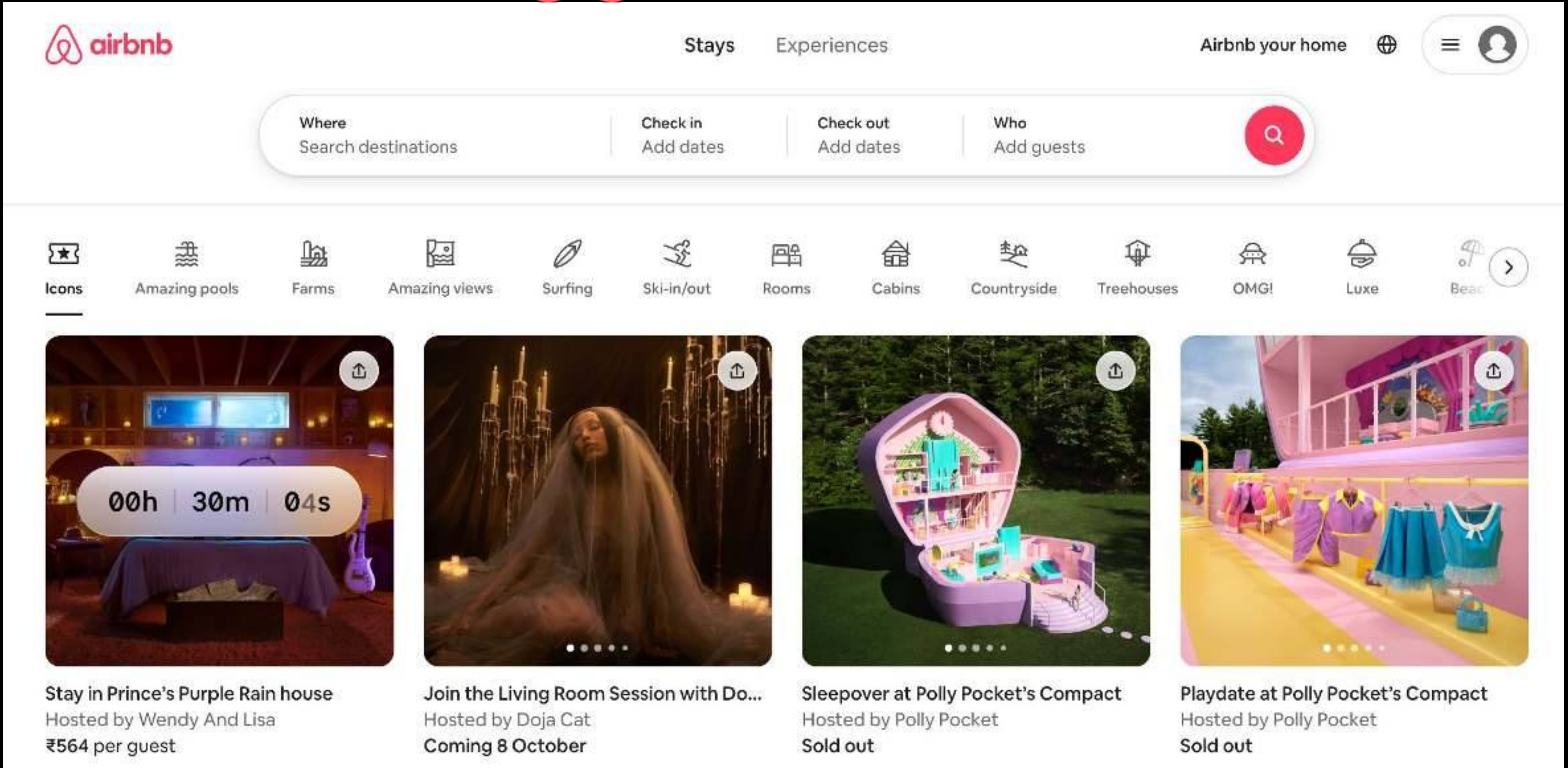
10. Express.js DeepDive

1. Parsing Requests
2. Express Router
3. Adding 404
4. Common Paths
5. Adding HTML Files
6. Serving HTML Files
7. Using File Helper

10.1 Parsing Requests

```
// External Modules
const express = require('express');
const bodyParser = require('body-parser');

const app = express();

app.use(bodyParser.urlencoded());

app.get('/details', (req, res, next) => {
  console.log("Hello World");
  res.send(`
    <h1>Enter Your Details:</h1>
    <form action="/submit-details" method="POST">
      <input type="text" name="username" placeholder="Enter your name"><br>
      <br><input type="submit" value="Submit">
    </form>
  `);
});

app.post('/submit-details', (req, res, next) => {
  console.log(req.url, req.method, req.body);
  res.redirect('/');
});
```

node js airbnb EXPRESS Js



10.2 Express Router

node > air-bnb > app.js > ...

```
// External Modules
const express = require('express');
// Local Modules
const hostRouter = require('./routes/host');
const userRouter = require('./routes/user');

const app = express();

app.use(express.urlencoded({extended:true}));
app.use(hostRouter);
app.use(userRouter);

const PORT = 3000;
app.listen(PORT, () => {
  console.log(`Server running at http://localhost:${PORT}/`);
});
```

10.2 Express Router

node > air-bnb > routes > host.js > ...

```
// External Modules
const express = require('express');

const router = express.Router();

router.get('/add-home', (req, res, next) => {
  res.send(`
    <h1>Register Your Home on AirBnB</h1>
    <form action="/submit-home" method="POST">
      <input type="text" name="houseName" placeholder="Enter your House Name"><br>
      <br><input type="submit" value="Submit">
    </form>
  `);
});

router.post('/submit-home', (req, res, next) => {
  console.log("Post Request details", req.url, req.method, req.body);
  res.redirect('/');
});

module.exports = router;
```

10.2 Express Router

node > air-bnb > routes > user.js > ...

```
// External Modules
const express = require('express');
const userRouter = express.Router();

userRouter.use('/', (req, res, next) => {
  console.log("Second Middleware", req.url, req.method);
  res.send(`
    <h2>Welcome to AirBnb</h2>
    <a href="/add-home">Register Your Home on AirBnb</a>
  `);
});

module.exports = userRouter;
```

10.3 Adding 404

```javascript
app.use(express.urlencoded({extended:true}));

app.use(hostRouter);
app.use(userRouter);

app.use((req, res, next) => {
  res.status(404).send('<h1>Page Not Found</h1>');
});

const PORT = 3000;
app.listen(PORT, () => {
  console.log(`Server running at http://localhost:${PORT}/`);
});
```

localhost:3000/hi

localhost:3000/hi

Page Not Found

10.4 Common Paths

```
// GET path
router.get('/host/add-home', (req, res, next) => {
  res.send(`
    <h1>Register Your Home on AirBnB</h1>
    <form action="/add-home" method="POST">
      <input type="text" name="houseName"
      placeholder="Enter your House Name"><br>
      <br><input type="submit" value="Submit">
    </form>
  `);
});

// POST path
router.post('/host/add-home', (req, res, next) => {
  console.log("Post Request details", req.url, req.
  method, req.body);
  res.redirect('/');
});
```

or

```
const app = express();

app.use(express.urlencoded({extended:true}));

app.use("/host", hostRouter);
app.use("/user", userRouter);

app.use((req, res, next) => {
  res.status(404).send('<h1>Page Not Found</h1>');
});
```

10.5 Adding HTML Files

EXPRESS Js

air-bnb > views > add-home.html > ...

```html
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width,
    initial-scale=1.0" />
    <title>airbnb</title>
  </head>
  <body>
    <header>
      <nav>
        <ul>
          <li><a href="/">Go to Home</a></li>
          <li><a href="/host/add-home">Add Home</a></li>
        </ul>
      </nav>
    </header>
    <main>
      <h1>Register Your Home on AirBnB</h1>
      <form action="/host/add-home" method="POST">
        <input
          type="text"
          name="houseName"
          placeholder="Enter your House Name"
        /><br />
        <br /><input type="submit" value="Submit" />
      </form>
    </main>
  </body>
</html>
```

air-bnb > views > user.html > ...

```html
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width,
    initial-scale=1.0" />
    <title>airbnb</title>
  </head>
  <body>
    <header>
      <nav>
        <ul>
          <li><a href="/">Go to Home</a></li>
          <li><a href="/host/add-home">Add Home</a></li>
        </ul>
      </nav>
    </header>
    <main>
      <h2>Welcome to AirBnb</h2>
    </main>
  </body>
</html>
```

```
air-bnb
├── node_modules
├── routes
│   ├── host.js
│   └── user.js
├── views
│   ├── 404.html
│   ├── add-home.html
│   └── user.html
├── app.js
├── package-lock.json
└── package.json
```

10.6 Serving HTML Files

air-bnb > routes > host.js > ...

```javascript
// Core Modules
const path = require('path');

// External Modules
const express = require('express');

const router = express.Router();

// GET path
router.get('/add-home', (req, res, next) => {
  res.sendFile(path.join(__dirname, '../', 'views', 'add-home.html'));
});
```

10.7 Using File Helper

air-bnb > util > path.js > ...

```
// Core Modules
const path = require('path');

module.exports = path.dirname(require.main.filename);
```

```
// Local Modules
const rootDir = require('../util/path');

userRouter.get('/', (req, res, next) => {
  console.log("Second Middleware", req.url, req.method);
  res.sendFile(path.join(rootDir, 'views', 'user.html'));
});
```

```
air-bnb
  node_modules
  routes
    host.js
    user.js
  util
    path.js
  views
  app.js
  package-lock.json
  package.json
```

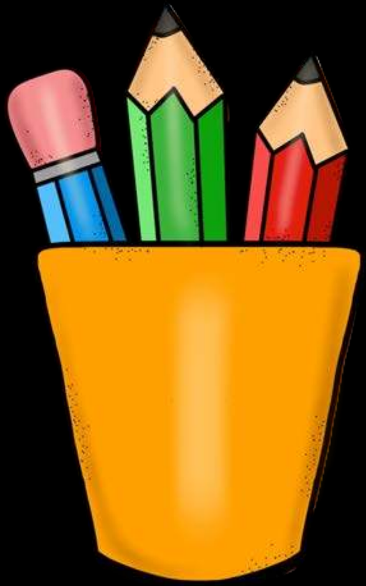
Practise Set

Reuse the app from the last assignment

1. Parse the body of the contact-us request and log it to console.
2. Move the code to separate local modules and use the Express router to import and use them in app.js
3. Move all the html code to html files and serve them using the file helper.
4. Also add a 404 page for this app.

Revision

1. Parsing Requests
2. Express Router
3. Adding 404
4. Common Paths
5. Adding HTML Files
6. Serving HTML Files
7. Using File Helper



11. Styling using tailwindcss

1. Serving Static Files
2. Introduction to Tailwind CSS
3. Utility Classes
4. Installing Extension
5. Including Tailwind CSS
6. Installing Tailwind CSS
7. Using Tailwind CSS
8. Building Tailwind Automatically

11.1 Serving Static Files

```html
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width,
    initial-scale=1.0" />
    <title>airbnb</title>
    <link rel="stylesheet" href="airbnb.css">
  </head>
  <body>
    <header>
      <nav>
        <ul>
          <li><a href="/">Go to Home</a></li>
          <li><a href="/host/add-home">Add Home</a></li>
        </ul>
      </nav>
    </header>
    <main>
      <h2>Welcome to AirBnb</h2>
    </main>
  </body>
</html>
```

```css
body {
  font-family: 'Arial', sans-serif;
  margin: 0;
  padding: 0;
  background-color: #f5f5f5;
}

header {
  background-color: #ff5a5f;
  padding: 1rem 2rem;
  box-shadow: 0px 2px 8px rgba(0, 0, 0, 0.1);
}

nav {
  display: flex;
  justify-content: space-between;
  align-items: center;
}

nav ul {
  list-style: none;
  margin: 0;
  padding: 0;
  display: flex;
}

nav li {
  margin-left: 20px;
}
```

11.1 Serving Static Files

```css
nav a {
  text-decoration: none;
  color: white;
  font-weight: bold;
  padding: 0.5rem 1rem;
  border-radius: 4px;
  transition: background-color 0.3s ease;
}

nav a:hover {
  background-color: #e54e52;
}

main {
  text-align: center;
  padding: 5rem 2rem;
  background-color: #fff;
  box-shadow: 0 4px 8px rgba(0, 0, 0, 0.1);
  margin: 2rem auto;
  max-width: 600px;
  border-radius: 8px;
}

h2 {
  color: #333;
  font-size: 2rem;
}
```


```javascript
// Granting access to public folder
app.use(express.static(path.join(rootDir, "public")));
```

11.2 Introduction to Tailwind CSS

1. Responsive: Mobile-first design for all device sizes.
2. Utility-First: Provides low-level utility classes for building custom designs.
3. Highly Customizable: Easily extendable through a config file.
4. Responsive Design: Built-in responsive utilities (e.g., sm:, md:).
5. No Predefined Components: Focuses on building custom components.
6. Purge CSS: Removes unused styles in production for smaller files.
7. Fast Development: Style elements directly in markup for speed.

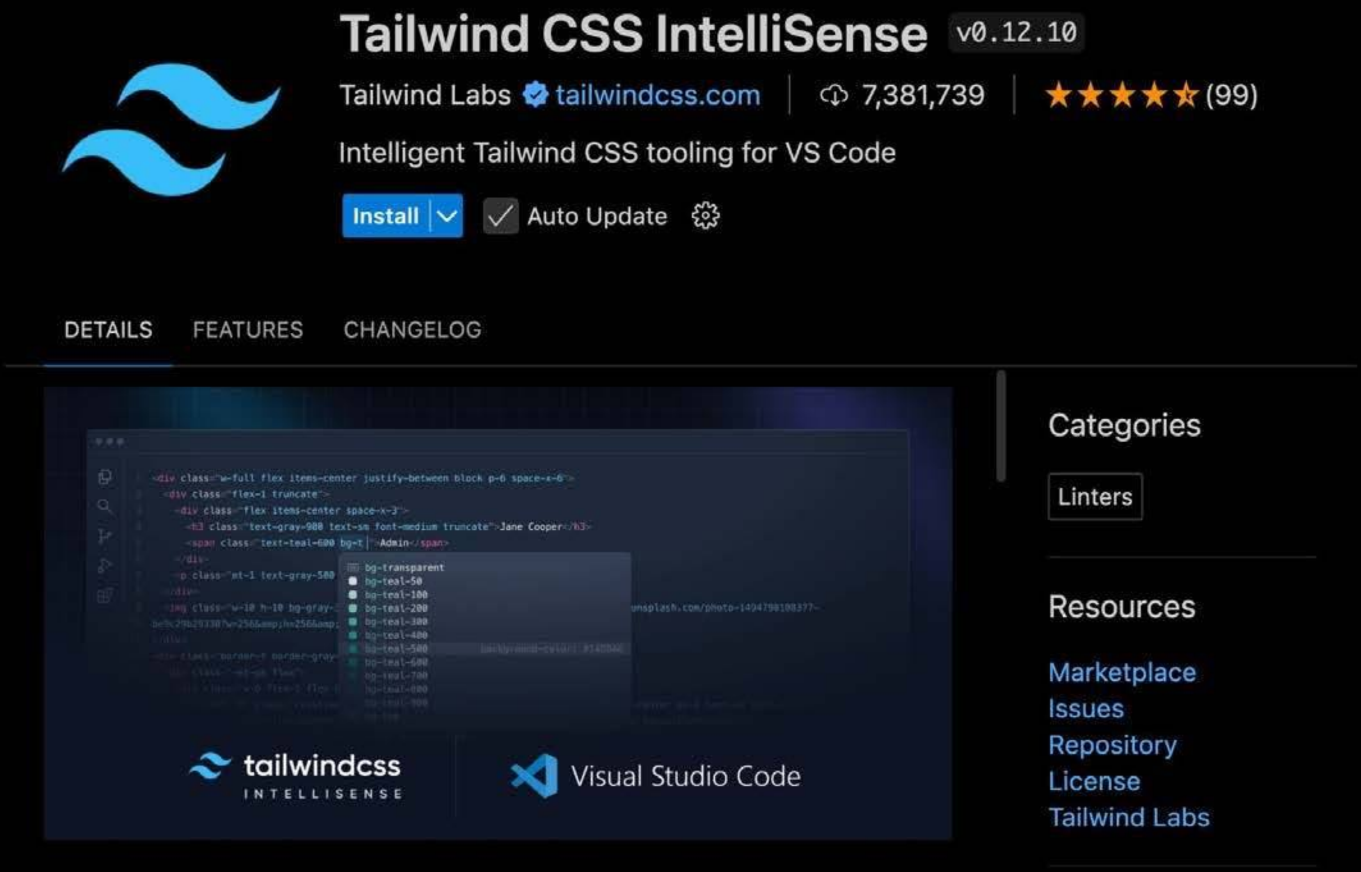
11.3 Utility Classes

```css
/* Colors */
.text-primary { color: #007bff; }
.bg-primary { background-color: #007bff; }

/* Sizing */
.w-full { width: 100%; }
.h-full { height: 100%; }

/* Typography */
.text-center { text-align: center; }
.font-bold { font-weight: bold; }

/* Spacing */
.m-1 { margin: 0.25rem; }
.m-2 { margin: 0.5rem; }
.p-1 { padding: 0.25rem; }
.p-2 { padding: 0.5rem; }
```

```css
/* Layout */
.d-flex { display: flex; }
.flex-col { flex-direction: column; }
.items-center { align-items: center; }
.justify-center { justify-content: center; }

/* Misc */
.rounded { border-radius: 0.25rem; }
.hidden { display: none; }
```

11.4 Installing Extension

11.4 Installing Extension



11.5 Including Tailwind CSS

Installation

Get started with Tailwind CSS

Tailwind CSS works by scanning all of your HTML files, JavaScript components, and any other templates for class names, generating the corresponding styles and then writing them to a static CSS file.

It's fast, flexible, and reliable — with zero-runtime.

Installation

Tailwind CLI Using PostCSS Framework Guides Play CDN

Use the Play CDN to try Tailwind right in the browser without any build step. The Play CDN is designed for development purposes only, and is not the best choice for production.

1 **Add the Play CDN script to your HTML**

Add the Play CDN script tag to the `<head>` of your HTML file, and start using Tailwind's utility classes to style your content.

index.html

```html
<!doctype html>
<html>
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale
  <script src="https://cdn.tailwindcss.com"></script>
</head>
<body>
  <h1 class="text-3xl font-bold underline">
    Hello world!
  </h1>
</body>
</html>
```

11.6 Installing Tailwind CSS

1. Install:

   npm init -y

   npm install -D tailwindcss postcss autoprefixer

2. Initialize Tailwind CSS Config

   npx tailwindcss init

11.6 Installing Tailwind CSS

Configure Tailwind in the Configuration Files (tailwind.config.js)

```
/** @type {import('tailwindcss').Config} */
module.exports = {
  content: ["./views/*.html"],
  theme: {
    extend: {},
  },
  plugins: [],
}
```

11.6 Installing Tailwind CSS

Add Tailwind Directives to views/input.css

```css
@tailwind base;
@tailwind components;
@tailwind utilities;
```

11.6 Installing Tailwind CSS

5. Include public/output.css into your index.html
6. Run Command
   npx tailwindcss -i ./views/input.css -o ./public/output.css --watch
7. Declare Shortcut:

```
"scripts": {
  "test": "echo \"Error: no test specified\" && exit 1",
  "tailwind": "npx tailwindcss -i ./src/input.css -o ./src/output.css --watch"
},
```

8. Run Command
   npm run tailwind

11.7 Using Tailwind CSS

```html
<body class="bg-gray-100 font-sans">
  <header class="bg-red-500 text-white p-4 shadow-md">
    <nav class="container mx-auto flex justify-between items-center">
      <ul class="flex space-x-4">
        <li>
          <a
            href="/"
            class="hover:bg-red-400 py-2 px-4 rounded transition duration-300"
            >Go to Home</a
          >
        </li>
        <li>
          <a
            href="/host/add-home"
            class="hover:bg-red-400 py-2 px-4 rounded transition duration-300"
            >Add Home</a
          >
        </li>
      </ul>
    </nav>
  </header>
  <main class="container mx-auto bg-white shadow-lg rounded-lg p-8 mt-10 max-w-xl">
    <h2 class="text-2xl text-gray-800 font-bold text-center mb-6">
      Welcome to Airbnb
    </h2>
  </main>
</body>
```

11.8 Building Tailwind Automatically

```
"scripts": {
  "test": "echo \"Error: no test specified\" && exit 1",
  "tailwind": "tailwindcss -i ./views/input.css -o ./public/output.css --watch",
  "start": "nodemon app.js & npm run tailwind"
},
```

Practise Set

Style other page using Tailwind.

Revision

1. Serving Static Files
2. Introduction to Tailwind CSS
3. Utility Classes
4. Installing Extension
5. Including Tailwind CSS
6. Installing Tailwind CSS
7. Using Tailwind CSS

12. Dynamic UI using EJS

1. Need of Dynamic UI
2. Sharing using Global Variables
3. What is EJS
4. Installing EJS
5. Using EJS Templates
6. Working with Partials

12.1 Need of Dynamic UI

- **Personalized Content:** Tailors responses based on user profiles, preferences, or behaviors to enhance user experience.
- **Dynamic Data Delivery:** Provides real-time information that updates with each request, such as live scores or stock prices.
- **Security and Access Control:** Delivers different content based on user authentication and authorization levels.
- **Localization and Internationalization:** Adjusts responses to accommodate different languages, cultures, or regional settings.
- **API Versatility:** Supports multiple client types (web, mobile, IoT) by providing appropriate data formats and structures.

12.2 Sharing using Global Variables

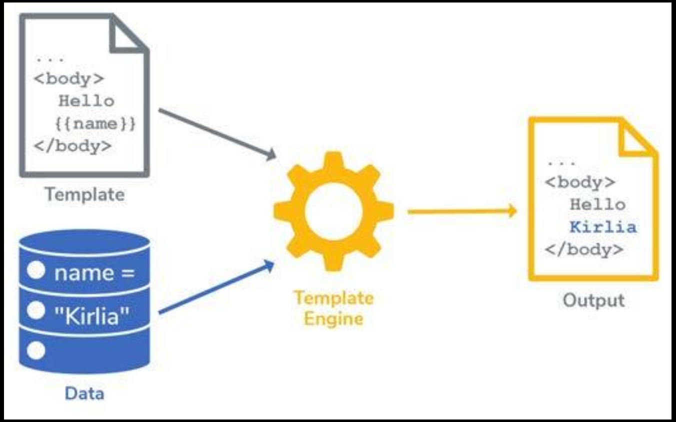
```
const homes = [];

// POST path
router.post('/add-home', (req, res, next) => {
  homes.push({houseName: req.body.houseName});
  res.redirect('/');
});

exports.hostRouter = router;
exports.homes = homes;
```

```
const {homes} = require('../routes/host');

userRouter.get('/', (req, res, next) => {
  console.log('homes', homes);
  res.sendFile(path.join(rootDir, 'views', 'user.html'));
});
```

Variables are shared across users

12.3 What is EJS

- HTML with JS: EJS lets you embed JavaScript code within HTML.
- Simple Syntax: Uses <% %> for control flow and <%= %> for output.
- Easy to Learn: Familiar to those who know HTML and JavaScript.
- Template Reuse: Supports partials for reusing code snippets.
- Flexible Logic: Allows full JavaScript expressions in templates.



12.4 Installing EJS

```
prashantjain@Prashants-Mac-mini air-bnb % npm install --save ejs

added 6 packages, and audited 232 packages in 879ms
```

```
const app = express();


app.set('view engine', 'ejs');
app.set('views', 'views');
```

12.4 Installing EJS

app.set(name, value)

Assigns setting `name` to `value`. You may store any value that you want, but certain names can be used to configure the behavior of the server. These special names are listed in the app settings table.

Calling `app.set('foo', true)` for a Boolean property is the same as calling `app.enable('foo')`. Similarly, calling `app.set('foo', false)` for a Boolean property is the same as calling `app.disable('foo')`.

Retrieve the value of a setting with `app.get()`.

```
app.set('title', 'My Site')
app.get('title') // "My Site"
```

`views`	String or Array	A directory or an array of directories for the application's views. If an array, the views are looked up in the order they occur in the array.	`process.cwd() + '/views'`
`view cache`	Boolean	Enables view template compilation caching. **NOTE**: Sub-apps will not inherit the value of this setting in production (when `NODE_ENV` is "production").	`true` in production, otherwise undefined.
`view engine`	String	The default engine extension to use when omitted. **NOTE**: Sub-apps will inherit the value of this setting.	N/A (undefined)

12.5 Using EJS Templates

- views
  - 404.html
  - add-home.html
  - input.css
  - user.ejs

```javascript
// Local Modules
const {homes} = require('../routes/host');

userRouter.get('/', (req, res, next) => {
  res.render('user', { homes: homes });
});
```

```html
<main class="container mx-auto bg-white shadow-lg rounded-lg p-8 mt-10 max-w-xl">
  <h2 class="text-5xl text-gray-800 font-bold text-center mb-6">
    Available Homes
  </h2>
  <ul class="list-disc list-inside space-y-2">
    <% homes.forEach(function(home) { %>
      <li class="text-gray-700 hover:text-red-500 transition-colors duration-300 text-2xl">
        <%= home.houseName %>
      </li>
    <% }); %>
  </ul>
</main>
```

12.6 Working with Partials

- views
  - partials
    - head.ejs
    - nav.ejs
  - 404.ejs
  - add-home.ejs
  - input.css
  - user.ejs

```html
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title><%= title %></title>
    <link rel="stylesheet" href="/output.css" />

```

12.6 Working with Partials

```html
<header class="bg-red-500 text-white p-4 shadow-md">
  <nav class="container mx-auto flex justify-between items-center">
    <ul class="flex space-x-4">
      <li>
        <a
          href="/"
          class="hover:bg-red-400 py-2 px-4 rounded transition duration-300"
          >Go to Home</a
        >
      </li>
      <li>
        <a
          href="/host/add-home"
          class="hover:bg-red-400 py-2 px-4 rounded transition duration-300"
          >Add Home</a
        >
      </li>
    </ul>
  </nav>
</header>
```

```
views
  partials
    head.ejs
    nav.ejs
  404.ejs
  add-home.ejs
  input.css
  user.ejs
```

12.6 Working with Partials

air-bnb > views > 404.ejs > ?

```
<%- include('partials/head') %>
  </head>
  <body class="bg-gray-100 font-sans">
    <%- include('partials/nav') %>
    <main class="container mx-auto mt-20 text-center">
      <h2 class="text-6xl font-bold text-red-500 mb-4">404</h2>
      <p class="text-2xl text-gray-700 mb-8">Oops! Page Not Found</p>
      <a href="/" class="bg-red-500 text-white py-2 px-6 rounded-lg
      hover:bg-red-600 transition duration-300">Go Back Home</a>
    </main>
  </body>
</html>
```

```
app.use((req, res, next) => {
  res.status(404).render('404', {title: 'Page Not Found'});
});
```

12.6 Working with Partials

```
userRouter.get('/', (req, res, next) => {
  res.render('user', { homes: homes, title: 'Available Homes' });
});
```

```
<%- include('partials/head') %>
  </head>
  <body class="bg-gray-100 font-sans">
    <%- include('partials/nav') %>
    <main class="container mx-auto bg-white shadow-lg rounded-lg p-8 mt-10
    max-w-xl">
      <h2 class="text-5xl text-gray-800 font-bold text-center mb-6">
        Available Homes
      </h2>
```

Practise Set

Reuse the app from the last assignment

1. Add more fields in the add home page like price per night, location, rating, photo.
2. Design the home card to show all of this information
3. Make the selected tab active on top.

Revision

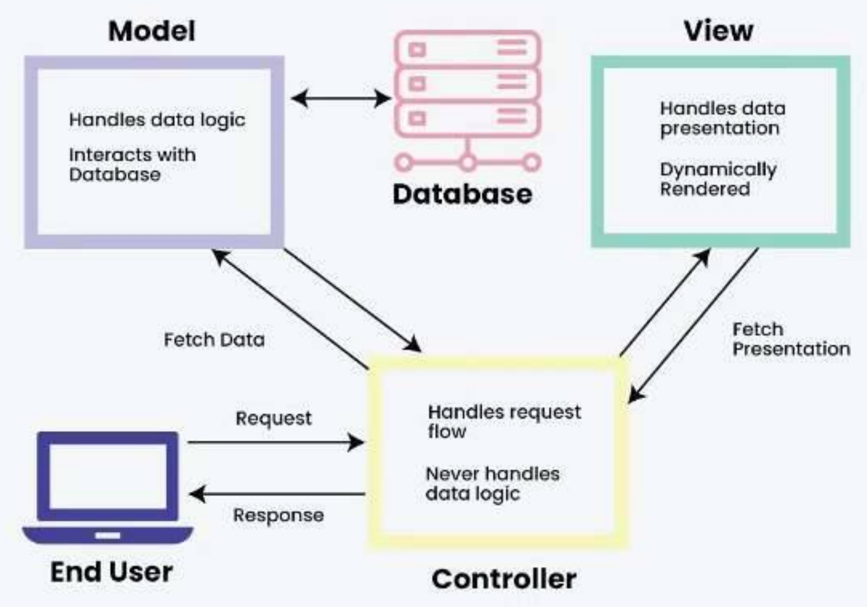
1. Need of Dynamic UI
2. Sharing using Global Variables
3. What is EJS
4. Installing EJS
5. Using EJS Templates
6. Working with Partials

13. Model View Controller

1. Separation of Concerns
2. Adding First Controller
3. Adding All Controllers
4. Adding 404 Controller
5. Adding Models
6. Writing data to Files
7. Nodemon causing problems
8. Reading data from Files

13.1 Separation of Concerns

- **MVC stands for Model-View-Controller**: A software architectural pattern for developing user interfaces.
- **Model**: Manages the data and business logic of the application.
- **View**: Handles the display and presentation of data to the user.
- **Controller**: Processes user input, interacts with the Model, and updates the View accordingly.
- Routes are a part of Controllers.
- **Purpose**: MVC separates concerns within an application, making it easier to manage and scale.



13.2 Adding First Controller

- controllers
  - homes.js
- node_modules
- public
- routes
  - hostRouter.js
  - userRouter.js
- utils
- views
  - partials
  - 404.ejs
  - addHome.ejs
  - home.ejs
  - homeAdded.ejs
  - input.css
- app.js
- package-lock.json
- package.json
- tailwind.config.js

homes.js

node > Chapter 12 - Dynamic > controllers > homes.js > ...

```
exports.getAddHome = (req, res, next) => {
  res.render("addHome", {
    pageTitle: "Add Home to airbnb",
    currentPage: "addHome",
  });
};
```

hostRouter.js

node > Chapter 12 - Dynamic > routes > hostRouter.js > ...

```
// External Module
const express = require('express');
const homesController = require('../controllers/homes');

const hostRouter = express.Router();

hostRouter.get("/add-home", homesController.getAddHome);
```

13.3 Adding All Controllers

homes.js

node > Chapter 12 - Dynamic > controllers > homes.js > ...

```js
const registeredHomes = [];

exports.getAddHome = (req, res, next) => {
  res.render("addHome", {
    pageTitle: "Add Home to airbnb",
    currentPage: "addHome",
  });
};

exports.postAddHome = (req, res, next) => {
  console.log("Home Registration successful for:", req.body);
  registeredHomes.push(req.body);
  res.render("homeAdded", {
    pageTitle: "Home Added Successfully",
    currentPage: "homeAdded",
  });
};

exports.getHomes = (req, res, next) => {
  console.log(registeredHomes);
  res.render("home", {
    registeredHomes: registeredHomes,
    pageTitle: "airbnb Home",
    currentPage: "Home",
  });
};
```

13.3 Adding All Controllers

hostRouter.js

node > Chapter 12 - Dynamic > routes > hostRouter.js > ...

```
// External Module
const express = require('express');
const homesController = require('../controllers/homes');

const hostRouter = express.Router();

hostRouter.get("/add-home", homesController.getAddHome);

hostRouter.post("/add-home", homesController.postAddHome);

exports.hostRouter = hostRouter;
```

13.3 Adding All Controllers

userRouter.js

node > Chapter 12 - Dynamic > routes > userRouter.js > ...

```javascript
// External Module
const express = require('express');
const userRouter = express.Router();
const homesController = require('../controllers/homes');

userRouter.get("/", homesController.getHomes);

module.exports = userRouter;
```

13.4 Adding 404 Controller

- controllers
  - error.js
  - homes.js
- node_modules
- public
- routes
  - hostRouter.js
  - userRouter.js
- utils
- views
  - partials
  - 404.ejs
  - addHome.ejs
  - home.ejs
  - homeAdded.ejs
  - input.css
- app.js
- package-lock.json
- package.json
- tailwind.config.js

error.js

node > Chapter 12 - Dynamic > controllers > error.js > ...

```
exports.get404 = (req, res, next) => {
  res.status(404).render('404', {pageTitle: 'Page Not Found', currentPage: '404'});
}
```

```
//Local Module
const userRouter = require("./routes/userRouter")
const {hostRouter} = require("./routes/hostRouter")
const rootDir = require("./utils/pathUtil");
const errorController = require("./controllers/error");

const app = express();

app.set('view engine', 'ejs');
app.set('views', 'views');

app.use(express.urlencoded());
app.use(userRouter);
app.use("/host", hostRouter);

app.use(express.static(path.join(rootDir, 'public')))

app.use(errorController.get404);
```

13.5 Adding Models

- controllers
  - error.js
  - homes.js
- models
  - home.js
- node_modules
- public
- routes
- utils
- views
- app.js
- package-lock.json
- package.json
- tailwind.config.js

home.js

node > Chapter 12 - Dynamic > models > home.js > ...

```javascript
// fake database
const registeredHomes = [];

module.exports = class Home {
  constructor(houseName, price, location, rating, photoUrl) {
    this.houseName = houseName;
    this.price = price;
    this.location = location;
    this.rating = rating;
    this.photoUrl = photoUrl;
  }

  save() {
    registeredHomes.push(this);
  }

  static fetchAll() {
    return registeredHomes;
  }
}
```

13.5 Adding Models

homes.js

node > Chapter 12 - Dynamic > controllers > homes.js > ...

```javascript
const Home = require("../models/home");

exports.getAddHome = (req, res, next) => {
  res.render("addHome", {
    pageTitle: "Add Home to airbnb",
    currentPage: "addHome",
  });
};

exports.postAddHome = (req, res, next) => {
  const { houseName, price, location, rating, photoUrl } = req.body;
  const home = new Home(houseName, price, location, rating, photoUrl);
  home.save();
  res.render("homeAdded", {
    pageTitle: "Home Added Successfully",
    currentPage: "homeAdded",
  });
};

exports.getHomes = (req, res, next) => {
  const homes = Home.fetchAll();
  res.render("home", {
    registeredHomes: homes,
    pageTitle: "airbnb Home",
    currentPage: "Home",
  });
};
```

13.6 Writing data to Files

- controllers
- data
  - homes.json
- models
  - home.js
- node_modules
- public
- routes
- utils
  - pathUtil.js
- views
- app.js
- package-lock.json
- package.json
- tailwind.config.js

```javascript
const fs = require('fs');
const path = require('path');
const rootDir = require('../utils/pathUtil');

// fake database
const registeredHomes = [];

module.exports = class Home {
  constructor(houseName, price, location, rating, photoUrl) {
    this.houseName = houseName;
    this.price = price;
    this.location = location;
    this.rating = rating;
    this.photoUrl = photoUrl;
  }

  save() {
    registeredHomes.push(this);
    const filePath = path.join(rootDir, 'data', 'homes.json');
    fs.writeFile(filePath, JSON.stringify(registeredHomes), (err) => {
      console.log(err);
    });
  }

  static fetchAll() {
    return registeredHomes;
  }
}
```

13.6 Writing data to Files

homes.json

node > Chapter 12 - Dynamic > data > homes.json > ...

```json
[
  {
    "houseName": "Utsav",
    "price": "999",
    "location": "Ghaziabad",
    "rating": "5",
    "photoUrl": "https://www.google.com/url?sa=i&url=https%3A%2F%2Fwww.airbnb.com%2Fstays%2Fdesign&psig=AOvVaw259nAPAVlXwRuraoxPS3u7&ust=1729404860631000&source=images&cd=vfe&opi=89978449&ved=0CBQQjRxqFwoTCOiViaXlmYkDFQAAAAAdAAAAABAE"
  }
]
```

13.7 Nodemon causing problems

- controllers
- data
  - homes.json
- models
- node_modules
- public
- routes
- utils
  - pathUtil.js
- views
- app.js
- nodemon.json
- package-lock.json
- package.json
- tailwind.config.js

nodemon.json

node > Chapter 12 - Dynamic > nodemon.json > ...

```json
{
  "watch": ["."],
  "ext": "js,json,ejs",
  "ignore": ["node_modules/", "data/"],
  "exec": "node app.js"
}
```

13.8 Reading data from Files

```
static fetchAll() {
  const filePath = path.join(rootDir, 'data', 'homes.json');
  const fileContent = fs.readFile(filePath, (err, data) => {
    if (err) {
      return [];
    }
    return JSON.parse(data);
  });
}
```

```
TypeError: /Users/prashantjain/workspace/Test Project/node/Chapter 12 - Dynamic/views/home.ejs:10
    8|       </h2>
    9|       <div class="flex flex-wrap justify-center gap-6">
 >> 10|         <% registeredHomes.forEach(home => { %>
    11|           <div class="bg-white rounded-lg shadow-md overflow-hidden hover:shadow-lg transi
    12|             <img src="https://scontent.fdel15-1.fna.fbcdn.net/v/t1.6435-9/95996760_5986942
    13|             <div class="p-4">

Cannot read properties of undefined (reading 'forEach')
    at eval (eval at compile (/Users/prashantjain/workspace/Test Project/node/Chapter 12 - Dynamic
    at home (/Users/prashantjain/workspace/Test Project/node/Chapter 12 - Dynamic/node_modules/ejs
    at tryHandleCache (/Users/prashantjain/workspace/Test Project/node/Chapter 12 - Dynamic/node_m
```

13.8 Reading data from Files

```javascript
save() {
  this.fetchAll((registeredHomes) => {
    registeredHomes.push(this);
    const filePath = path.join(rootDir, 'data', 'homes.json');
    fs.writeFile(filePath, JSON.stringify(registeredHomes), (err) => {
      console.log(err);
    });
  });
}

static fetchAll(callback) {
  const filePath = path.join(rootDir, 'data', 'homes.json');
  fs.readFile(filePath, (err, data) => {
    let homes = [];
    if (!err) {
      homes = JSON.parse(data);
    }
    callback(homes);
  });
}
```

```javascript
exports.getHomes = (req, res, next) => {
  Home.fetchAll((homes) => {
    res.render("home", {
      registeredHomes: homes,
      pageTitle: "airbnb Home",
      currentPage: "Home",
    });
  });
};
```

Practise Milestone

Take your airbnb forward:

1. Structure the views folder into host & store and move the respective view files there.
2. Add more views to store like: home-list, home-detail, favourite-list, reserve, bookings. And to host view: edit-home, host-home-list
3. Improve the header with navigation to all pages.
4. Register all the new routes and add dummy views there.
5. Change controllers to store and host setup.
6. Add Edit and Delete buttons to the host-home-list view.
7. Keep the logic for Edit, Delete, Favourite pending.

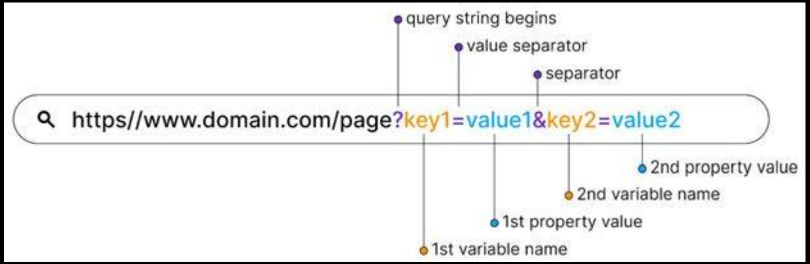
Revision

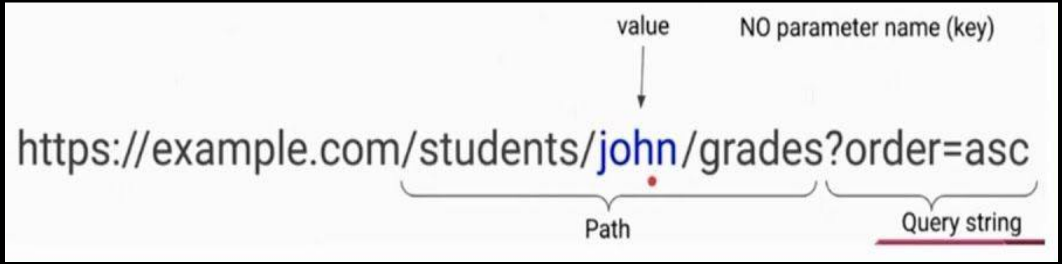
1. Separation of Concerns
2. Adding First Controller
3. Adding All Controllers
4. Adding 404 Controller
5. Adding Models
6. Writing data to Files
7. Nodemon causing problems
8. Reading data from Files

14. Dynamic Path & Model deep-dive

1. What are dynamic paths
2. Adding Home Details Page
3. Showing Real Home Data
4. Adding Favourites Feature
5. Adding Favourites Model
6. Edit-Home: Adding Feature Skeleton
7. Edit-Home: Showing Existing Data
8. Edit-Home: Handing Edit Request
9. Adding Delete Feature
10. Removing Home from Favourites

14.1 What are dynamic paths





- **Path parameters** are variables embedded directly in the URL path to capture dynamic values, like /users/:userId where :userId is replaced with a specific user ID.
- **Query parameters** are key-value pairs appended to the URL after a ?, used to send additional data, like /search?query=nodejs where query=nodejs specifies the search term.

14.2 Adding Home Details Page

1. Add a details button in /homes-list to go to link path /homes/:home-id
2. Add a random id to each home in the data file.
3. Add a random id on home object before saving in the home model.
4. Add a route in the store router for /homes/:home-id
5. Add a method in store controller to get the home-id using req.params and log it, before sending out a dummy response with home id.

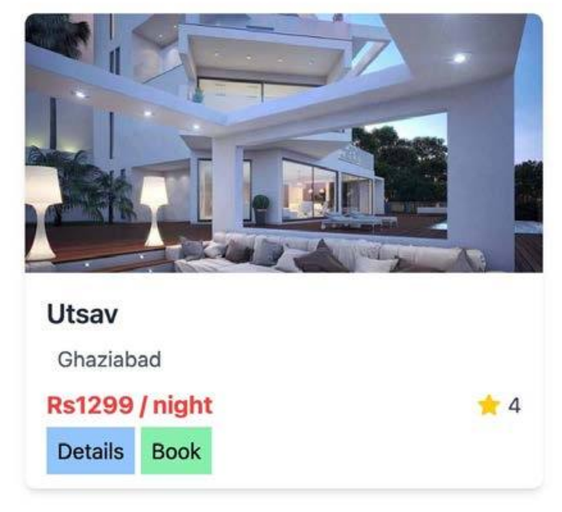
14.2 Adding Home Details Page

1.

home-list.ejs × | homes.json | home.js

views > store > home-list.ejs > ?

```
<%- include('../partials/head') %>
  <body>
    <%- include('../partials/nav') %>
    <main class="container mx-auto bg-white shadow-lg rounded-lg p-8 mt-10 max-w-6xl">
      <div class="flex flex-wrap justify-center gap-6">
              <a href="/homes/<%= home.id %>" class="bg-blue-300 p-2">Details</a>
              <a href="#" class="bg-green-300 p-2">Book</a>
            </div>
```



2.

home-list.ejs | homes.json × | home.js

data > homes.json > ...

```
[{"id":"1", "houseName":"Utsav","price":"1299","location":"Ghaziabad","rating":"4",
"photoUrl":"asdf"},{"id":"2", "houseName":"Second House","price":"999",
"location":"Ghaziabad","rating":"4","photoUrl":"asdf"}]
```

3.

home-list.ejs | homes.json | home.js ×

models > home.js > ...

```
module.exports = class Home {

  save() {
    this.id = Math.random().toString();
    Home.fetchAll((registeredHomes) => {
```

14.2 Adding Home Details Page

4.

storeRouter.js × | storeController.js

routes > storeRouter.js > ...

```javascript
storeRouter.get("/", homesController.getIndex);
storeRouter.get("/homes", homesController.getHomes);
// homes/1
storeRouter.get("/homes/:homeId", homesController.getHome);
storeRouter.get("/bookings", homesController.getBookings);
```

localhost:3000/homes/1

Details of home 1

5.

storeRouter.js | storeController.js ×

controllers > storeController.js > ...

```javascript
exports.getHome = (req, res, next) => {
  const homeId = req.params.homeId;
  console.log(homeId);
  res.send("Details of home " + homeId);
};
```

14.3 Showing Real Home Data

1. Add a static findById method in Home model that takes a callback.
2. Use this findById method in the controller to load home details and log them.
3. Now change the controller:
   a. Redirect to /homes in case the home is not found, logging an error.
   b. Rendering the /home-detail page with home data.
4. Create a home-detail page to use the entire page to show all home data.
5. Download from Github:
   a. The styled home-detail file.
   b. 10 image of homes that you can put locally and use.

14.3 Showing Real Home Data

1.

storeController.js   home.js

models > home.js > ...

```js
module.exports = class Home {

  static findById(id, callback) {
    this.fetchAll((homes) => {
      const home = homes.find((home) => home.id === id);
      callback(home);
    });
  }
};
```

```
Server running on address http://localhost:3000
{
  id: '1',
  houseName: 'Utsav',
  price: '1299',
  location: 'Ghaziabad',
  rating: '4',
  photoUrl: 'asdf'
}
```

2.

```js
exports.getHome = (req, res, next) => {
  const homeId = req.params.homeId;
  Home.findById(homeId, (home) => {
    console.log(home);
  });
};
```

14.3 Showing Real Home Data

3.
```
exports.getHome = (req, res, next) => {
  const homeId = req.params.homeId;
  Home.findById(homeId, (home) => {
    if (!home) {
      return res.redirect('/homes'); // Redirect if home not found
    }
    res.render("store/home-detail", {
      home: home,
      pageTitle: home.houseName,
      currentPage: "homes",
    });
  });
};
```

14.3 Showing Real Home Data

4.

5.

- images
  - house1.png
  - house2.png
  - house3.png
  - house4.png
  - house5.png
  - house6.png
  - house7.png
  - house8.png

14.4 Adding Favourites Feature

1. Create a partial with a form and a button that submits to /favourites path with a hidden input having home-id value.
2. Add this partial to home-detail page
3. Add this partial to home-list page.
4. Add router for handling POST request to /favourites path.
5. Add a method in store controller to add a home id as favourites, logging the id for now, before redirecting to /favourites path.

14.4 Adding Favourites Feature

1.

home-detail.ejs | storeController.js | storeRouter.js | home-list.ejs | favourite.ejs

views > partials > favourite.ejs > form

```
<form action="/favourites" method="post">
  <button type="submit" class="bg-red-500 text-white px-8 py-3 rounded-lg
    Add to Favorites
  </button>
  <input type="hidden" name="homeId" value="<%= home.id %>">
</form>
```

2.

home-detail.ejs | storeController.js | storeRouter.js | home-list.ejs

views > store > home-detail.ejs > ? > body > ? > main.container.mx-auto.bg-white.shadow-lg.rounded-lg.p-8.mt-10.max-w-6xl > div.grid.grid-cols-1.md:gr

```
<%- include('../partials/head') %>
  <body>
      <%- include('../partials/nav') %>
      <main class="container mx-auto bg-white shadow-lg rounded-lg p-8 mt-10 max-w-6xl">
          <div class="grid grid-cols-1 md:grid-cols-2 gap-8">
                  <div class="mt-8">
                    <%- include('../partials/favourite') %>
                  </div>
              </div>
          </div>
      </main>
  </body>
</html>
```

14.4 Adding Favourites Feature

3.

storeController.js storeRouter.js home-list.ejs

views > store > home-list.ejs > ?

```
<%- include('../partials/head') %>
  <body>
    <%- include('../partials/nav') %>
    <main class="container mx-auto bg-white shadow-lg rounded-lg p-8 mt-10 max-w-6xl">
      <div class="grid grid-cols-1 md:grid-cols-2 lg:grid-cols-3 gap-8">
              <div class="flex gap-3">
                <a href="/homes/<%= home.id %>" class="flex-1 bg-white text-red-500 py
                hover:text-white border border-red-500">View Details</a>
                <%- include('../partials/favourite', { home: home }) %>
              </div>
            </div>
          </div>
        <% }) %>
      </div>
    </main>
  </body>
</html>
```

14.4 Adding Favourites Feature

4.

```
storeRouter.post("/favourites", homesController.postFavourites);
storeRouter.get("/favourites", homesController.getFavouriteList);
```

5.

storeController.js ×

controllers > storeController.js > ...

```
exports.postFavourites = (req, res, next) => {
  const homeId = req.body.homeId;
  console.log(homeId);
  res.redirect('/favourites');
};
```

14.5 Adding Favourites Model

1. Create a new Model to handle Favourites:
   a. A static method getFavourites that reads the favourites.json file and return the ids of all homes marked favourite.
   b. A static method addFavourites that adds the home-id to favourites-id array if not already there, then updates the file.
2. Use this model in addFavourites controller.
3. Use this model in getFavourites controller while also fetching details of all homes from Homes Model.
4. Change the UI of favourites page to show new content.

14.5 Adding Favourites Model

1.

```
const fs = require('fs');
const path = require('path');

const rootDir = require('../utils/pathUtil');
const favouritesPath = path.join(rootDir, "data", "favourites.json");

module.exports = class Favourites {
  static addFavourite(homeId) {
    this.getFavourites((favourites) => {
      if (favourites.includes(homeId)) {
        console.log("Home already in favourites");
      } else {
        favourites.push(homeId);
        fs.writeFile(favouritesPath, JSON.stringify(favourites), (err) => {
          console.log("File Writing Concluded", err);
        });
      }
    });
  }

  static getFavourites(callback) {
    fs.readFile(favouritesPath, (err, data) => {
      callback(!err ? JSON.parse(data) : []);
    });
  }
};
```

14.5 Adding Favourites Model

2.

favourites.js | storeController.js | favourites.json

controllers > storeController.js > ...

```
exports.postFavourites = (req, res, next) => {
  const homeId = req.body.homeId;
  Favourites.addFavourite(homeId);
  res.redirect('/favourites');
};
```

favourites.js | storeController.js | favourites.json

data > favourites.json > ...

```
["1","2"]
```

14.5 Adding Favourites Model

3.

```
exports.getFavouriteList = (req, res, next) => {
  Favourites.getFavourites((favourites) => {
    Home.fetchAll((registeredHomes) => {
      const favouritesWithDetails = favourites.map(homeId => registeredHomes.find(home => home.id === homeId));
      res.render("store/favourite-list", {
        favourites: favouritesWithDetails,
        pageTitle: "My Favourites",
        currentPage: "favourites",
      });
    });
  });
};
```

14.5 Adding Favourites Model

4.

14.6 Edit-Home: Adding Feature Skeleton

1. Rename the add-home.ejs to edit-home.ejs
2. Fix it's usage in the add-home controller.
3. Change the path of edit button everywhere.
4. Add a new router for GET /edit-home/:home-id
5. Add a new controller method in host controller, optionally logging query param of editing=true, while passing editing flag to view.

14.6 Edit-Home: Adding Feature Skeleton

1.

```
host
  edit-home.ejs
  home-added.ejs
  host-home-list.ejs
```

2.

```
exports.getAddHome = (req, res, next) => {
  res.render("host/edit-home", {
    pageTitle: "Add Home to airbnb",
    currentPage: "addHome",
  });
};
```

14.7 Edit-Home: Showing Existing Data

3.

```
<%- include('../partials/head') %>
  <body>
    <%- include('../partials/nav') %>
    <main class="container mx-auto bg-white shadow-lg rounded-lg p-8 mt-10 max-w-6xl">
      <div class="flex flex-wrap justify-center gap-6">
              </div>
              <a href="/host/edit-home/<%= home.id %>?editing=true" class="bg-blue-500 p-2">Edit</a>
              <button class="bg-red-500 p-2">Delete</button>
            </div>
          </div>
        <% }) %>
      </div>
    </main>
  </body>
</html>
```

14.6 Edit-Home: Adding Feature Skeleton

4.

```
hostRouter.get("/edit-home/:homeId", hostController.getEditHome);
```

5.

```
exports.getEditHome = (req, res, next) => {
  const editing = req.query.editing === "true";
  const homeId = req.params.homeId;
  console.log(editing, homeId);
  res.render("host/edit-home", {
    pageTitle: "Edit Home",
    currentPage: "editHome",
    editing: editing,
    homeId: homeId,
  });
};
```

localhost:3000/host/edit-home/1?editing=true

```
Server running on address http://localhost:3000
false 1
true 1
```

14.7 Edit-Home: Showing Existing Data

1. Change the controller to now fetch the home details, if not found redirect to /host/host-home-list otherwise passing the data to view.
2. Change the edit-home.ejs to show dynamic content based on values:
    a. Different submit path.
    b. Different button text.
    c. Pre-filled values if present.

14.7 Edit-Home: Showing Existing Data

1.

```
exports.getEditHome = (req, res, next) => {
  const editing = req.query.editing === "true";
  const homeId = req.params.homeId;
  Home.findById(homeId, (home) => {
    if (!home) {
      return res.redirect("/host/host-home-list");
    }
    res.render("host/edit-home", {
      pageTitle: "Edit Home",
      currentPage: "editHome",
      editing: editing,
      home: home,
    });
  });
};
```

14.7 Edit-Home: Showing Existing Data

2.

```
<%- include('../partials/head') %>
</head>
<body>
  <%- include('../partials/nav') %>
  <main class="container mx-auto mt-8 p-8 bg-white rounded-lg shadow-md">
    <h1 class="text-3xl font-bold mb-6 text-center text-gray-800">Register Your Home on AirBnB</h1>
    <form action="/host/<%= editing ? 'edit-home' : 'add-home' %>" method="POST" class="max-w-md mx-auto">
      <input
        type="text"
        name="houseName"
        value="<%= home ? home.houseName : '' %>"
        placeholder="Enter your House Name"
        class="w-full px-4 py-2 mb-4 border rounded-md focus:outline-none focus:ring-2 focus:ring-red-500"/>
      <input
        type="text"
        name="price"
        value="<%= home ? home.price : '' %>"
        placeholder="Price Per Night"
        class="w-full px-4 py-2 mb-4 border rounded-md focus:outline-none focus:ring-2 focus:ring-red-500"/>
      <input
        type="text"
        name="location"
        value="<%= home ? home.location : '' %>"
        placeholder="Location"
        class="w-full px-4 py-2 mb-4 border rounded-md focus:outline-none focus:ring-2 focus:ring-red-500"/>
      <input
        type="text"
        name="rating"
        value="<%= home ? home.rating : '' %>"
        placeholder="Rating"
        class="w-full px-4 py-2 mb-4 border rounded-md focus:outline-none focus:ring-2 focus:ring-red-500"/>
      <input
        type="text"
        name="photoUrl"
        value="<%= home ? home.photoUrl : '' %>"
        placeholder="Enter Photo URL"
        class="w-full px-4 py-2 mb-4 border rounded-md focus:outline-none focus:ring-2 focus:ring-red-500"/>
      <input type="submit" value="<%= editing ? 'Update' : 'Add' %> Home" class="w-full bg-red-500 text-white py-
      transition duration-300"/>
    </form>
  </main>
</body>
</html>
```

14.8 Edit-Home: Handing Edit Request

1. Add a router for POST /edit-home
2. Add a controller for /edit-home, which creates a home model object and saves it, before redirecting to /host/host-home-list
3. Add hidden id input field in the edit-home.ejs to get the id as part of POST request.
4. Change the save method in Model to handle creation of new as well as updation of existing Home.

14.8 Edit-Home: Handing Edit Request

1.

```
hostRouter.post("/edit-home", hostController.postEditHome);
```

2.

```
exports.postEditHome = (req, res, next) => {
  const { id, houseName, price, location, rating, photoUrl } = req.body;
  const home = new Home(houseName, price, location, rating, photoUrl);
  home.id = id;
  home.save();
  res.redirect("/host/host-home-list");
};
```

3.

```
<%- include('../partials/head') %>
  <body>
    <%- include('../partials/nav') %>
    <main class="container mx-auto mt-8 p-8 bg-white rounded-lg shadow-md">
      <h1 class="text-3xl font-bold mb-6 text-center text-gray-800">Register
      <form action="/host/<%= editing ? 'edit-home' : 'add-home' %>" method="P
        <input type="hidden" name="id" value="<%= home ? home.id : '' %>">
        <input
          type="text"
          name="houseName"
```

14.8 Edit-Home: Handing Edit Request

4.

```
module.exports = class Home {

  save() {
    Home.fetchAll((registeredHomes) => {
      console.log(registeredHomes, this);
      if (this.id) {
        registeredHomes = registeredHomes.map((home) => {
          console.log(home.id, this.id);
          if (home.id === this.id) {
            return this;
          }
          return home;
        });
      } else {
        console.log("New Home");
        this.id = Math.random().toString();
        registeredHomes.push(this);
      }
      const homeDataPath = path.join(rootDir, "data", "homes.json");
      fs.writeFile(homeDataPath, JSON.stringify(registeredHomes), (error) => {
        console.log("File Writing Concluded", error);
      });
    });
  }
```

14.9 Adding Delete Feature

1. Surround the delete button with a form that submits to path /host/delete-home/:homeId
2. Add a route in the host routes.
3. Add a static delete method to the Home model that takes an id and deletes the home.
4. Add a method in host controller to handle the request, delete the home and redirect to /host/host-homes-list page.
5. Pending: deleted homes might still be in favourites list.

14.9 Adding Delete Feature

1.
```html
<a href="/host/edit-home/<%= home.id %>?editing=true" class="bg-blue-500 p-2 rounded
text-white hover:bg-blue-600">Edit</a>
<form action="/host/delete-home/<%= home.id %>" method="POST" class="inline">
  <button type="submit" class="bg-red-500 p-2 rounded text-white
  hover:bg-red-600">Delete</button>
</form>
```

2.
```javascript
hostRouter.post("/delete-home/:homeId", hostController.postDeleteHome);
```

3.
```javascript
static deleteById(id, callback) {
  this.fetchAll((homes) => {
    const updatedHomes = homes.filter((home) => home.id !== id);
    fs.writeFile(path.join(rootDir, "data", "homes.json"), JSON.stringify(updatedHomes), callback);
  });
}
```

14.9 Adding Delete Feature

4.

```
exports.postDeleteHome = (req, res, next) => {
  const homeId = req.params.homeId;
  Home.deleteById(homeId, (error) => {
    if (error) {
      console.log("Error Deleting Home", error);
    }
    res.redirect("/host/host-home-list");
  });
};
```

14.10 Removing Home from Favourites

1. Add a static method to the favourites model to delete the home.
2. Add a form to the favourites-list in each home to path /favourites/delete/:homeId
3. Add a route for POST delete-favourites in the store routes.
4. Add a method in store controller to use the model's static method and then redirect to favourites-list.
5. Use this model method in home-delete to make sure any deleted home is also removed from favourites.

14.10 Removing Home from Favourites

1.

```
static deleteFavourite(homeId, callback) {
  this.getFavourites((favourites) => {
    const updatedFavourites = favourites.filter((id) => id !== homeId);
    fs.writeFile(favouritesPath, JSON.stringify(updatedFavourites), callback);
  });
}
```

2.

```
<form action="/favourites/delete/<%= home.id %>" method="POST">
  <button type="submit" class="bg-red-500 p-2 rounded text-white
  hover:bg-red-600">Delete</button>
</form>
```

3.

```
storeRouter.post("/favourites/delete/:homeId", storeController.deleteFavourite);
```

14.10 Removing Home from Favourites

4.

```
exports.deleteFavourite = (req, res, next) => {
  const homeId = req.params.homeId;
  Favourites.deleteFavourite(homeId, (error) => {
    if (error) {
      console.log("Error Deleting Favourite", error);
    }
    res.redirect('/favourites');
  });
};
```

5.

```
static deleteById(id, callback) {
  this.fetchAll((homes) => {
    const updatedHomes = homes.filter((home) => home.id !== id);
    fs.writeFile(path.join(rootDir, "data", "homes.json"), JSON.stringify(updatedHomes), (error) => {
      Favourites.deleteFavourite(id, callback);
    });
  });
}
```

Revision

1. What are dynamic paths
2. Adding Home Details Page
3. Showing Real Home Data
4. Adding Favourites Feature
5. Adding Favourites Model
6. Edit-Home: Adding Feature Skeleton
7. Edit-Home: Showing Existing Data
8. Edit-Home: Handing Edit Request
9. Adding Delete Feature
10. Removing Home from Favourites

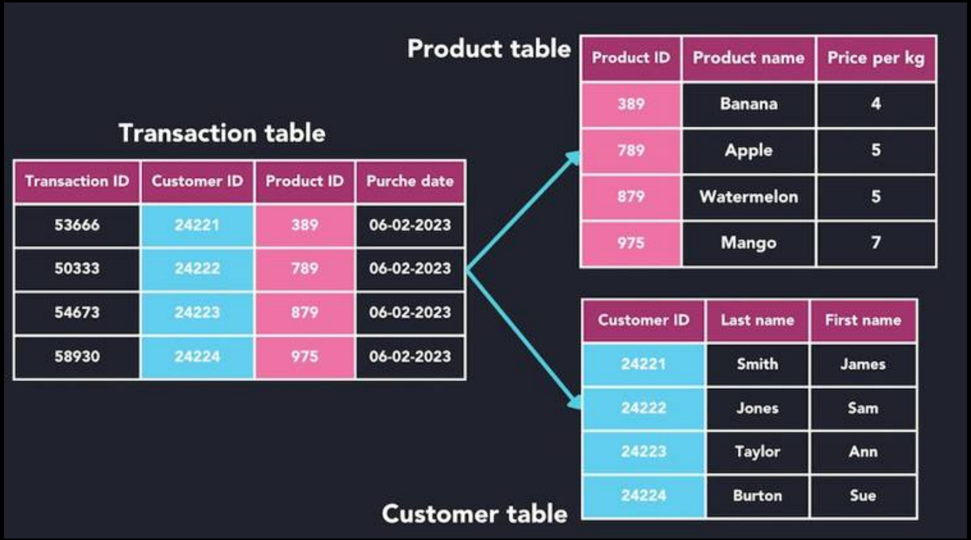
15. Introduction to SQL

1. What is a DB (Database)
2. Introduction to SQL DB
3. Introduction to NoSQL DB
4. SQL vs NoSQL
5. Installing MySQL
6. Connecting App to DB
7. Creating homes Table
8. Querying homes in App
9. Adding DB in Models
10. Adding Home in Model
11. Implementing Model using Where

15.1 What is a DB (Database)

1. Store Data: Keep large amounts of data in a structured format.
2. Enable Data Management: Allow for adding, updating, and deleting data easily.
3. Facilitate Quick Access: Provide fast retrieval of data through queries.
4. Ensure Data Integrity: Maintain accuracy and consistency of data over time.
5. Support Multiple Users: Handle concurrent access by many users simultaneously.
6. Secure Data: Protect information through access controls and authentication.

15.2 Introduction to SQL DB



Product table

Product ID	Product name	Price per kg
389	Banana	4
789	Apple	5
879	Watermelon	5
975	Mango	7

Transaction table

Transaction ID	Customer ID	Product ID	Purche date
53666	24221	389	06-02-2023
50333	24222	789	06-02-2023
54673	24223	879	06-02-2023
58930	24224	975	06-02-2023

Customer table

Customer ID	Last name	First name
24221	Smith	James
24222	Jones	Sam
24223	Taylor	Ann
24224	Burton	Sue

- Vertical Scalability: Typically scaled by increasing the resources of a single server (scaling up).
- Relationships: Tables can have multiple types of relationships.

- Relational Model: Organize data into tables with rows and columns.
- Fixed Schema: Require a predefined schema; the structure of the data must be known in advance.

15.2 Introduction to SQL DB

Table: Customers

customer_id	first_name	last_name	age	country
1	John	Doe	31	USA
2	Robert	Luna	22	USA
3	David	Robinson	22	UK
4	John	Reinhardt	25	UK
5	Betty	Doe	28	UAE

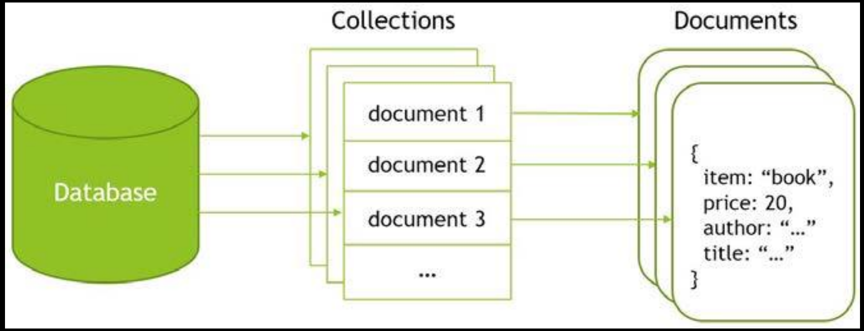
```
SELECT age, country
FROM Customers
WHERE country = 'USA';
```

age	country
31	USA
22	USA

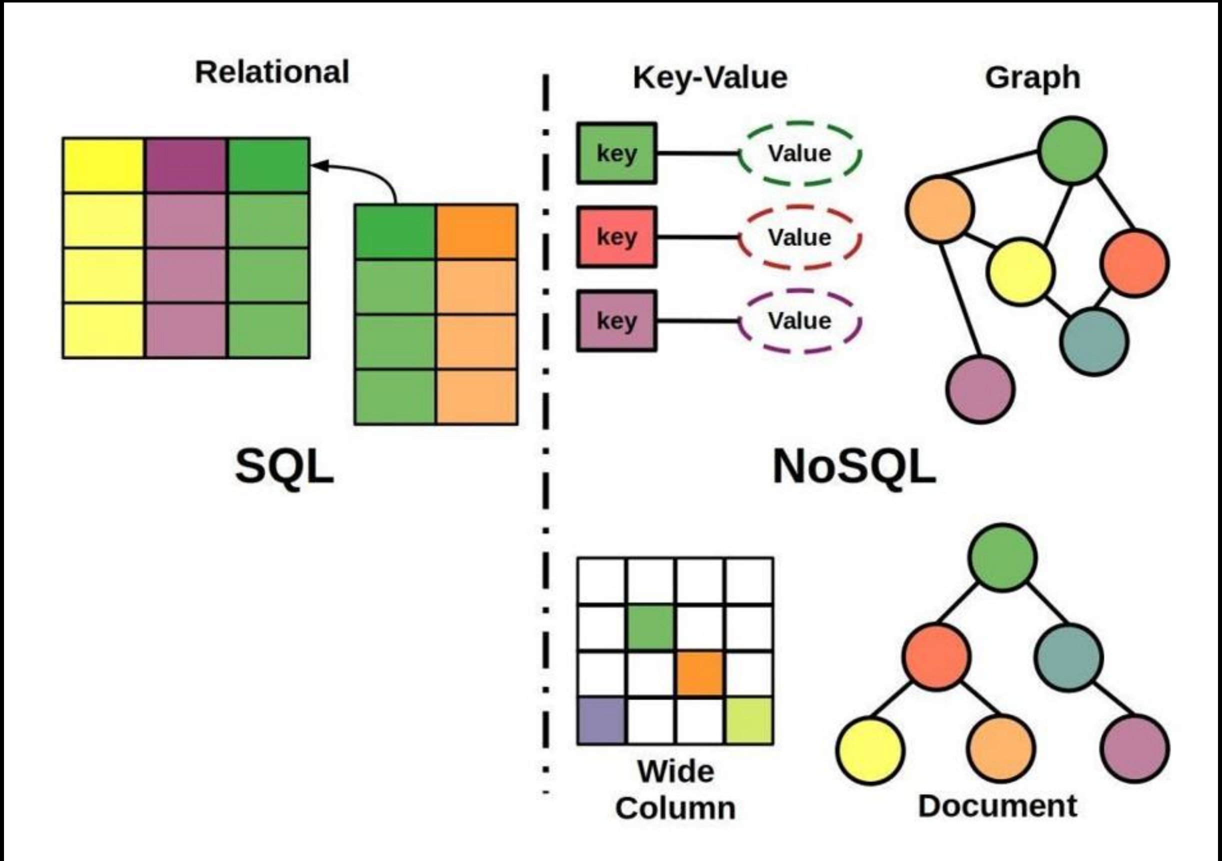
- Relational Model Use of SQL: Utilize SQL for querying and managing data, which is a standardized and widely-used language.
- ACID Compliance: Support transactions that are Atomic, Consistent, Isolated, and Durable.
- Complex Queries: Excel at handling complex queries and relationships between data.

15.3 Introduction to NoSQL DB

- Flexible Schema: Allow for dynamic schemas, accommodating unstructured or semi-structured data without predefined structures.
- Duplicacy over Relations: Duplicates data across records (denormalization) to enhance performance and scalability, rather than relying on complex relationships and joins as in relational databases.
- Horizontal Scalability: Designed to scale out by adding more servers, handling large volumes of data efficiently.
- Performance: Optimized for high throughput and low latency, suitable for real-time applications.



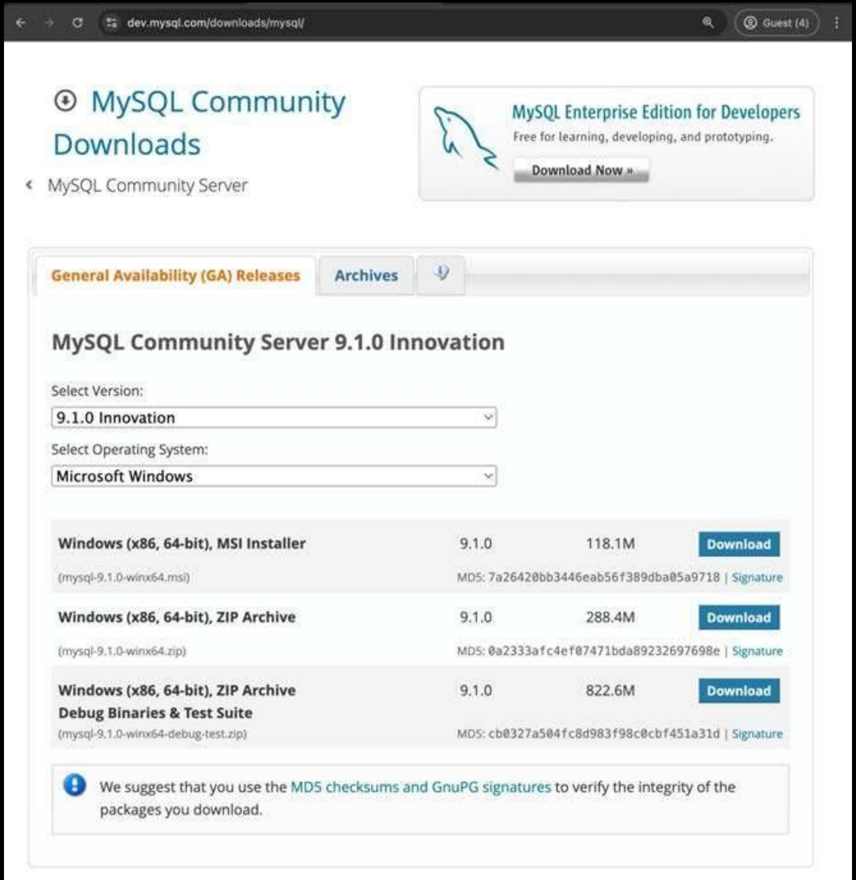
15.4 SQL vs NoSQL



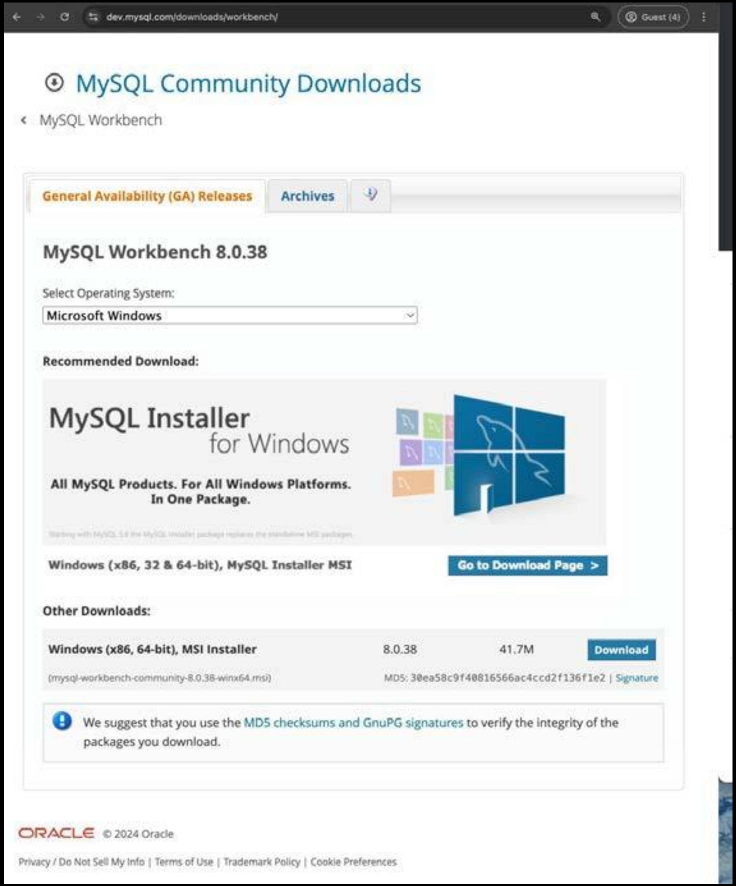
15.4 SQL vs NoSQL

Feature	SQL Databases	NoSQL Databases
Data Model	Relational (tables with rows and columns)	Non-relational (document, key-value, graph, etc.)
Schema	Fixed schema (predefined structure)	Flexible schema (dynamic structure)
Scalability	Vertically scalable (scale up)	Horizontally scalable (scale out)
Query Language	SQL (Structured Query Language)	Various query languages and APIs
ACID Compliance	Strong ACID compliance	Varies; often prioritizes performance over ACID
Use Cases	Structured data and complex queries	Unstructured data and real-time applications

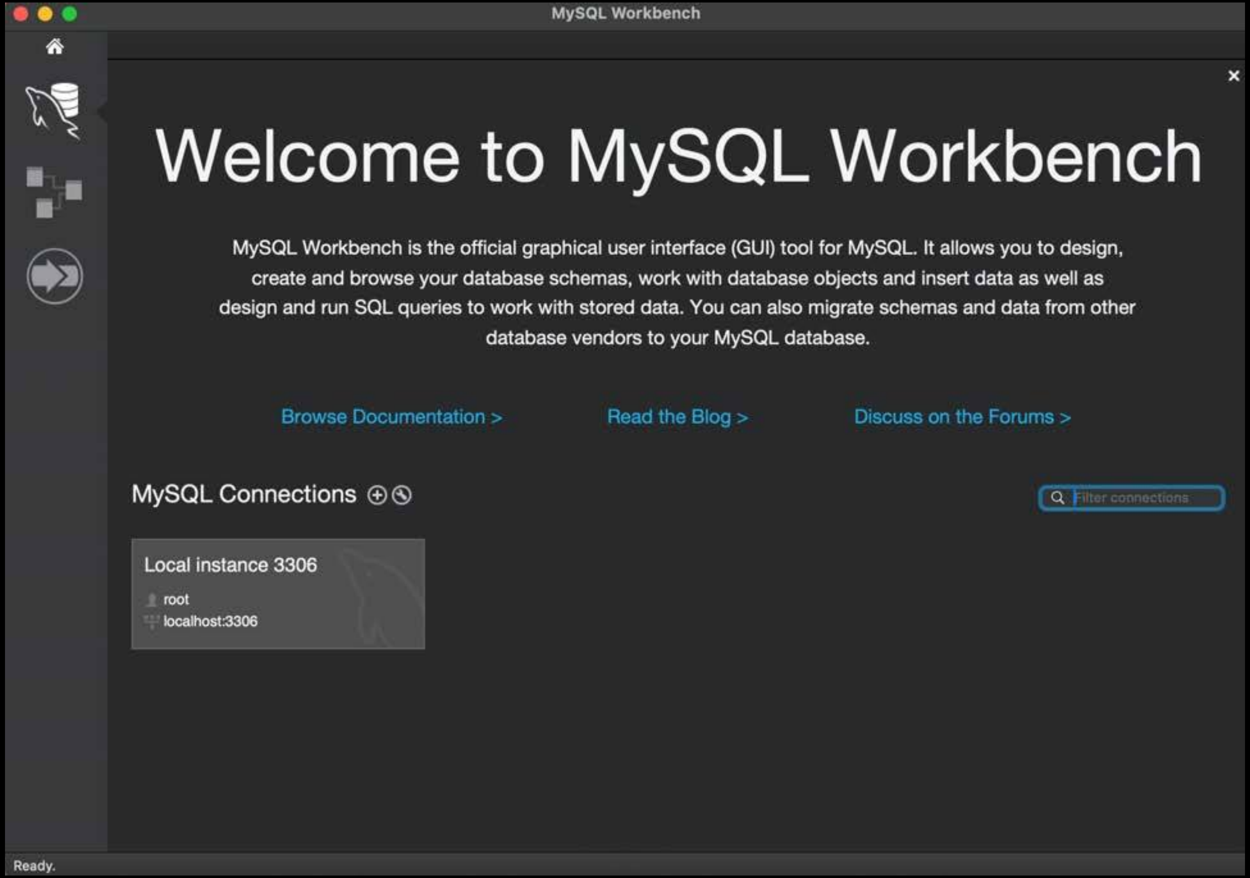
15.5 Installing MySQL



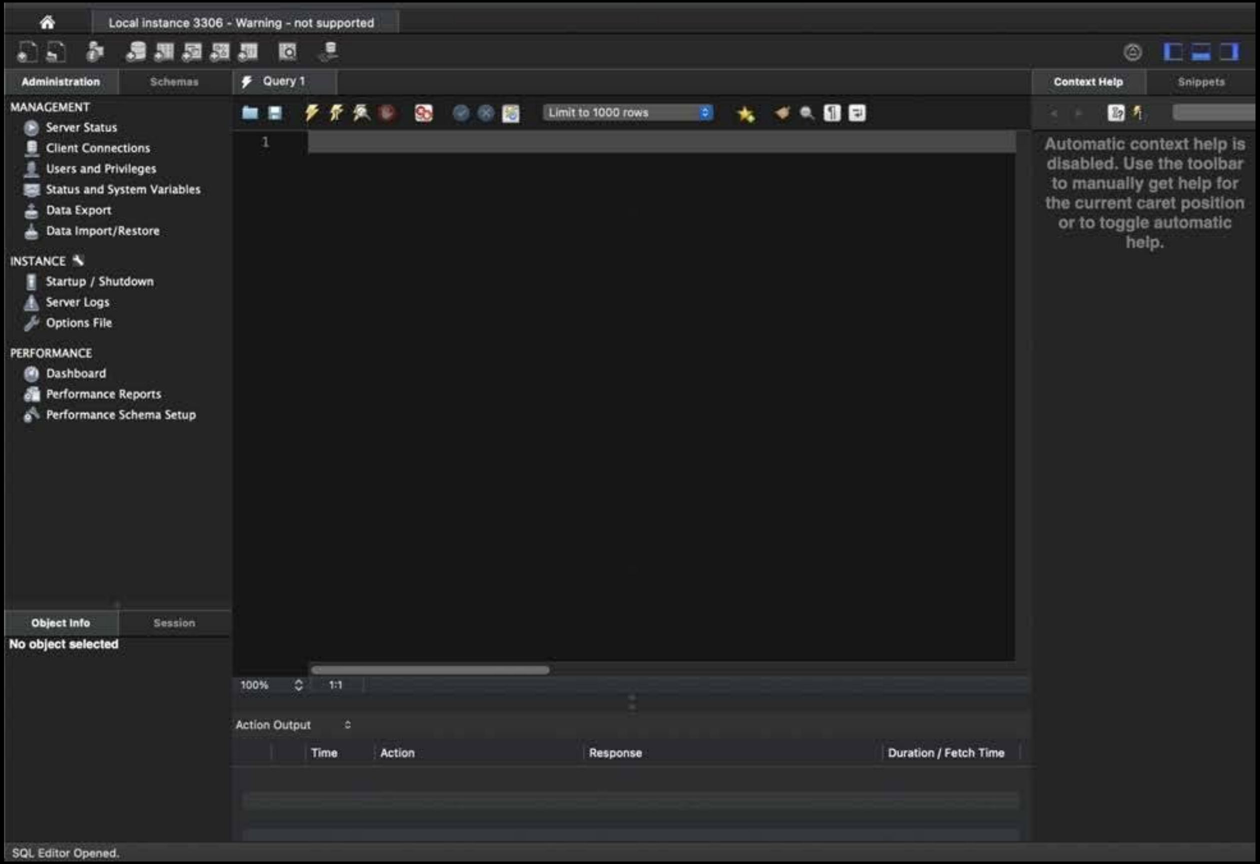
15.5 Installing MySQL



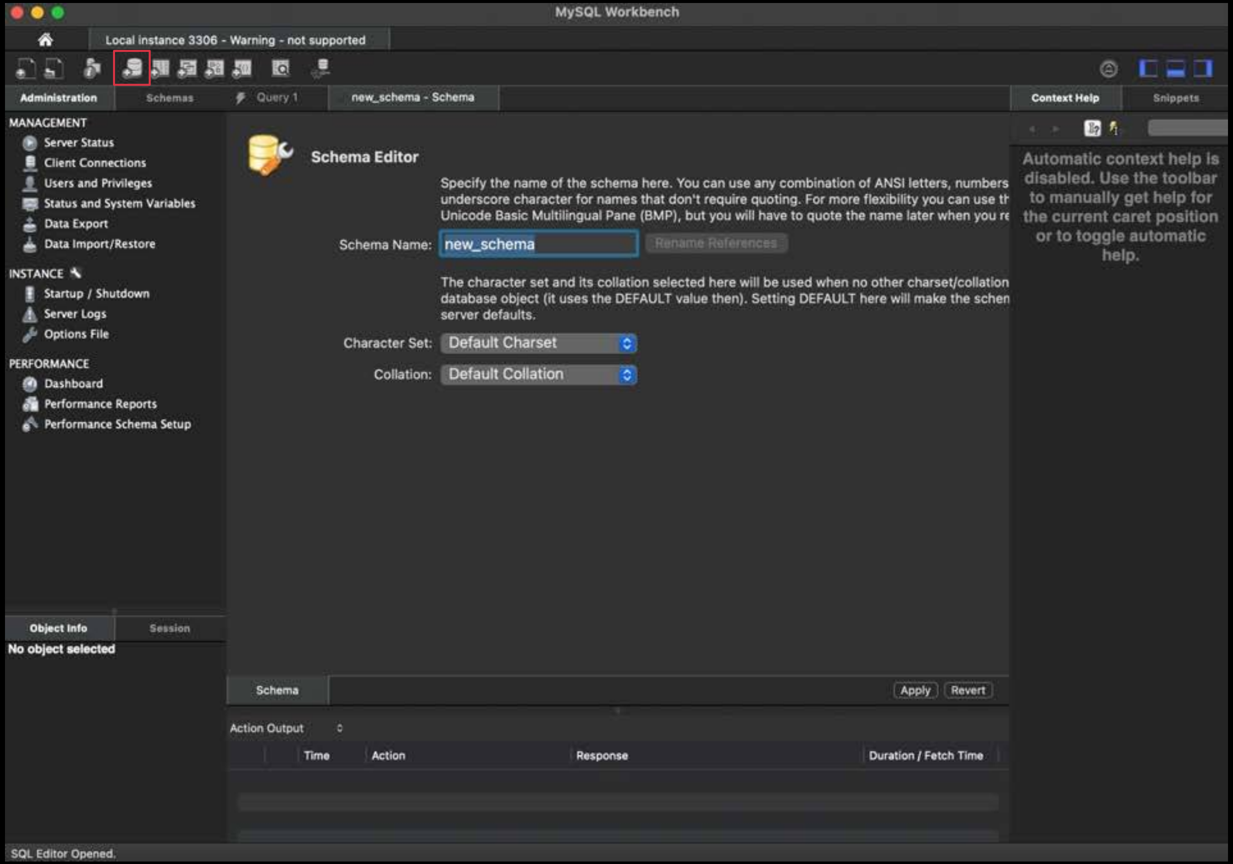
15.5 Installing MySQL



15.5 Installing MySQL



15.5 Installing MySQL



15.6 Connecting App to DB

- utils
  - database.js
  - pathUtil.js

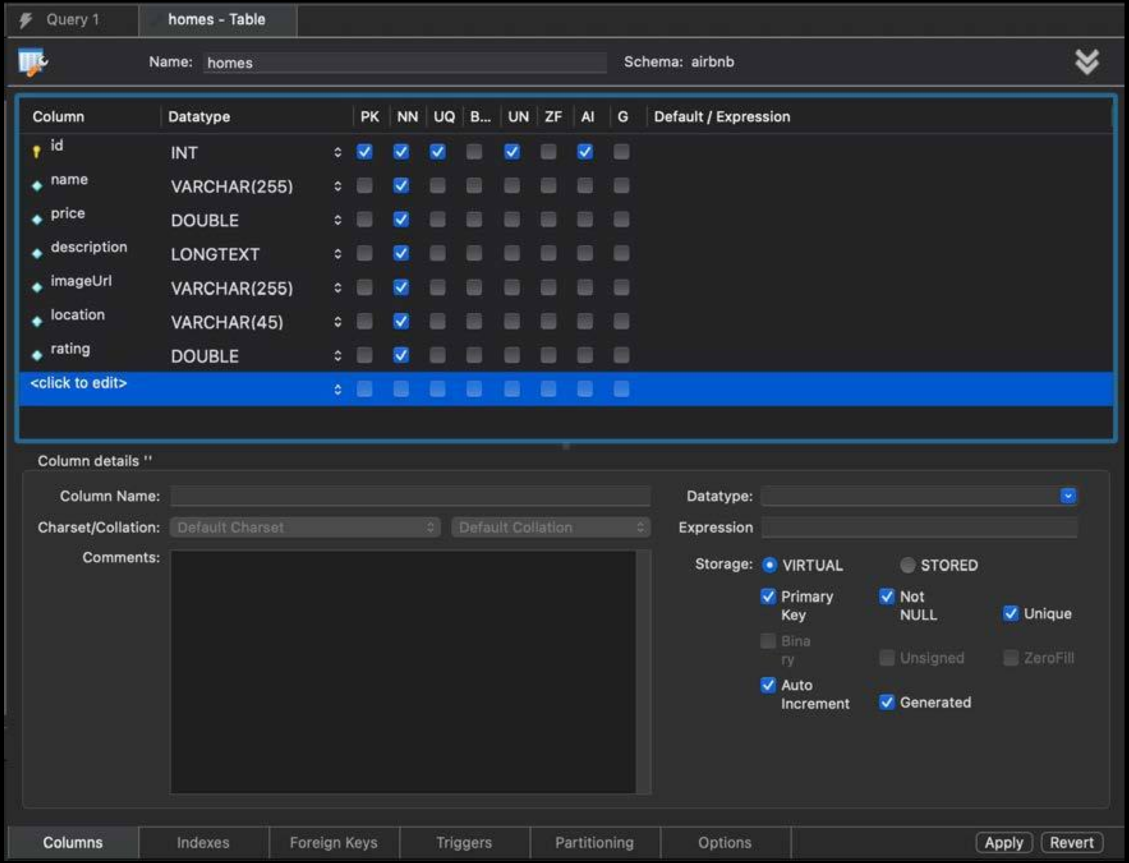
```
prashantjain@Prashants-Mac-mini Chapter 13 - MVC % npm install --save mysql2

added 12 packages, and audited 222 packages in 586ms

49 packages are looking for funding
  run `npm fund` for details
```

database.js

utils > database.js > ...

```js
const mysql = require("mysql2");

const pool = mysql.createPool({
  host: "localhost",
  user: "root",
  password: "CompleteCoding@01",
  database: "airbnb",
});

module.exports = pool.promise();
```

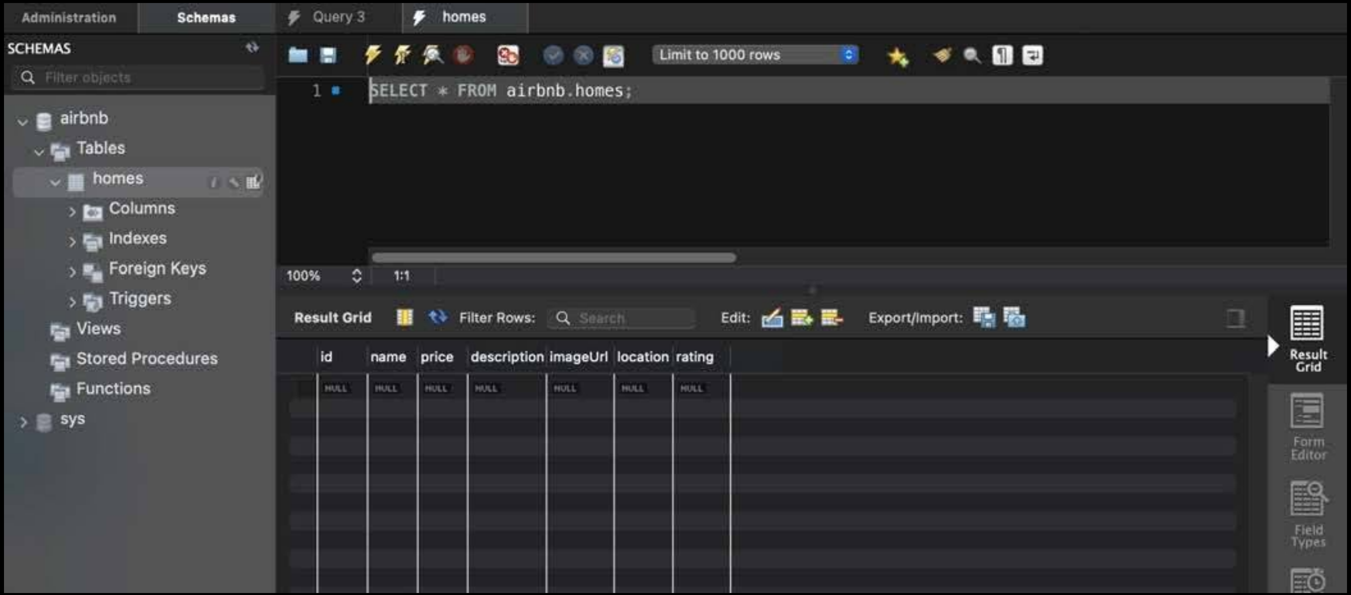
15.7 Creating homes Table



15.7 Creating homes Table

```sql
CREATE TABLE `airbnb`.`homes` (
  `id` INT UNSIGNED NOT NULL AUTO_INCREMENT,
  `name` VARCHAR(255) NOT NULL,
  `price` DOUBLE NOT NULL,
  `description` LONGTEXT NOT NULL,
  `imageUrl` VARCHAR(255) NOT NULL,
  `location` VARCHAR(45) NOT NULL,
  `rating` DOUBLE NOT NULL,
  PRIMARY KEY (`id`),
  UNIQUE INDEX `id_UNIQUE` (`id` ASC) VISIBLE);
```

15.7 Creating homes Table



15.8 Querying homes in App

```
const db = require("./utils/database");

db.execute("SELECT * FROM homes").then(([rows, fields]) => {
  console.log(rows);
  console.log(fields);
}).catch((error) => {
  console.log("Error Fetching Homes", error);
});
```

```
Server running on address http://localhost:3000
[
  {
    id: 1,
    name: 'Utsav',
    price: 999,
    description: 'the best holiday home',
    imageUrl: '/images/house1.png',
    location: 'delhi',
    rating: 4.5
  }
]
[
  `id` INT UNSIGNED NOT NULL PRIMARY KEY UNIQUE_KEY AUTO_INCREMENT,
  `name` VARCHAR(255) NOT NULL,
  `price` DOUBLE NOT NULL,
  `description` LONGTEXT NOT NULL,
  `imageUrl` VARCHAR(255) NOT NULL,
  `location` VARCHAR(45) NOT NULL,
  `rating` DOUBLE NOT NULL
]
```

15.9 Adding DB in Models

1. Remove the test code from app.js
2. Change the Home.js model file to remove all code related to file operations.
3. Import the DB from the utils.
4. Change photoUrl to imageUrl and houseName to name in the entire project.
5. Implement fetchAll:
    a. Using the query we used while testing.
    b. fetchAll will not take a callback but return a promise.
6. Go to StoreController and use the promise to get the data here.
7. Fix all the usages of fetchAll.

15.9 Adding DB in Models

2,3.

```
const db = require("../utils/database");
const Favourites = require("./favourites");

module.exports = class Home {
  constructor(houseName, price, location, rating, photoUrl) {
    this.houseName = houseName;
    this.price = price;
    this.location = location;
    this.rating = rating;
    this.photoUrl = photoUrl;
  }

  save() { ⇥ tab
  }

  static fetchAll() {
  }

  static findById(id) {
  }

  static deleteById(id) {
  }
};
```

15.9 Adding DB in Models

5.

```
static fetchAll() {
  return db.execute("SELECT * FROM homes");
}
```

6.

```
exports.getHomes = (req, res, next) => {
  Home.fetchAll()
    .then(([rows, fields]) => {
      res.render("store/home-list", {
        registeredHomes: rows,
        pageTitle: "Homes List",
        currentPage: "Home",
      });
    })
    .catch((error) => {
      console.log("Error Fetching Homes", error);
    });
};
```

15.9 Adding DB in Models

7.

```
exports.getIndex = (req, res, next) => {
  Home.fetchAll()
  .then(([rows, fields]) => {
    res.render("store/index", {
      registeredHomes: rows,
      pageTitle: "airbnb Home",
      currentPage: "index",
    })
  })
  .catch((error) => {
    console.log("Error Fetching Homes", error);
  });
};
```

```
exports.getFavouriteList = (req, res, next) => {
  Favourites.getFavourites((favourites) => {
    Home.fetchAll().then(([registeredHomes, fields]) => {
      const favouritesWithDetails = favourites.map((homeId) =>
        registeredHomes.find((home) => home.id === homeId)
      );
      res.render("store/favourite-list", {
        favourites: favouritesWithDetails,
        pageTitle: "My Favourites",
        currentPage: "favourites",
      });
    }).catch((error) => {
      console.log("Error Fetching Homes", error);
    });
  });
};
```

```
exports.getHostHomes = (req, res, next) => {
  Home.fetchAll().then(([rows, fields]) => {
    res.render("host/host-home-list", {
      registeredHomes: rows,
      pageTitle: "Host Homes",
      currentPage: "hostHomes",
    });
  }).catch((error) => {
    console.log("Error Fetching Homes", error);
  });
};
```

15.10 Adding Home in Model

1. Add the description field in home. Change constructor and usage.
2. Make changes in UI to input and show it everywhere.
3. Implement the save method using the insert query.
4. Change the usages of save method to use the promise.

15.10 Adding Home in Model

1.

```
module.exports = class Home {
  constructor(name, description, price, location, rating, imageUrl) {
    this.name = name;
    this.description = description;
    this.price = price;
    this.location = location;
    this.rating = rating;
    this.imageUrl = imageUrl;
  }
```

```
exports.postAddHome = (req, res, next) => {
  const { name, description, price, location, rating, imageUrl } = req.body;
  const home = new Home(name, description, price, location, rating, imageUrl);
```

```
exports.postEditHome = (req, res, next) => {
  const { id, name, description, price, location, rating, imageUrl } = req.body;
  const home = new Home(name, description, price, location, rating, imageUrl);
```

15.10 Adding Home in Model

2.

```
<input
  type="text"
  name="description"
  value="<%= home ? home.description : '' %>"
  placeholder="Enter your House Description"
  class="w-full px-4 py-2 mb-4 border rounded-md focus:outline-none focus:ring-2
  focus:ring-red-500"/>

<div class="border-b pb-4">
    <h3 class="text-2xl font-semibold mb-2">Description</h3>
    <p class="text-gray-600"><%= home.description %></p>
</div>

<div class="border-b pb-4">
    <h3 class="text-2xl font-semibold mb-2">Location</h3>
    <p class="text-gray-600"><%= home.location %></p>
</div>
```

15.10 Adding Home in Model

3.

```
save() {
  return db.execute(
    "INSERT INTO homes (name, price, location, rating, imageUrl) VALUES (?, ?, ?, ?, ?)",
    [this.name, this.price, this.location, this.rating, this.imageUrl]
  );
}
```

4.

```
exports.postAddHome = (req, res, next) => {
  const { name, description, price, location, rating, imageUrl } = req.body;
  const home = new Home(name, description, price, location, rating, imageUrl);
  home.save().then(() => {
    res.render("host/home-added", {
      pageTitle: "Home Added Successfully",
      currentPage: "homeAdded",
    });
  }).catch((error) => {
    console.log("Error Adding Home", error);
  });
};

exports.postEditHome = (req, res, next) => {
  const { id, name, description, price, location, rating, imageUrl } = req.body;
  const home = new Home(name, description, price, location, rating, imageUrl);
  home.id = id;
  home.save().then(() => {
    res.redirect("/host/host-home-list");
  }).catch((error) => {
    console.log("Error Editing Home", error);
  });
};
```

15.11 Implementing Model using Where

```
static findById(id) {
  return db.execute("SELECT * FROM homes WHERE id = ?", [id]);
}

static deleteById(id) {
  return db.execute("DELETE FROM homes WHERE id = ?", [id]);
}
```

```
exports.postDeleteHome = (req, res, next) => {
  const homeId = req.params.homeId;
  Home.deleteById(homeId).then(() => {
    res.redirect("/host/host-home-list");
  }).catch((error) => {
    console.log("Error Deleting Home", error);
  });
};
```

```
exports.getHome = (req, res, next) => {
  const homeId = req.params.homeId;
  Home.findById(homeId).then(([rows]) => {
    const home = rows[0];
    if (!home) {
      return res.redirect("/homes");
    }
    res.render("store/home-detail", {
      home: home,
      pageTitle: home.name,
      currentPage: "homes",
    });
  }).catch((error) => {
    console.log("Error Fetching Home", error);
  });
};
```

Practise Milestone

Take your airbnb forward:

- Change the save method to support both edit and insert functionality.

Practise Milestone (Solution)

```
.id

    airbnbDb.
'UPDATE homes SET houseName=?, price=?, location=?, rating=?, photoUrl=?, description=? WHERE id=?',
[.houseName, .price, .location, .rating, .photoUrl, .description, .id]
);

    airbnbDb.
'INSERT INTO homes (houseName, price, location, rating, photoUrl, description) VALUES
(?, ?, ?, ?, ?, ?)',
[.houseName, .price, .location, .rating, .photoUrl, .description]
);
```

Practise Milestone (Solution)

```javascript
exports.postEditHome = (req, res, next) => {
  const { id, houseName, price, location, rating, photoUrl, description } =
    req.body;
  const newHome = new Home(
    houseName,
    price,
    location,
    rating,
    photoUrl,
    description
  );
  newHome.id = id;
  newHome
    .save()
    .then(() => {
      res.redirect("/host/host-homes");
    })
    .catch((error) => {
      console.log("Error while updating home", error);
    });
};
```

Revision

1. What is a DB (Database)
2. Introduction to SQL DB
3. Introduction to NoSQL DB
4. SQL vs NoSQL
5. Installing MySQL
6. Connecting App to DB
7. Creating homes Table
8. Querying homes in App
9. Adding DB in Models
10. Adding Home in Model
11. Implementing Model using Where

16. Introduction to MongoDB

1. What is MongoDB
2. Setting up MongoDB
3. Installing MongoDB Driver
4. Creating MongoDB Connection
5. Saving a Home
6. Install MongoDB Compass
7. Install MongoDB for VSCode
8. Fetching all Homes
9. Fetching one Home
10. Supporting Edit & Delete
11. Adding MongoDB to Favourite

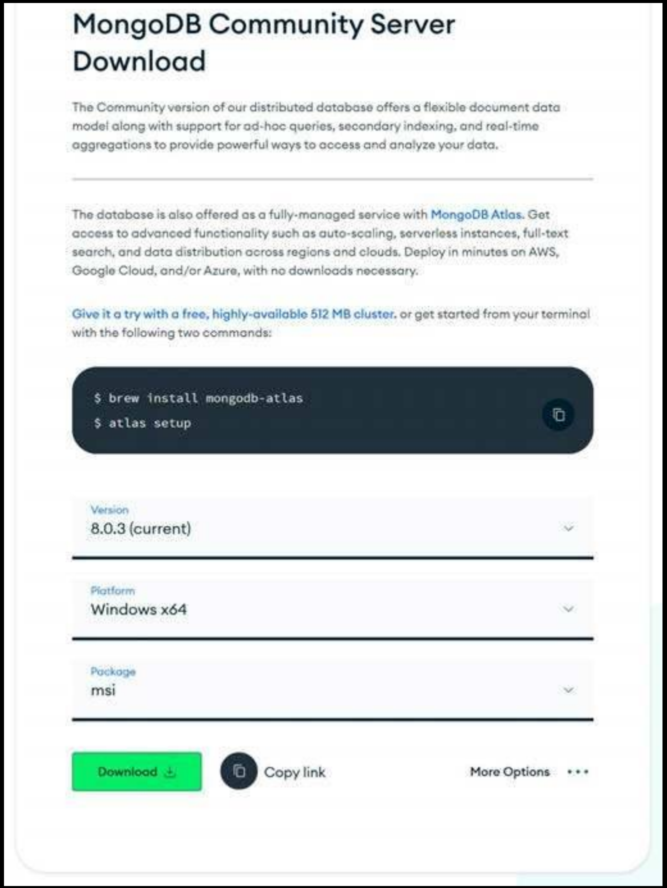
16.1 What is MongoDB

1. MongoDB is the product and the company that builds it.
2. The name comes from the work Humongous.
3. NoSQL Document Database: Stores data in flexible, JSON-like documents.
4. Dynamic Schema: Allows fields to vary across documents without predefined schemas.
5. High Performance: Optimized for fast read and write operations.
6. Scalability: Supports horizontal scaling through sharding.
7. High Availability: Provides replication with automatic failover.
8. Rich Query Capabilities: Offers powerful querying, indexing, and aggregation.
9. Geospatial and Text Search: Includes support for location-based and full-text queries.
10. Cross-Platform Compatibility: Works with various operating systems and programming languages.
11. Easy Integration: Integrates smoothly with modern development stacks.


```
{
  "_id": "4f5b5c85-d8d3-4f58-8acf-3f5e5e4e59ea",
  "Items": [
    {
      "ProductId": 1,
      "ProductName": "Elden Ring",
      "Price": "49.97",
      "Quantity": 1
    },
    {
      "ProductId": 2,
      "ProductName": "FIFA 23",
      "Price": "69.97",
      "Quantity": 1
    }
  ]
}
```

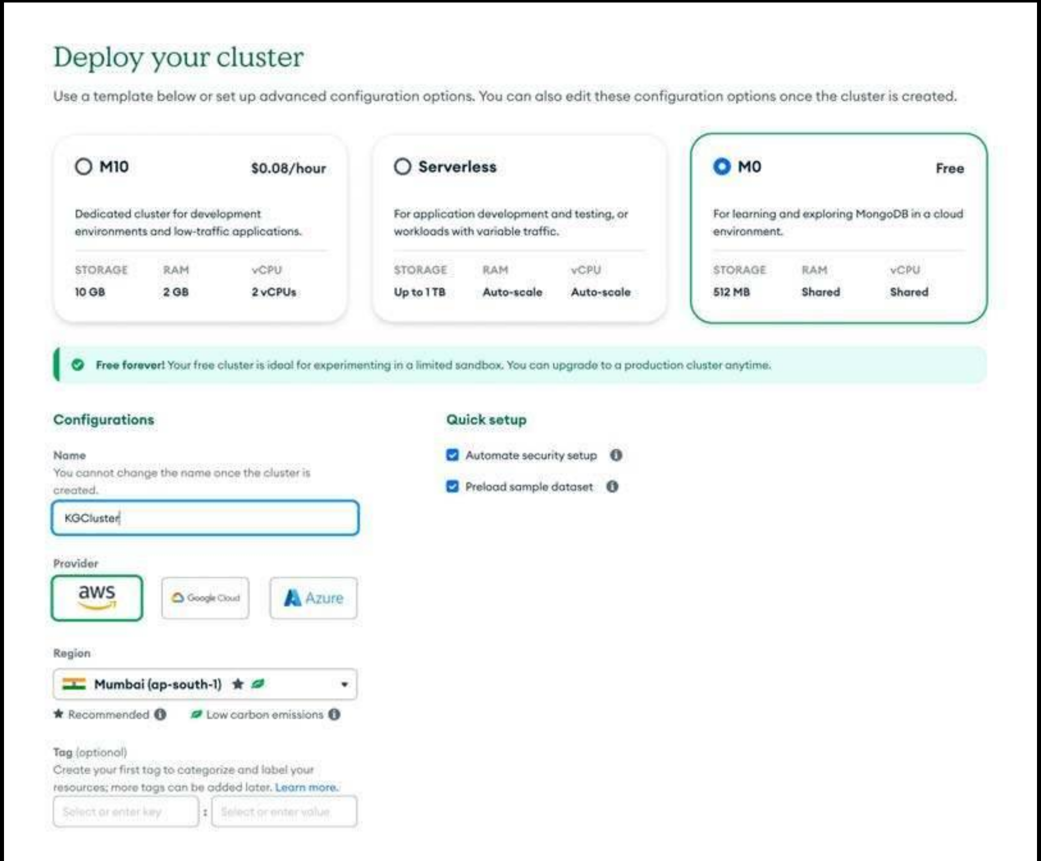

MongoDB.

16.2 Setting up MongoDB

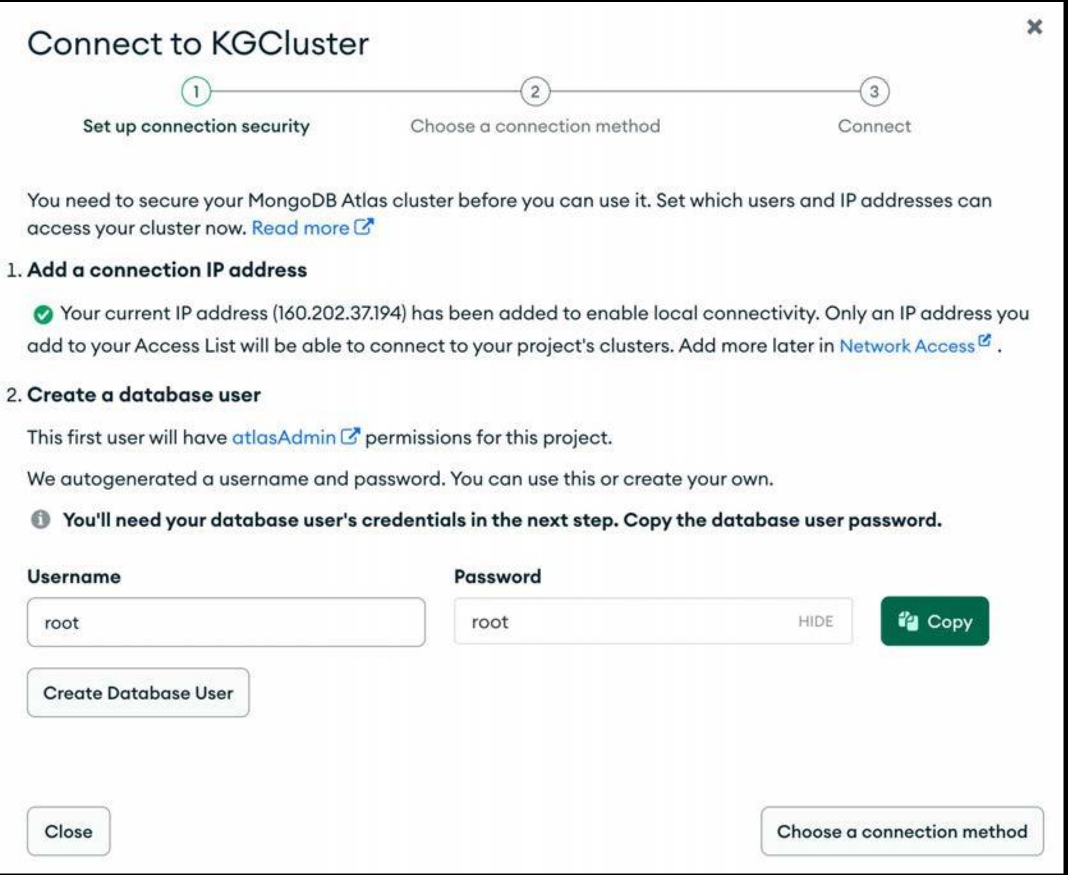


VS

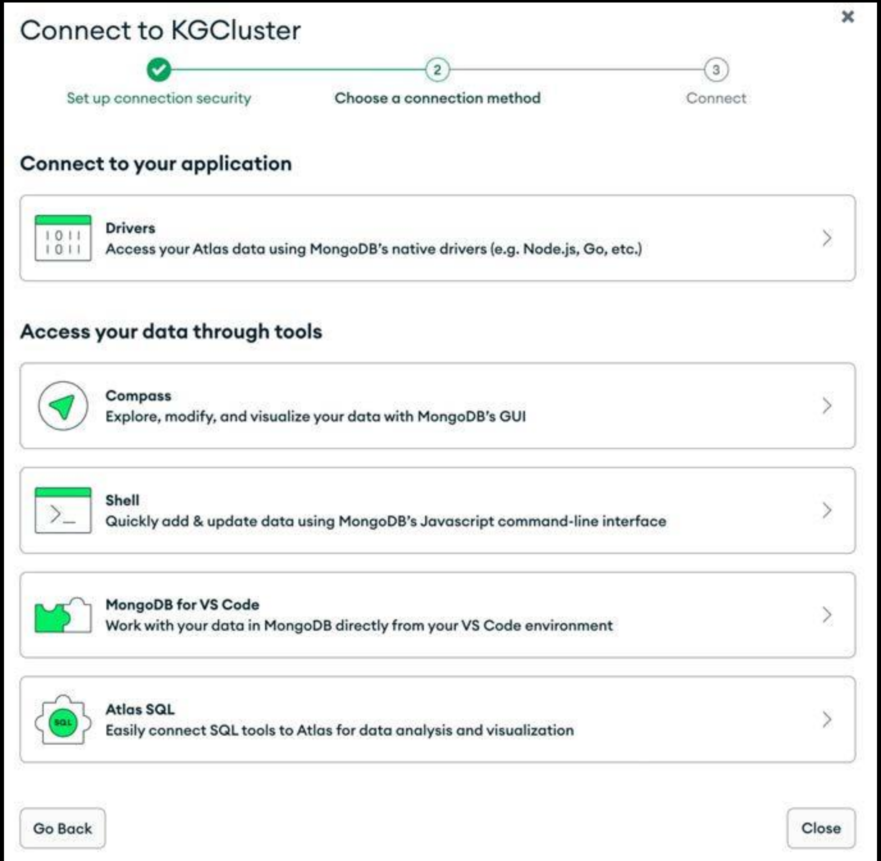
16.2 Setting up MongoDB

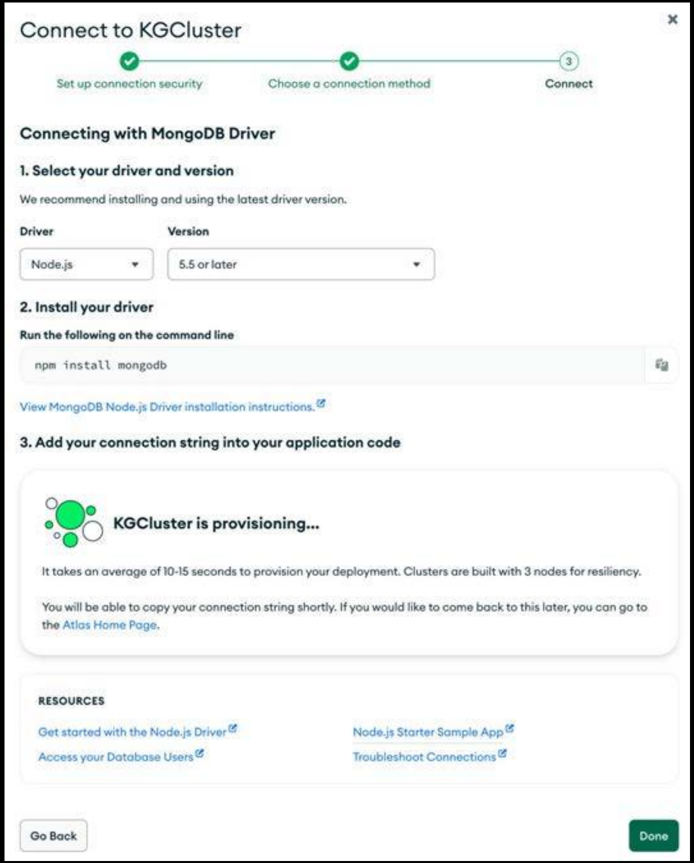


16.2 Setting up MongoDB

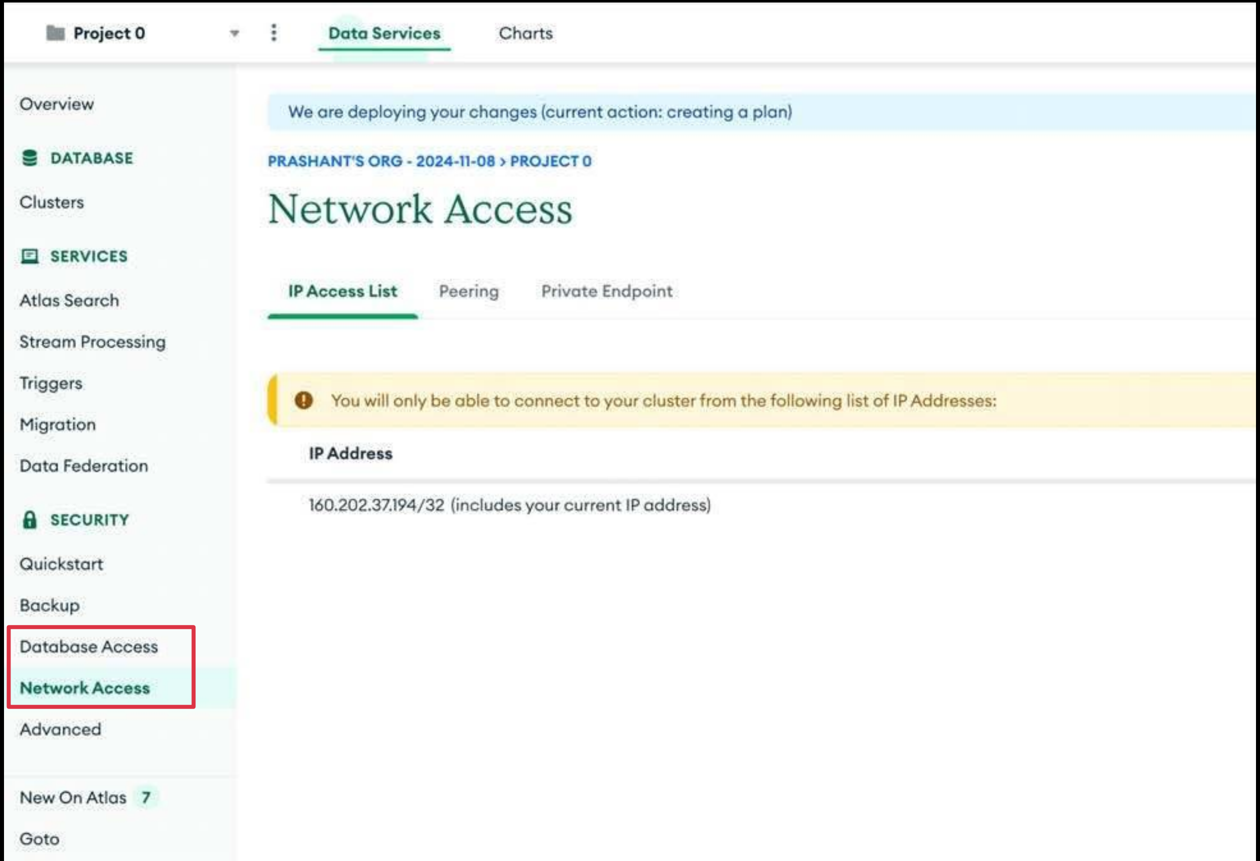


16.2 Setting up MongoDB





16.2 Setting up MongoDB



16.3 Installing MongoDB Driver

TERMINAL

```
prashantjain@Prashants-Mac-mini Chapter 13 - MVC % npm install mongodb

added 12 packages, and audited 234 packages in 2s

49 packages are looking for funding
  run `npm fund` for details

2 low severity vulnerabilities

To address all issues, run:
  npm audit fix

Run `npm audit` for details.
prashantjain@Prashants-Mac-mini Chapter 13 - MVC %
```

16.3 Installing MongoDB Driver

database.js M

Test Project > node > Chapter 13 - MVC > utils > database.js > ...

```
const mongodb = require("mongodb");

const MongoClient = mongodb.MongoClient;

const url = "mongodb+srv://root:root@kgcluster.ie6mb.mongodb.net/?retryWrites=true&w=majority&appName=KGCluster";

const mongoConnect = (callback) => {
  MongoClient.connect(url)
  .then((client) => {
    console.log("Connected to MongoDB");
    callback(client);
  })
  .catch((err) => {
    console.log(err);
  });
};

module.exports = mongoConnect;
```

16.3 Installing MongoDB Driver

database.js M   app.js M ×

Test Project > node > Chapter 13 - MVC > app.js > ...

```javascript

const mongoConnect = require("./utils/database");
const PORT = 3000;
mongoConnect((client) => {
  console.log(client);
  app.listen(PORT, () => {
    console.log(`Server running on address http://localhost:${PORT}`);
  });
});
```

```
Connected to MongoDB
<ref *1> MongoClient {
  _events: [Object: null prototype] {},
  _eventsCount: 0,
  _maxListeners: undefined,
  mongoLogger: undefined,
  s: {
    url: 'mongodb+srv://root:root@kgcluster.ie6mb.mongodb.net/?retryWrites=true&w=majority&appName=KGCluster',
    bsonOptions: {
      raw: false,
      useBigInt64: false,
      promoteLongs: true,
      promoteValues: true,
      promoteBuffers: false,
      ignoreUndefined: false,
      bsonRegExp: false,
      serializeFunctions: false,
      fieldsAsRaw: {},
      enableUtf8Validation: true
    },
    namespace: MongoDBNamespace { db: 'admin', collection: undefined },
    hasBeenClosed: false,
    sessionPool: ServerSessionPool { client: [Circular *1], sessions: [List] },
    activeSessions: Set(0) {},
    authProviders: MongoClientAuthProviders { existingProviders: [Map] },
    options: [Getter],
    readConcern: [Getter],
    writeConcern: [Getter],
    readPreference: [Getter],
    isMongoClient: [Getter]
  },
  connectionLock: undefined,
  topology: Topology {
    _events: [Object: null prototype] {
      topologyDescriptionChanged: [Array],
      connectionPoolCreated: [Function (anonymous)],
```

```
Server running on address http://localhost:3000
TypeError: db.execute is not a function
    at Home.fetchAll (/Users/prashantjain/workspace/stuff/Test Project/no
2:15)
    at exports.getIndex (/Users/prashantjain/workspace/stuff/Test Project
toreController.js:5:8)
    at Layer.handle [as handle_request] (/Users/prashantjain/workspace/st
VC/node_modules/express/lib/router/layer.js:95:5)
    at next (/Users/prashantjain/workspace/stuff/Test Project/node/Chapte
router/route.js:149:13)
    at Route.dispatch (/Users/prashantjain/workspace/stuff/Test Project/n
press/lib/router/route.js:119:3)
    at Layer.handle [as handle_request] (/Users/prashantjain/workspace/st
VC/node_modules/express/lib/router/layer.js:95:5)
    at /Users/prashantjain/workspace/stuff/Test Project/node/Chapter 13 -
/index.js:284:15
    at Function.process_params (/Users/prashantjain/workspace/stuff/Test
odules/express/lib/router/index.js:346:12)
    at next (/Users/prashantjain/workspace/stuff/Test Project/node/Chapte
router/index.js:280:10)
    at Function.handle (/Users/prashantjain/workspace/stuff/Test Project/
xpress/lib/router/index.js:175:3)
```

16.3 Installing MongoDB Driver

```
module.exports = class Home {
  constructor(name, description, price, location, rating,
  imageUrl) {
    this.name = name;
    this.description = description;
    this.price = price;
    this.location = location;
    this.rating = rating;
    this.imageUrl = imageUrl;
  }

  save() {
  }

  static fetchAll() {
  }

  static findById(id) {
  }

  static deleteById(id) {
  }
};
```

16.4 Creating MongoDB Connection

database.js M × app.js M home.js M

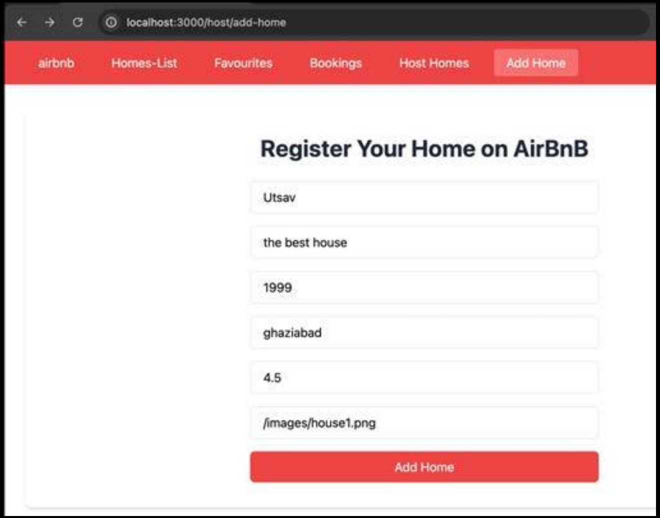
Test Project > node > Chapter 13 - MVC > utils > database.js > ...

```
const mongodb = require("mongodb");

const MongoClient = mongodb.MongoClient;

const url = "mongodb+srv://root:root@kgcluster.ie6mb.mongodb.net/?retryWrites=true&w=majority&appName=KGCluster";

let _db;

const mongoConnect = (callback) => {
  MongoClient.connect(url)
  .then((client) => {
    console.log("Connected to MongoDB");
    _db = client.db("airbnb");
    callback();
  })
  .catch((err) => {
    console.log(err);
    throw err;
  });
};

const getDb = () => {
  if (!_db) {
    throw new Error("Database not connected");
  }
  return _db;
};

exports.mongoConnect = mongoConnect;
exports.getDb = getDb;
```

16.5 Saving a Home

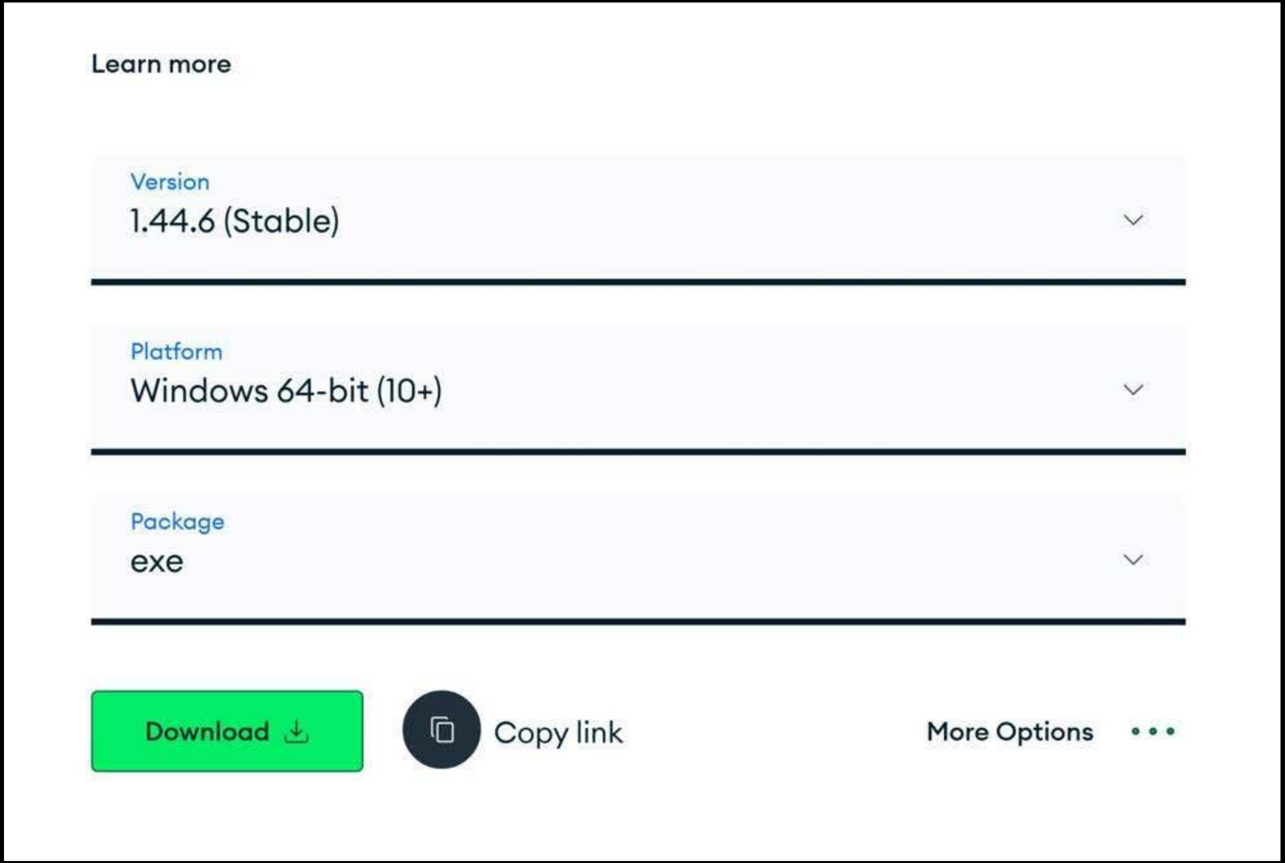
```
save() {
  const db = getDb();
  // insert Many takes an array of objects
  return db.collection("homes").insertOne(this).then((result) => {
    console.log(result);
  });
}
```

```
{
  acknowledged: true,
  insertedId: new ObjectId('672df86c3d20696a2b523c79')
}
```

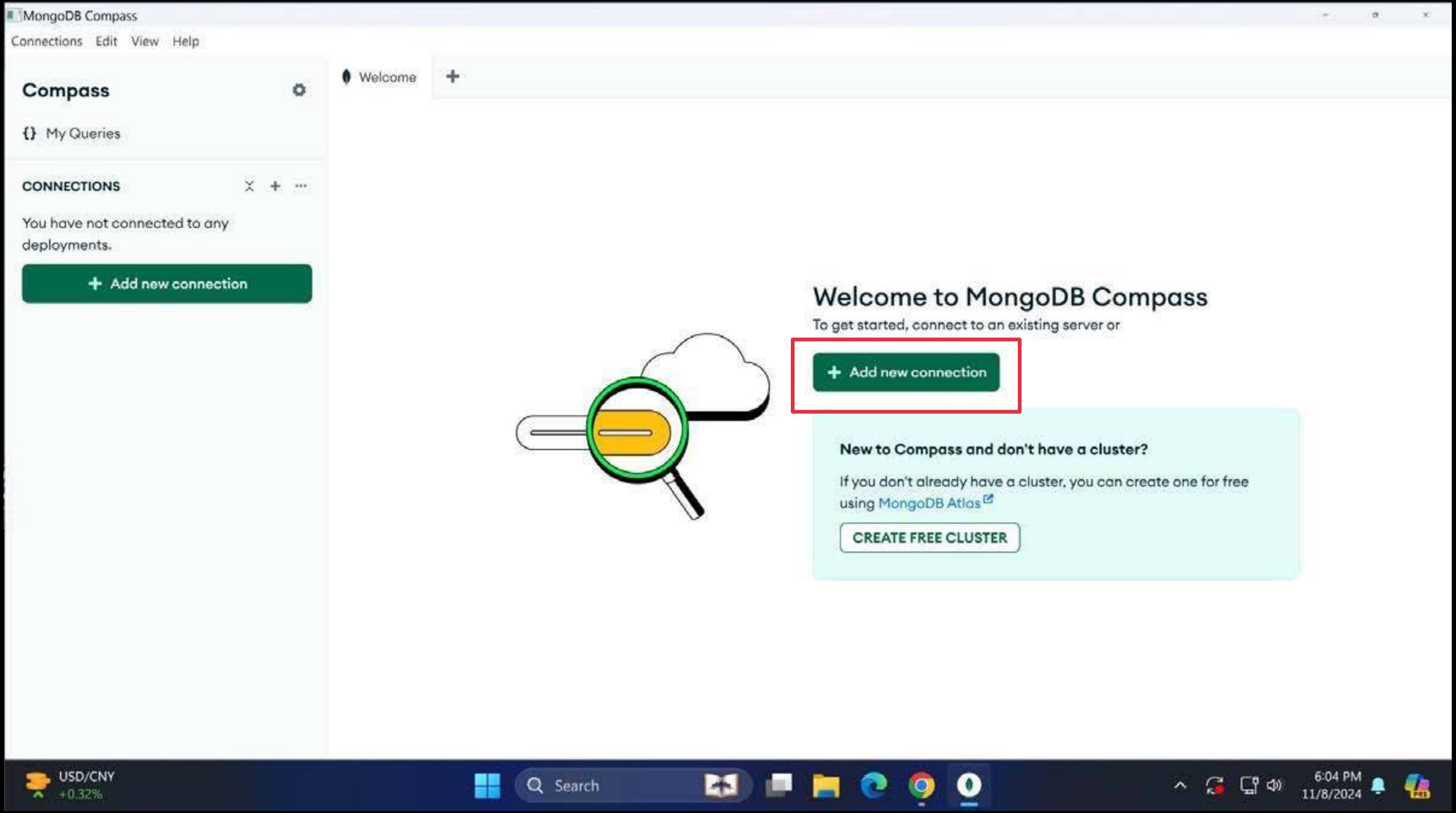


16.6 Install MongoDB Compass

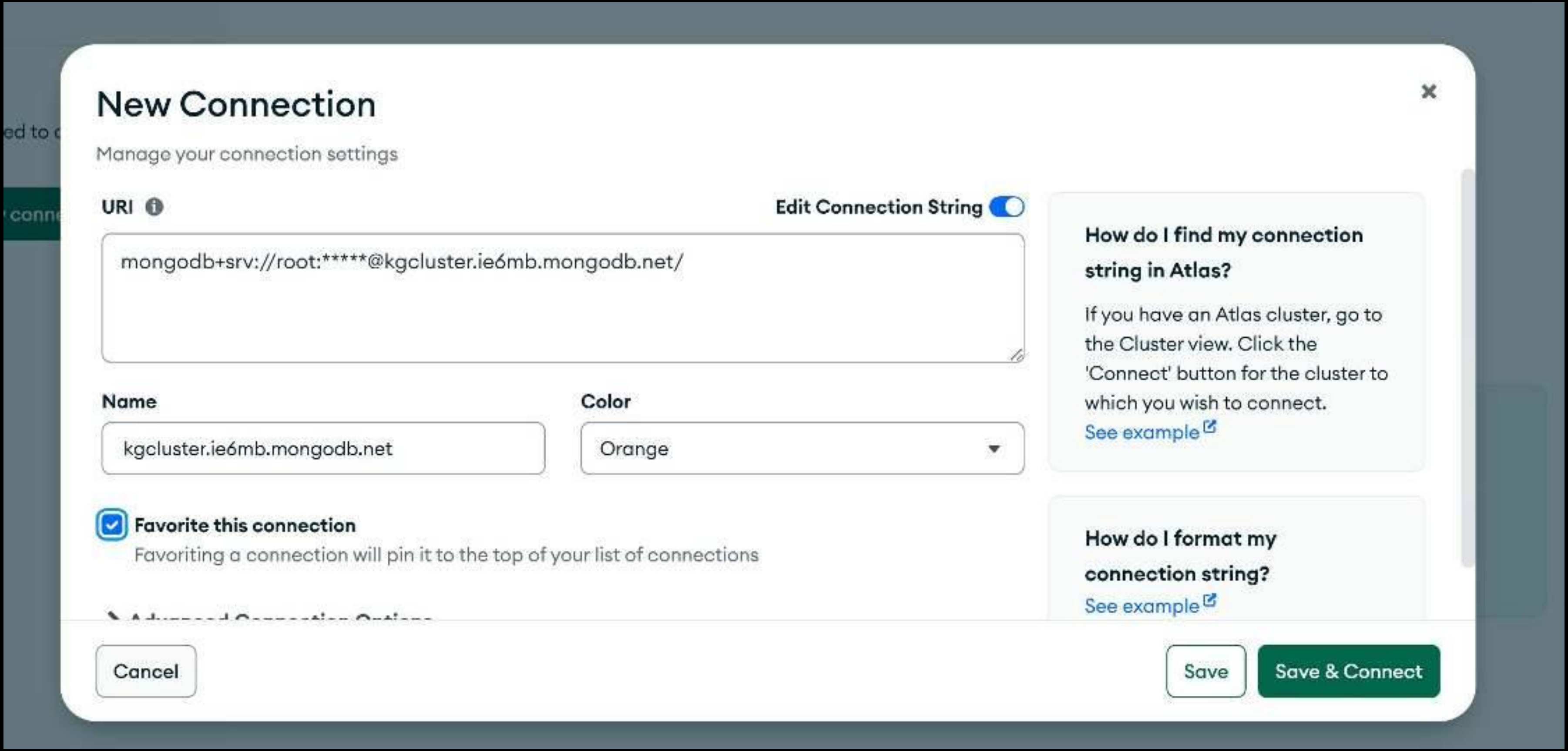
16.6 Install MongoDB Compass



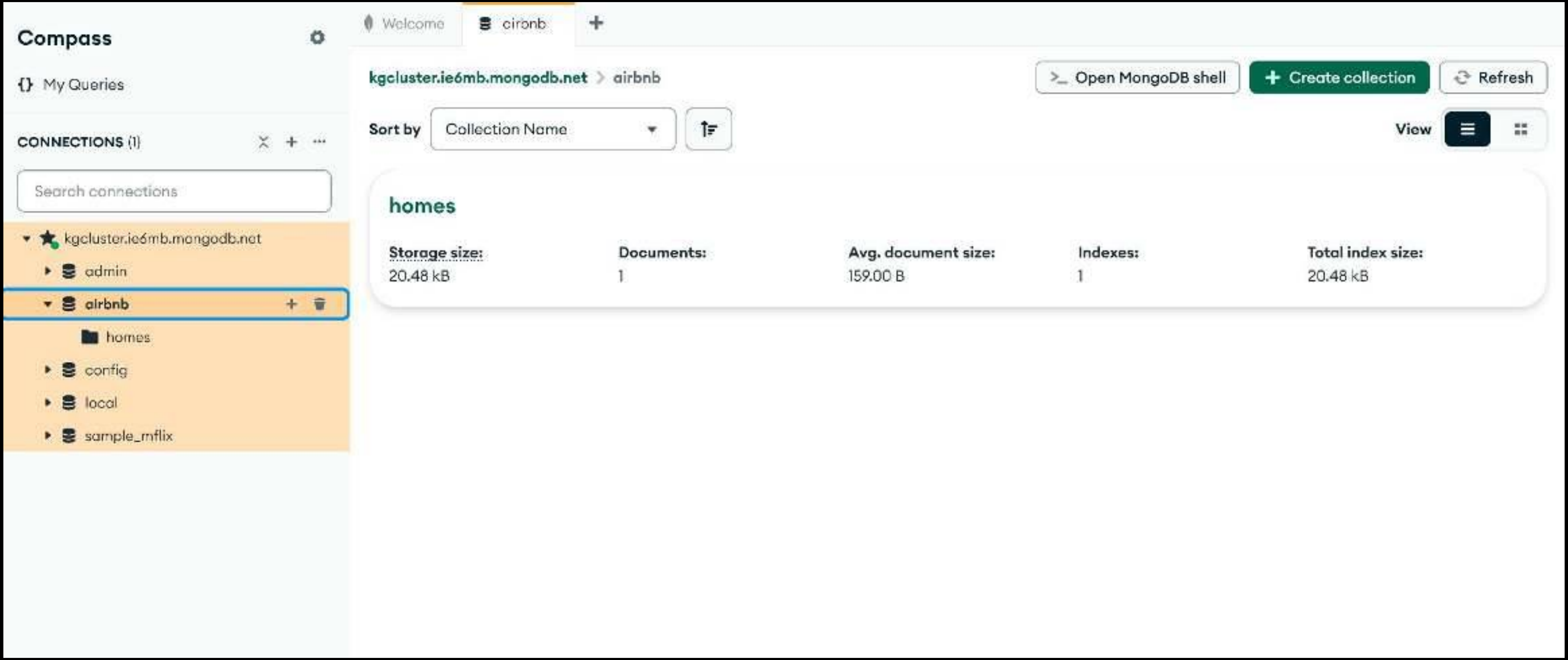
16.6 Install MongoDB Compass



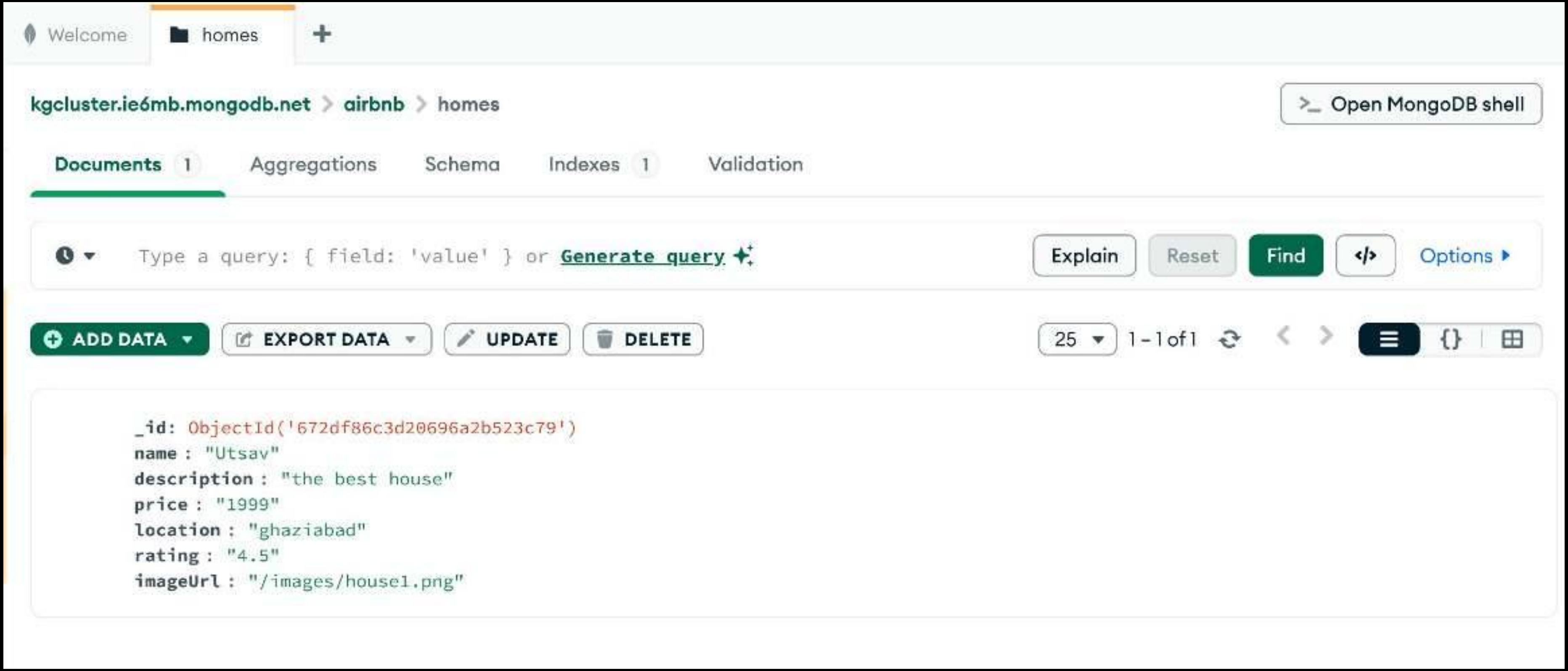
16.6 Install MongoDB Compass



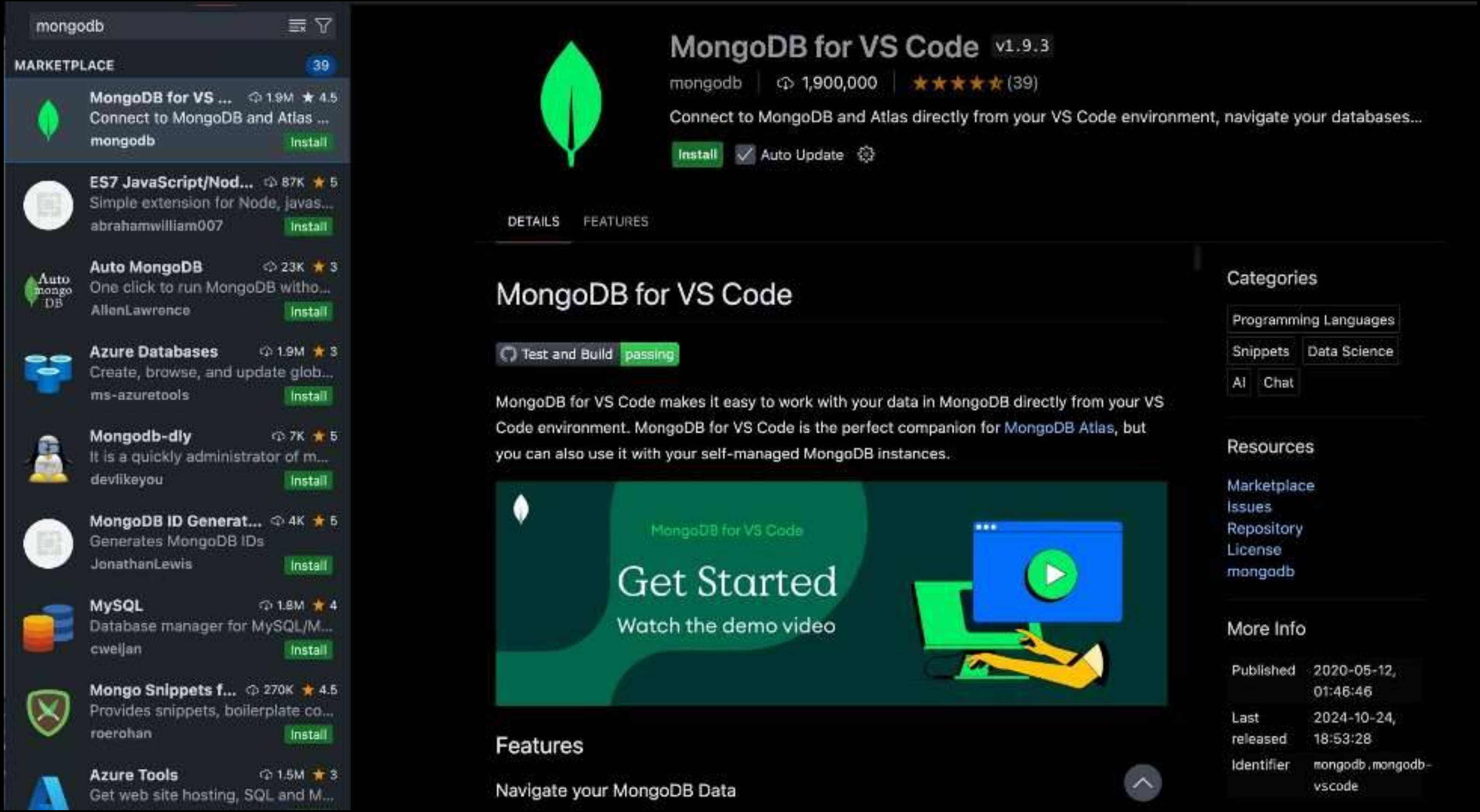
16.6 Install MongoDB Compass



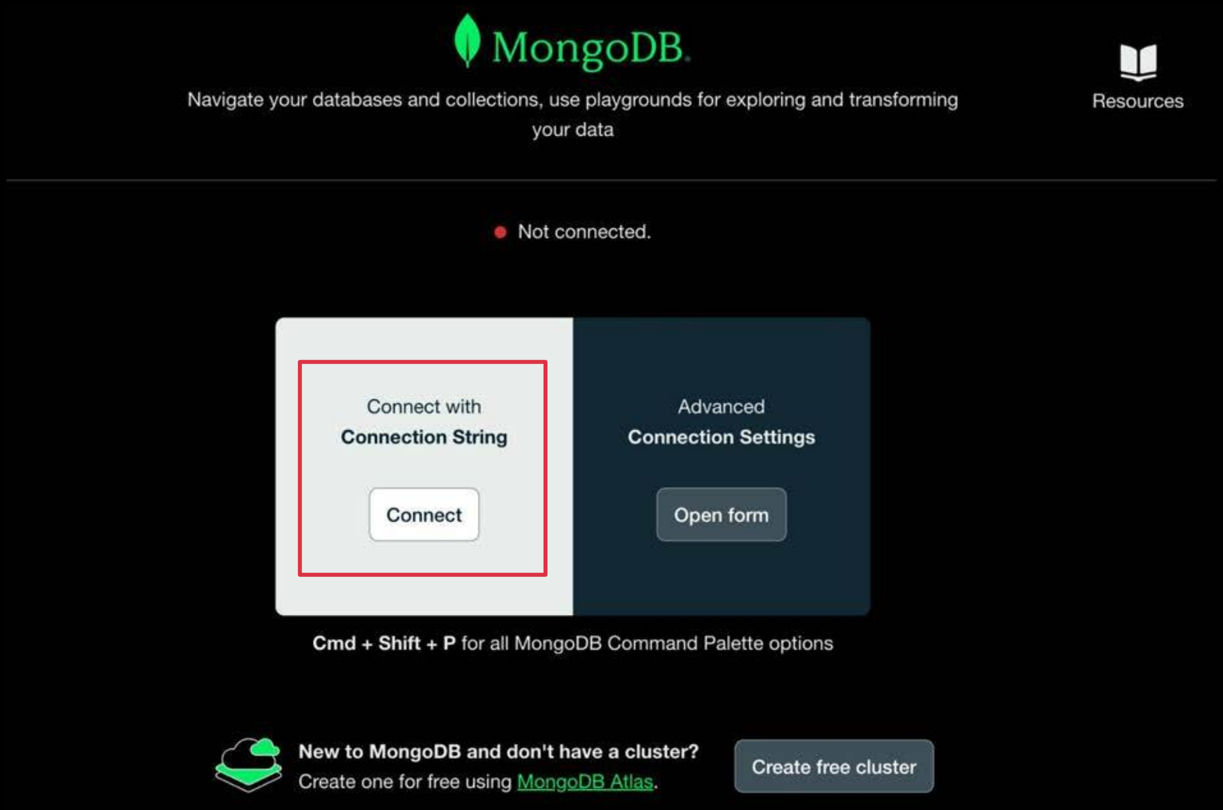
16.6 Install MongoDB Compass



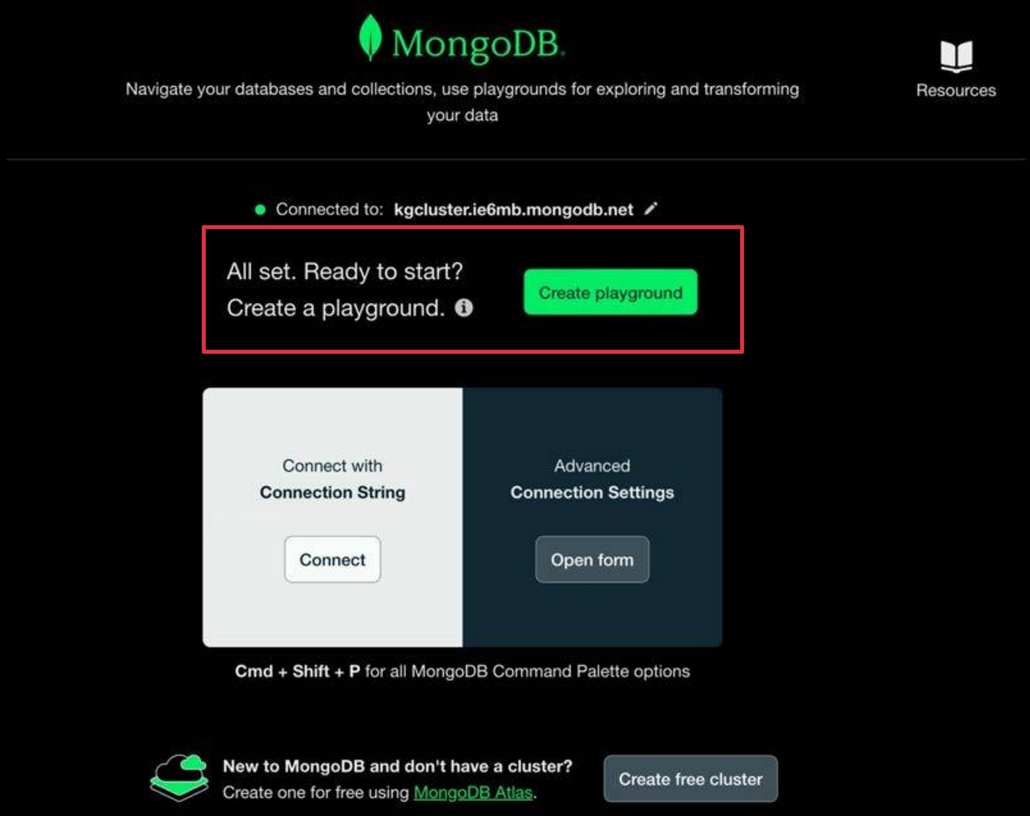
16.7 Install MongoDB for VSCode



16.7 Install MongoDB for VSCode



16.7 Install MongoDB for VSCode



16.7 Install MongoDB for VSCode

Currently connected to kgcluster.ie6mb.mongodb.net. Click here to change connection.

```
/* global use, db */
// MongoDB Playground
// To disable this template go to Settings | MongoDB | Use Default Template For Playground.
// Make sure you are connected to enable completions and to be able to run a playground.
// Use Ctrl+Space inside a snippet or a string literal to trigger completions.
// The result of the last command run in a playground is shown on the results panel.
// By default the first 20 documents will be returned with a cursor.
// Use 'console.log()' to print to the debug output.
// For more documentation on playgrounds please refer to
// https://www.mongodb.com/docs/mongodb-vscode/playgrounds/

// Select the database to use.
use('airbnb');

// Get all documents from homes collection
const homes = db.getCollection('homes').find({}).toArray();

// Print the results
console.log('All homes in collection:');
console.log(JSON.stringify(homes, null, 2));

// Print total count
const totalHomes = db.getCollection('homes').countDocuments();
console.log(`Total number of homes: ${totalHomes}`);
```

16.7 Install MongoDB for VSCode

Currently connected to kgcluster.ie6mb.mongodb.net. Click here to change connection.

```javascript
/* global use, db */
// MongoDB Playground
// To disable this template go to Settings | MongoDB | Use Default Template For Playground.
// Make sure you are connected to enable completions and to be able to run a playground.
// Use Ctrl+Space inside a snippet or a string literal to trigger completions.
// The result of the last command run in a playground is shown on the results panel.
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// Use 'console.log()' to print to the debug output.
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// Select the database to use.
use('airbnb');

// Get all documents from homes collection
const homes = db.getCollection('homes').find({}).toArray();

// Print the results
console.log('All homes in collection:');
console.log(JSON.stringify(homes, null, 2));

// Print total count
const totalHomes = db.getCollection('homes').countDocuments();
console.log(`Total number of homes: ${totalHomes}`);
```


```
All homes in collection:
[
  {
    "_id": "672df86c3d20696a2b523c79",
    "name": "Utsav",
    "description": "the best house",
    "price": "1999",
    "location": "ghaziabad",
    "rating": "4.5",
    "imageUrl": "/images/house1.png"
  }
]
Total number of homes: 1
```

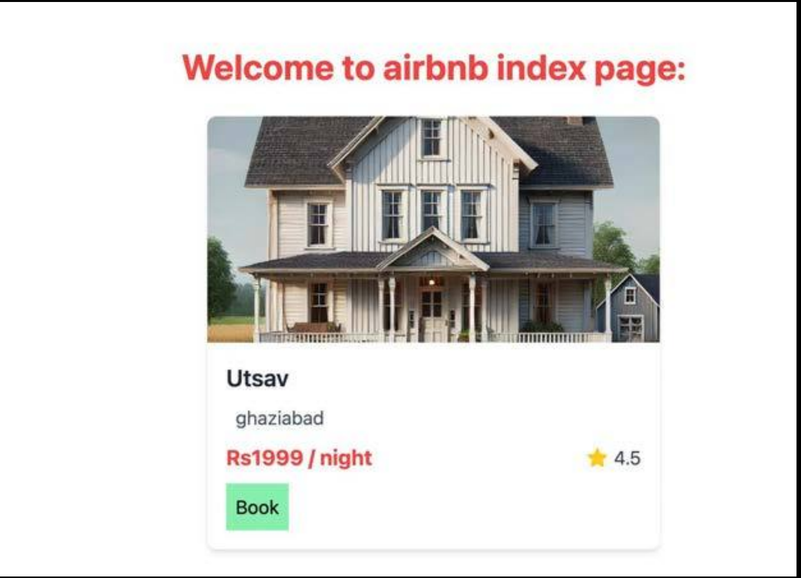
16.8 Fetching all Homes

```
static fetchAll() {
  const db = getDb();
  return db.collection("homes")
  .find()
  .toArray()
  .then((homes) => {
    console.log(homes);
    return homes;
  }).catch((err) => {
    console.log(err);
  });
}
```

1. Basic Usage: collection.find(query) retrieves documents matching the query criteria.
2. Returns a Cursor: The method returns a cursor, an iterator over the result set.

16.8 Fetching all Homes

```javascript
exports.getHomes = (req, res, next) => {
  Home.fetchAll()
    .then(rows => {
      res.render("store/home-list", {
        registeredHomes: rows,
        pageTitle: "Homes List",
        currentPage: "Home",
      });
    })
    .catch((error) => {
      console.log("Error Fetching Homes", error);
    });
};
```

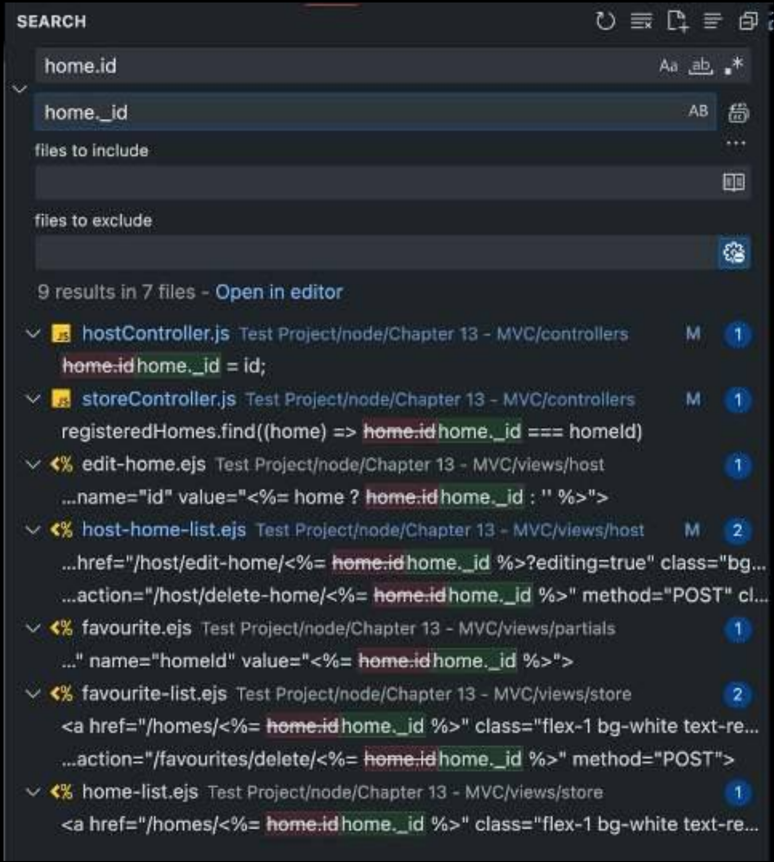


16.9 Fetching one Home

```
static findById(homeId) {
  const db = getDb();
  return db.collection("homes")
  .find({ _id: homeId })
  .next()
  .then((home) => {
    console.log(home);
    return home;
  }).catch((err) => {
    console.log(err);
  });
}
```

```
exports.getHome = (req, res, next) => {
  const homeId = req.params.homeId;
  Home.findById(homeId).then(home => {
    if (!home) {
      return res.redirect("/homes");
    }
    res.render("store/home-detail", {
      home: home,
      pageTitle: home.name,
      currentPage: "homes",
    });
  }).catch((error) => {
    console.log("Error Fetching Home", error);
  });
};
```

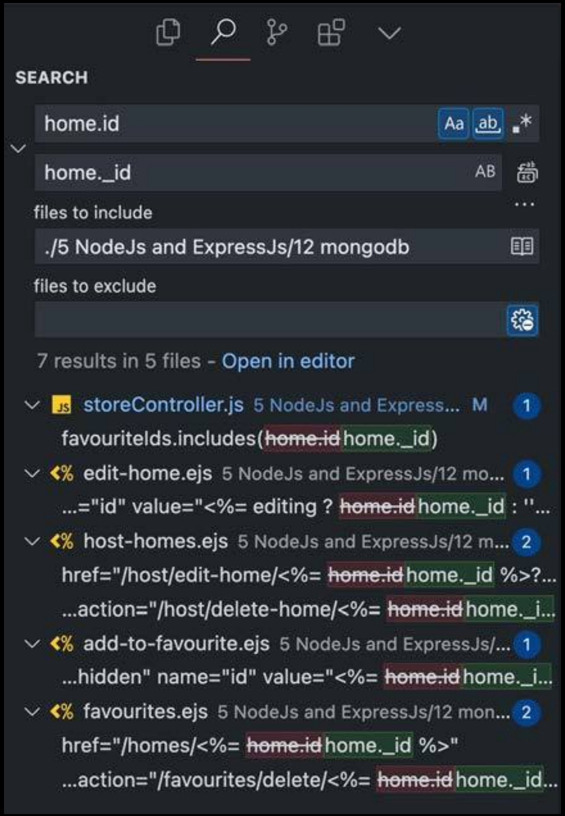
16.9 Fetching one Home



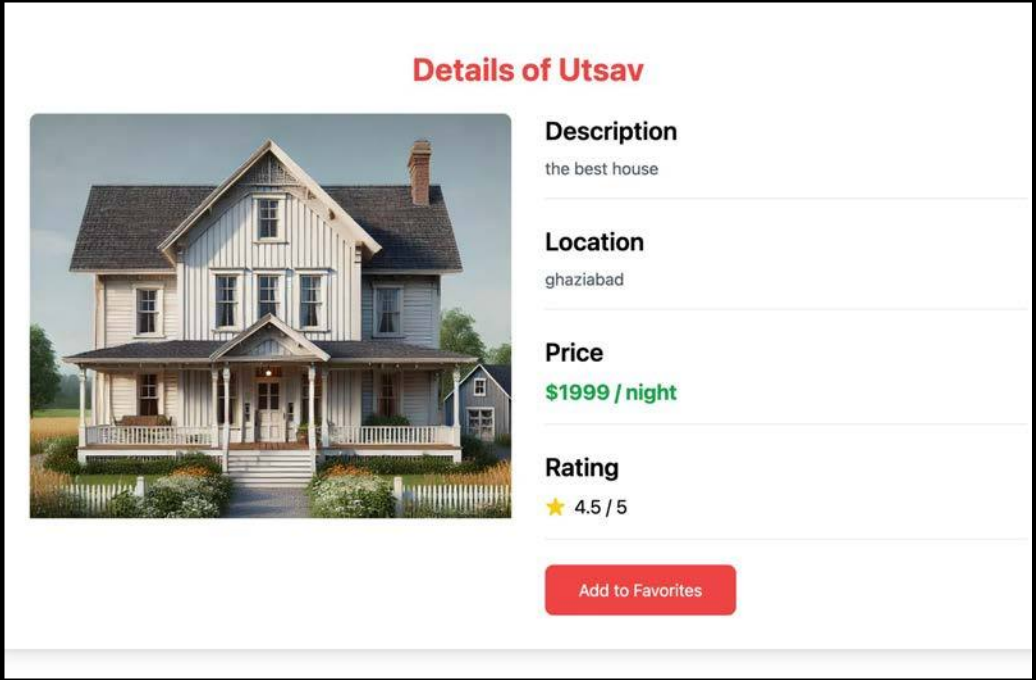

```
[
  {
    _id: new ObjectId('672df86c3d20696a2b523c79'),
    name: 'Utsav',
    description: 'the best house',
    price: '1999',
    location: 'ghaziabad',
    rating: '4.5',
    imageUrl: '/images/house1.png'
  }
]
```

16.9 Fetching one Home

```
static findById(homeId) {
  const db = getDb();
  return db.collection("homes")
  .find({ _id: new ObjectId(String(homeId)) })
  .next()
  .then((home) => {
    console.log(home);
    return home;
  }).catch((err) => {
    console.log(err);
  });
}
```



16.9 Fetching one Home



16.10 Supporting Edit & Delete

```javascript
save() {
  const db = getDb();
  if (this.id) {
    // Update existing home
    return db
      .collection("homes")
      .updateOne(
        { _id: new ObjectId(String(this.id)) },
        { $set: this}
      );
  } else {
    // Insert new home
    return db
      .collection("homes")
      .insertOne(this)
      .then((result) => {
        console.log(result);
      });
  }
}
```

```javascript
static deleteById(homeId) {
  const db = getDb();
  return db
    .collection("homes")
    .deleteOne({ _id: new ObjectId(String(homeId)) });
}
```

16.11 Adding MongoDB to Favourite

1. Remove all the file handling related code from Favourite Model.
2. Delete the data folder.
3. Change the following methods in Favourite model to use mongo:
    a. fetchAll
    b. Change addToFavourites to save method
    c. deleteById
4. Change the usages of Favourite model in StoreController to use the promise syntax
    a. getFavourites
    b. postAddFavourites
    c. postRemoveFavourite

16.11 Adding MongoDB to Favourite

1.

```
module.exports = class Favourite {
  constructor(homeId) {
    this.homeId = homeId;
  }

  save() {
  }

  static fetchAll() {
  }

  static deleteById(homeId) {
  }
}
```

2.

- > controllers
- > models
- > node_modules
- > public
- > routers
- > util
- > views
- app.js
- nodemon.json
- package-lock.json
- package.json
- tailwind.config.js

16.11 Adding MongoDB to Favourite

3.
```
save() {
  const db = getDb();
  return db.collection("favourites").insertOne(this);
}

static fetchAll() {
  const db = getDb();
  return db.collection('favourites').find().toArray();
}

static deleteById(homeId) {
  const db = getDb();
  return db.collection('favourites').deleteOne({homeId});
}
```

16.11 Adding MongoDB to Favourite

4.a.

```
exports.getFavourites = (req, res, next) => {
  Favourite.fetchAll().then((favouriteIds) => {
    Home.fetchAll().then((registeredHomes) => {
      console.log(favouriteIds);
      console.log(registeredHomes);
      const favouriteHomes = registeredHomes.filter((home) =>
        favouriteIds.includes(home._id.toString())
      );
      res.render("store/favourites", {
        homes: favouriteHomes,
        pageTitle: "Favourites",
      });
    });
  });
};
```

16.11 Adding MongoDB to Favourite

4.b.

```
exports.postAddFavourites = (req, res, next) => {
  const homeId = req.body.id;
  const fav = new Favourite(homeId);
  fav
    .save()
    .then(() => {
      res.redirect("/favourites");
    })
    .catch((err) => {
      console.log("Error while adding to favourites", err);
      res.redirect("/favourites");
    });
};
```

16.11 Adding MongoDB to Favourite

4.c.

```
exports.postRemoveFavourite = (req, res, next) => {
  const homeId = req.params.homeId;
  Favourite.deleteById(homeId)
    .then(() => {
      res.redirect("/favourites");
    })
    .catch((err) => {
      console.log("Error while remove from favourites ", err);
    });
};
```

Practise Milestone

Take your airbnb forward:

- Change the favourite model to fix the problem of having duplicate favourite records.

Practise Milestone (Solution)

```
save() {
  const db = getDb();
  return db.collection('favourites').findOne({homeId: this.homeId})
  .then(existingFav => {
    if (!existingFav) {
      return db.collection("favourites").insertOne(this);
    }
    return Promise.resolve();
  });
}
```

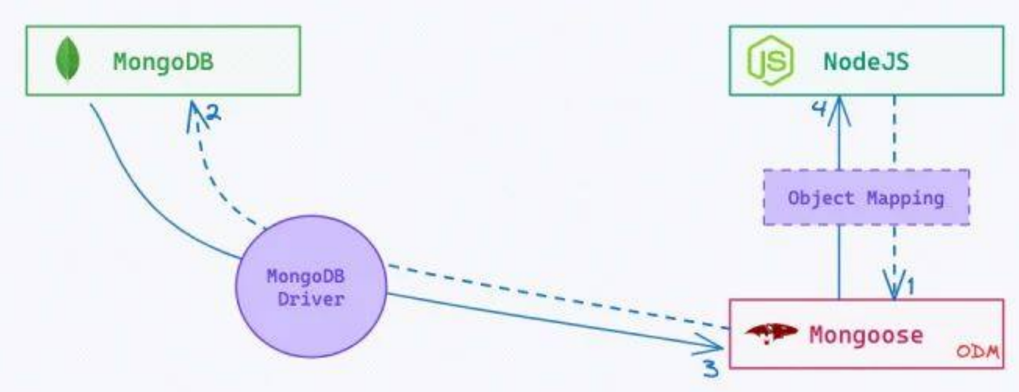
Revision

1. What is MongoDB
2. Setting up MongoDB
3. Installing MongoDB Driver
4. Creating MongoDB Connection
5. Saving a Home
6. Install MongoDB Compass
7. Install MongoDB for VSCode
8. Fetching all Homes
9. Fetching one Home
10. Supporting Edit & Delete
11. Adding MongoDB to Favourite

17. Introduction to Mongoose

1. What is Mongoose
2. Setting up Mongoose
3. Create Home Schema
4. Saving Home using Mongoose
5. Fetching Homes
6. Fetching one Home
7. Editing a Home
8. Deleting a Home
9. Using Mongoose for Favourite
10. Fetching Relations

17.1 What is Mongoose



1. Mongoose is an Object Data Modeling (ODM) library for MongoDB and Node.js.
2. Provides a schema-based solution to model application data.
3. Simplifies data validation and type casting in Node.js applications.
4. Enables easy interaction with MongoDB through intuitive methods.
5. Supports middleware for pre and post-processing of data.
6. Helps manage relationships between data with built-in functions.

17.2 Setting up Mongoose

mongoose

elegant mongodb object modeling for node.js

read the docs | discover plugins

Star | Version 8.8.0 | Fork

Let's face it, **writing MongoDB validation, casting and business logic boilerplate is a drag**. That's why we wrote Mongoose.

```
const mongoose = require('mongoose');
mongoose.connect('mongodb://127.0.0.1:27017/test');

const Cat = mongoose.model('Cat', { name: String });

const kitty = new Cat({ name: 'Zildjian' });
kitty.save().then(() => console.log('meow'));
```

Mongoose provides a straight-forward, schema-based solution to model your application data. It includes built-in type casting, validation, query building, business logic hooks and more, out of the box.

Get Professionally Supported Mongoose

17.2 Setting up Mongoose

1. Install Mongoose package.
2. Import and use mongoose in app.js
3. Delete the database-util file.
4. Remove the usage of db-util from everywhere.

1.

```
prashantjain@Prashants-Mac-mini 12 mongodb % npm install mongoose

added 8 packages, and audited 258 packages in 1s

48 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
```

17.2 Setting up Mongoose

2.

```
const PORT = 3001;
mongoose.connect("mongodb+srv://root:root@kgcluster.ie6mb.mongodb.net/airbnb?
retryWrites=true&w=majority&appName=KGCluster").then(() => {
  console.log("Connected to MongoDB");
  app.listen(PORT, () => {
    console.log(`Server running at: http://localhost:${PORT}`);
  });
}).catch((err) => {
  console.log("Error while connecting to MongoDB", err);
});
```

17.3 Create Home Schema

1. Delete complete airbnb db from mongo.
2. Delete the existing Home Model code.
3. Create the new Home Schema in the Home Model File.

```
const mongoose = require("mongoose");

// _id is automatically added by mongoose
const homeSchema = new mongoose.Schema({
  houseName: {type: String, required: true},
  price: {type: Number, required: true},
  location: {type: String, required: true},
  rating: {type: Number, required: true},
  photoUrl: String,
  description: String
});
```

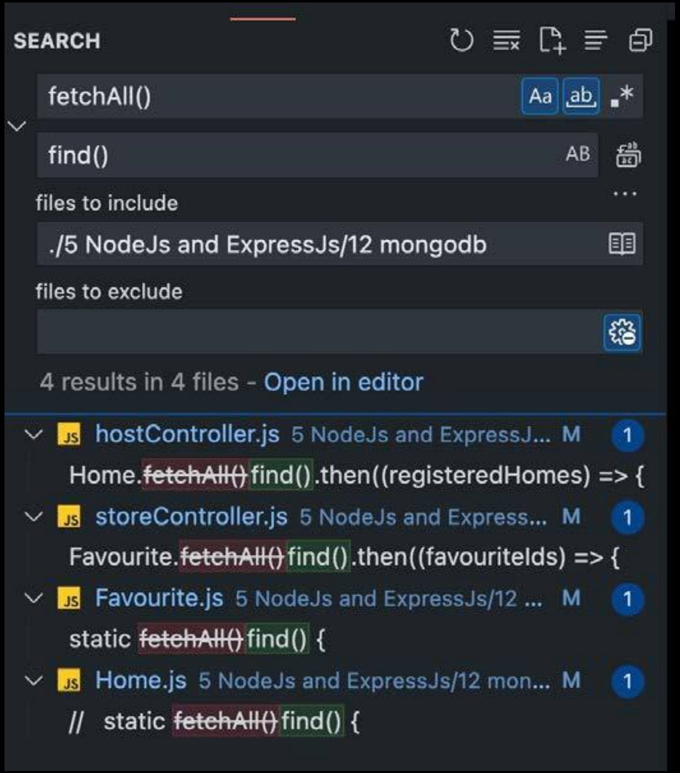
17.4 Saving Home using Mongoose

```
module.exports = mongoose.model("Home", homeSchema);
```

```
exports.postAddHome = (req, res, next) => {
  const { houseName, price, location, rating, photoUrl, description } =
    req.body;
  const newHome = new Home({
    houseName,
    price,
    location,
    rating,
    photoUrl,
    description});

  newHome.save().then((rows) => {
    res.render("host/home-added", { pageTitle: "Home Hosted" });
  });
};
```

17.5 Fetching Homes




```
exports.getIndex = (req, res, next) => {
  Home.find().then((registeredHomes) => {
    res.render("store/index", {
      homes: registeredHomes,
      pageTitle: "Tumahara airbnb",
    });
  });
};
```

17.6 Fetching one Home

It already works !!!

17.7 Editing a Home

```
exports.postEditHome = (req, res, next) => {
  const { id, houseName, price, location, rating, photoUrl, description } = req.body;
  Home.findById(id).then((home) => {
    if (!home) {
      console.log("Home not found for editing");
      return res.redirect("/host/host-homes");
    }
    home.houseName = houseName;
    home.price = price;
    home.location = location;
    home.rating = rating;
    home.photoUrl = photoUrl;
    home.description = description;
    return home.save();
  }).then(() => {
    res.redirect("/host/host-homes");
  })
  .catch((err) => {
    console.log("Error while updating home", err);
  });
};
```

17.8 Deleting a Home

```
exports.postDeleteHome = (req, res, next) => {
  const homeId = req.params.homeId;
  console.log("Came to delete ", homeId);
  Home.findByIdAndDelete(homeId).then(() => {
    res.redirect("/host/host-homes");
  });
};
```

17.9 Using Mongoose for Favourite

1. Delete the existing Favourite Model code.
2. Create the new Favourite Schema in the Favourite Model File.
3. Fix the following functionalities:
    a. Getting all Favourite
    b. Adding A Favourite
    c. Deleting a Favourite
4. Removing Favourite while removing home.

17.9 Using Mongoose for Favourite

1, 2.

```
const mongoose = require('mongoose');

const favouriteSchema = new mongoose.Schema({
  homeId: {
    type: mongoose.Schema.Types.ObjectId,
    ref: 'Home',
    required: true,
    unique: true
  }
});

module.exports = mongoose.model('Favourite', favouriteSchema);
```

17.9 Using Mongoose for Favourite

3.a.

```
exports.getFavourites = (req, res, next) => {
  Favourite.find().then((favouriteIds) => {
    favouriteIds = favouriteIds.map((favourite) => favourite.homeId.toString());
    Home.find().then((registeredHomes) => {
      console.log(favouriteIds);
      console.log(registeredHomes);
      const favouriteHomes = registeredHomes.filter((home) =>
        favouriteIds.includes(home._id.toString())
      );
      res.render("store/favourites", {
        homes: favouriteHomes,
        pageTitle: "Favourites",
      });
    });
  });
};
```

17.9 Using Mongoose for Favourite

3.b.

```
exports.postAddFavourites = (req, res, next) => {
  const homeId = req.body.id;
  Favourite.findOne({ homeId: homeId })
    .then(existingFav => {
      if (existingFav) {
        return res.redirect("/favourites");
      }
      const fav = new Favourite({ homeId: homeId });
      return fav.save();
    })
    .then(() => {
      res.redirect("/favourites");
    })
    .catch((err) => {
      console.log("Error while adding to favourites", err);
    });
};
```

17.9 Using Mongoose for Favourite

3.c.

```
exports.postRemoveFavourite = (req, res, next) => {
  const homeId = req.params.homeId;
  Favourite.findOneAndDelete({ homeId: homeId })
    .then(() => {
      res.redirect("/favourites");
    })
    .catch((err) => {
      console.log("Error while remove from favourites ", err);
    });
};
```

17.9 Using Mongoose for Favourite

4.

```
homeSchema.pre('findOneAndDelete', async function(next) {
  const homeId = this.getQuery()["_id"];
  await Favourite.deleteMany({ homeId: homeId });
  next();
});
```

17.10 Fetching Relations

```
exports.getFavourites = (req, res, next) => {
  Favourite.find()
    .populate("homeId")
    .then((favourites) => {
      const favouriteHomes = favourites.map((favourite) => favourite.homeId);
      res.render("store/favourites", {
        homes: favouriteHomes,
        pageTitle: "Favourites",
      });
    });
};
```

Revision

1. What is Mongoose
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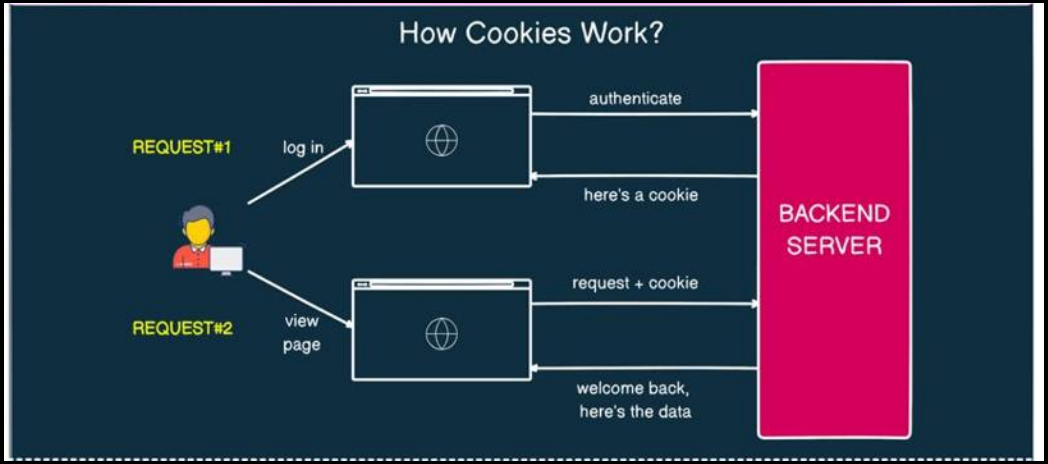
5. Fetching Homes
6. Fetching one Home
7. Editing a Home
8. Deleting a Home
9. Using Mongoose for Favourite
10. Fetching Relations

18. Cookies & Sessions

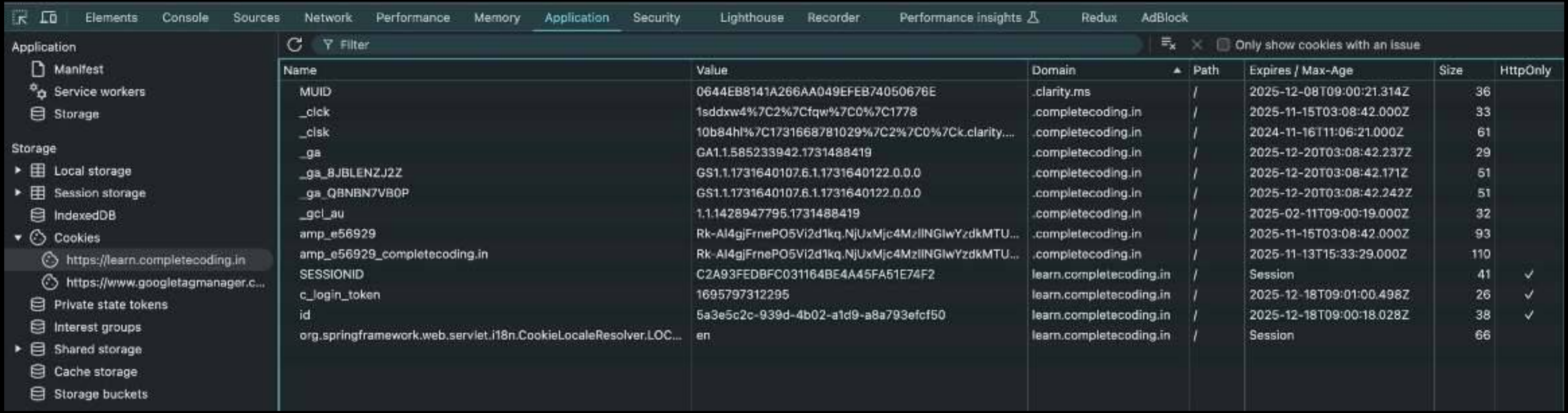
1. What are Cookies
2. Adding Login Functionality
3. Checking Login State
4. Using a Cookie
5. Define the Logout Feature
6. Problem with Cookies
7. What are Sessions
8. Installing Session Package
9. Creating a Sessions
10. Saving Session in DB

18.1 What are Cookies



1. Cookies are small pieces of data stored in the user's browser by server.
2. They help websites remember user information and preferences between page loads or visits.
3. Cookies can manage user sessions and store data for personalized experiences.

18.1 What are Cookies



Name	Value	Domain	Path	Expires / Max-Age	Size	HttpOnly
MUID	0644EB8141A266AA049EFEB74050676E	.clarity.ms	/	2025-12-08T09:00:21.314Z	36	
_clck	1sddxw4%7C2%7Cfqw%7C0%7C1778	.completecoding.in	/	2025-11-15T03:08:42.000Z	33	
_clsk	10b84hl%7C1731668781029%7C2%7C0%7Ck.clarity....	.completecoding.in	/	2024-11-16T11:06:21.000Z	61	
_ga	GA1.1.585233942.1731488419	.completecoding.in	/	2025-12-20T03:08:42.237Z	29	
_ga_8JBLENZJ2Z	GS1.1.1731640107.6.1.1731640122.0.0.0	.completecoding.in	/	2025-12-20T03:08:42.171Z	51	
_ga_QBNBN7VB0P	GS1.1.1731640107.6.1.1731640122.0.0.0	.completecoding.in	/	2025-12-20T03:08:42.242Z	51	
_gcl_au	1.1.1428947795.1731488419	.completecoding.in	/	2025-02-11T09:00:19.000Z	32	
amp_e56929	Rk-Al4gjFrnePO5Vi2d1kq.NjUxMjc4MzllNGIwYzdkMTU...	.completecoding.in	/	2025-11-15T03:08:42.000Z	93	
amp_e56929_completecoding.in	Rk-Al4gjFrnePO5Vi2d1kq.NjUxMjc4MzllNGIwYzdkMTU...	.completecoding.in	/	2025-11-13T15:33:29.000Z	110	
SESSIONID	C2A93FEDBFC031164BE4A45FA51E74F2	learn.completecoding.in	/	Session	41	✓
c_login_token	1695797312295	learn.completecoding.in	/	2025-12-18T09:01:00.498Z	26	✓
id	5a3e5c2c-939d-4b02-a1d9-a8a793efcf50	learn.completecoding.in	/	2025-12-18T09:00:18.028Z	38	✓
org.springframework.web.servlet.i18n.CookieLocaleResolver.LOC...	en	learn.completecoding.in	/	Session	66	

18.2 Adding Login Functionality

1. Add a login button to the nav bar pointing to /login
2. Create a auth router to handle login related routes, register the new router in app.js
3. Create a auth controller to handle GET request for /login and return a UI that does the following:
    a. Accepts email and password
    b. Has a Login button that submits the form to /login with a POST request.
4. Add a POST login path in router and handler in controller.
5. Assume the person logged in and redirect them to the home page.

18.2 Adding Login Functionality

1.

```html
<div>
  <a href="/login" class="bg-blue-600 hover:bg-blue-700 text-white font-semibold py-2.5
   px-6 rounded-lg transition duration-300 ease-in-out transform hover:scale-105 shadow-md">
    Login
  </a>
</div>
```

18.2 Adding Login Functionality

2.

```
const express = require("express");
const authRouter = express.Router();
const authController = require("../controllers/authController");

authRouter.get("/login", authController.getLogin);

exports.authRouter = authRouter;
```

```
app.use("/host", hostRouter);
app.use(authRouter);
```

18.2 Adding Login Functionality

3.

```
exports.getLogin = (req, res, next) => {
  res.render("auth/login", { pageTitle: "Login" });
};
```

18.2 Adding Login Functionality

3.

```
<%- include('../partials/head') %>
  [illegible]
  [illegible]
   <%- include('../partials/nav') %>
      [illegible] Login Here [illegible]
      [illegible]
```

18.2 Adding Login Functionality

4, 5.

```
authRouter.get("/login", authController.getLogin);
authRouter.post("/login", authController.postLogin);
```

```
exports.postLogin = (req, res, next) => {
  console.log(req.body);
  res.redirect("/");
};
```

18.3 Checking Login State

1. Add a isLoggedIn field in the req object and use it everywhere else.
2. Add a condition in the navigation ejs file that no path other than home and login should be visible until a user has logged in.
3. Fix all the render calls to send the flag
4. Also add a middleware for host routes that if the user is not logged in they should be redirected to the login page.

18.3 Checking Login State

1.

```
exports.postLogin = (req, res, next) => {
  console.log(req.body);
  req.isLoggedIn = true;
  res.redirect("/");
};
```

18.3 Checking Login State

2.

```
<% if (isLoggedIn) { %>
  <a href="/favourites" class="bg-blue-600 hover:bg-blue-700 text-white
    Favourites
  </a>
  <a href="/host/host-homes" class="bg-blue-600 hover:bg-blue-700 text-w
    Host Homes
  </a>
  <a href="/host/add-home" class="bg-blue-600 hover:bg-blue-700 text-whi
    Add Home
  </a>
<% } %>
```

18.3 Checking Login State

3.

```
res.render("host/edit-home", {
  home: home,
  editing: editing,
  pageTitle: "Edit Your Home",
  isLoggedIn: req.isLoggedIn,
});
```

18.3 Checking Login State

4.

```
app.use(storeRouter);
app.use("/host", (req, res, next) => {
  if (!req.isLoggedIn) {
    return res.redirect("/login");
  }
  next();
});
app.use("/host", hostRouter);
```

18.4 Using a Cookie

1. Understand why a global variable would not work.
2. Set a cookie on successful login. See it in storage, also on the next request.
3. Read the cookie using syntax and Define a middleware to set this value to the request object.

```
console.log(req.get('Cookie').split('=')[1]);
```

4. Change the Cookie from the browser and see the result.

18.4 Using a Cookie

2.

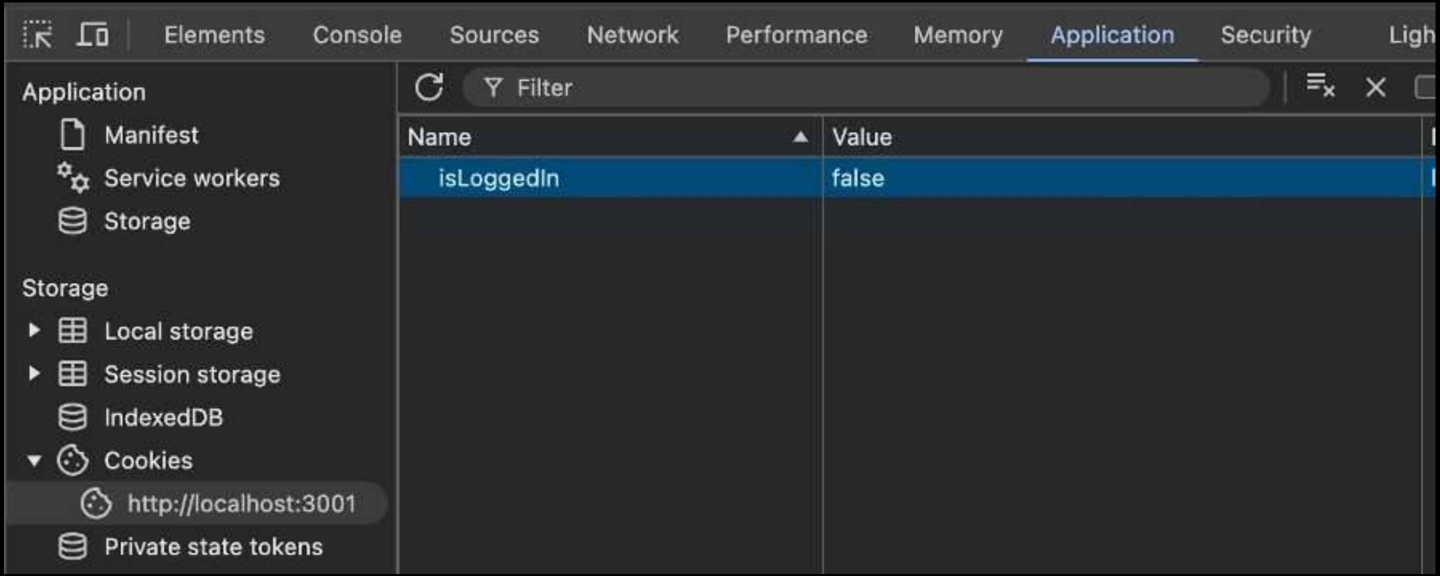
```
exports.postLogin = (req, res, next) => {
  console.log(req.body);
  res.cookie("isLoggedIn", true);
  res.redirect("/");
};
```

3.

```
app.use((req, res, next) => {
  req.isLoggedIn = req.get('Cookie')?.split('=')[1] || false;
  console.log(req.isLoggedIn);
  next();
});
```

18.4 Using a Cookie

4.



18.5 Define the Logout Feature

1. Define a logout button in nav bar that should come only when user is logged in. Button should be a form that submits to link /logout.
2. Hide the login button in case user is logged in.
3. Handle the logout path and set the isLoggedIn cookie to false.

18.5 Define the Logout Feature

1, 2.

```
<div class="flex space-x-4">
  <% if (!isLoggedIn) { %>
    <a href="/login" class="bg-blue-600 hover:bg-blue-700 tex
      Login
    </a>
  <% } else { %>
    <form action="/logout" method="POST">
      <button type="submit" class="bg-red-600 hover:bg-red-70
        Logout
      </button>
    </form>
  <% } %>
</div>
```

18.5 Define the Logout Feature

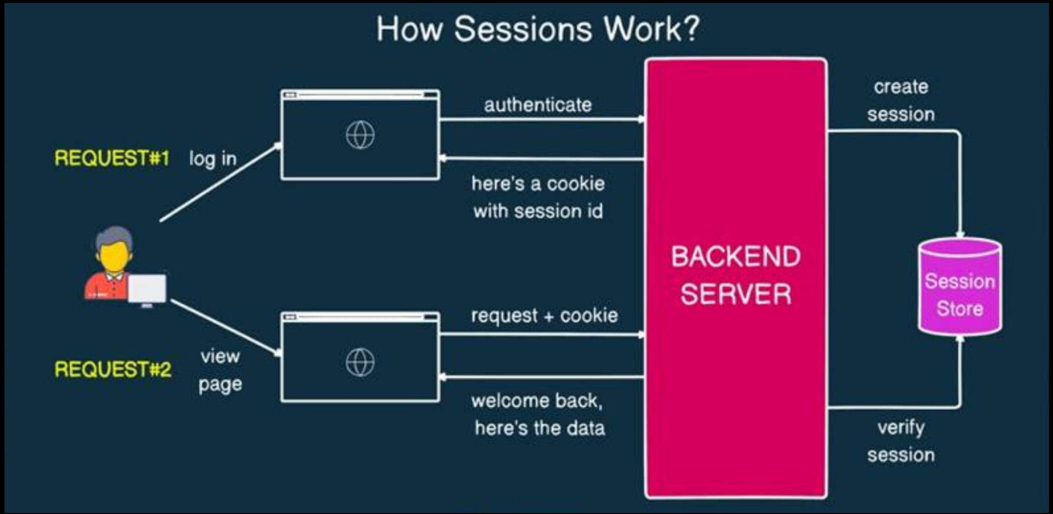
3.

```
exports.postLogout = (req, res, next) => {
  res.cookie("isLoggedIn", false);
  res.redirect("/login");
};
```

18.6 Problem with Cookies

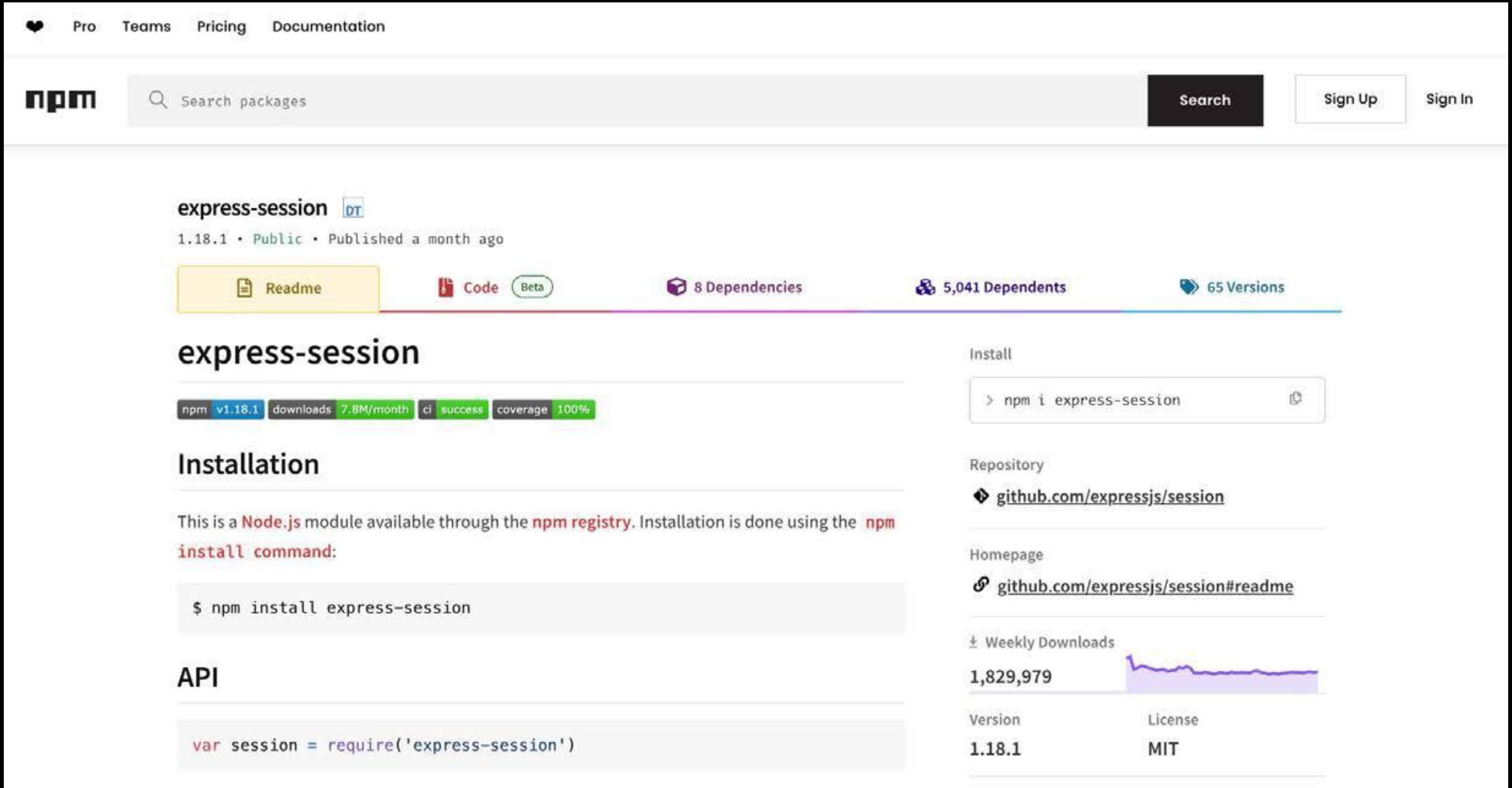
1. Cookies can be intercepted or stolen, posing security risks.
2. They have limited storage capacity (about 4KB).
3. Users can delete or modify cookies, leading to data loss or tampering.
4. Data in cookies is not encrypted, making sensitive information vulnerable.
5. Storing important info in cookies exposes it to client-side attacks.

18.7 What are Sessions



1. Sessions are server-side storage mechanisms that track user interactions with a website.
2. They maintain user state and data across multiple requests in a web application.
3. Sessions enable persistent user experiences by maintaining state between the client and server over stateless HTTP.

18.8 Installing Session Package



18.8 Installing Session Package

```
prashantjain@Prashants-MacBook-Pro 13 mongoose % npm install express-session

added 6 packages, and audited 264 packages in 2s

48 packages are looking for funding
  run `npm fund` for details

found   vulnerabilities
```

```
const session = require('express-session');
```

```
app.use(bodyParser.urlencoded({ extended: true }));
app.use(session({
  // Secret key used to sign the session ID cookie and encrypt session data
  secret: 'Complete Coding Secret',
  // Forces session to be saved back to the session store, even if not modified
  resave: false,
  // Forces a session that is "uninitialized" to be saved to the store
  saveUninitialized: true,
}));
```

18.9 Creating a Session

1. Sensitive info is stored on server.
2. Same session is valid for all requests from one user, using cookies.
3. Remove setting the cookie and now save the flag in session.
4. Check the browser for cookie changes.
5. Log Session in some get request.
6. Try to use a different browser and show that session is different.
7. Sessions are stored in memory so they reset when server restarts.

18.9 Creating a Sessions

3.

```
exports.postLogin = (req, res, next) => {
  console.log(req.body);
  req.session.isLoggedIn = true;
  res.redirect("/");
};
```

4.

18.9 Creating a Sessions

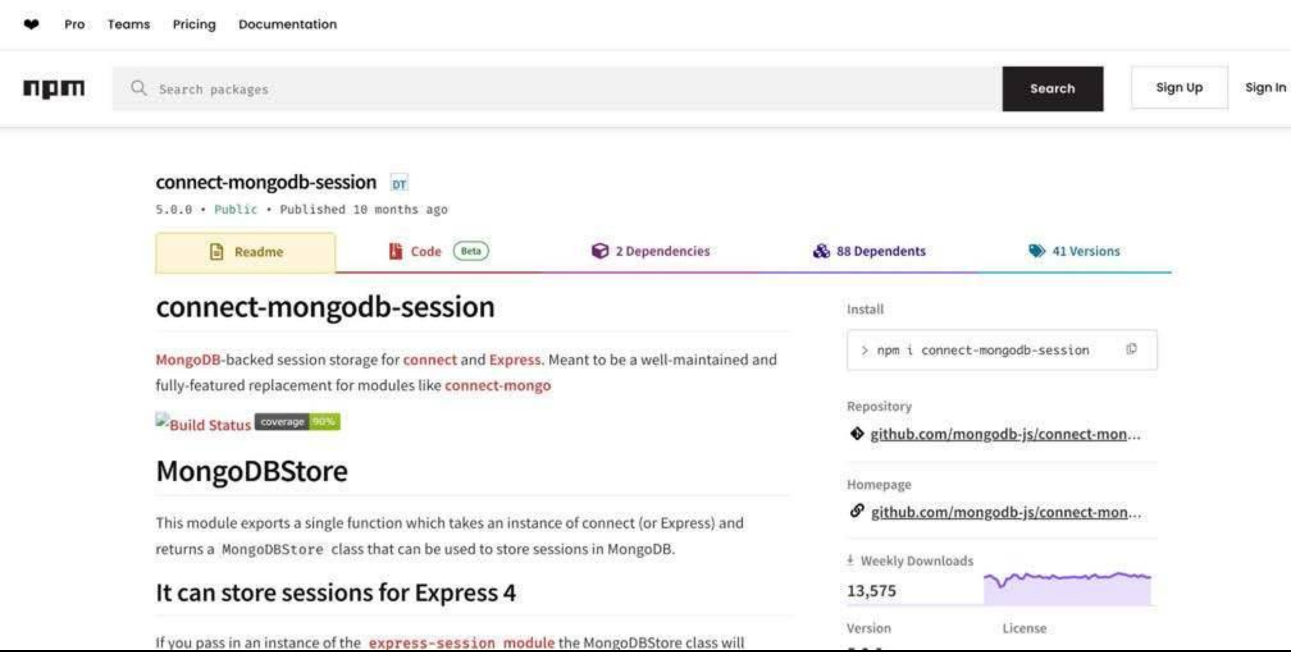
5.

```
exports.getIndex = (req, res, next) => {
  console.log(req.session, req.session.isLoggedIn);
  Home.find().then((registeredHomes) => {
```

```
Session {
  cookie: { path: '/', _expires: null, originalMaxAge: null, httpOnly: true },
  isLoggedIn: true
} true
```

18.10 Saving Session in DB

★ 176 connect-mongodb-session Lightweight MongoDB-based session store built and maintained by MongoDB.



18.10 Saving Session in DB

```
prashantjain@Prashants-MacBook-Pro 13 mongoose % npm install connect-mongodb-session

added 3 packages, and audited 267 packages in 2s

48 packages are looking for funding
  run `npm fund` for details

found   vulnerabilities
```

```
const MongoDBStore = require('connect-mongodb-session')(session);

const MONGO_DB_URL = "mongodb+srv://root:root@kgcluster.ie6mb.mongodb.ne

const store = new MongoDBStore({
  uri: MONGO_DB_URL,
  collection: 'sessions'
});
```

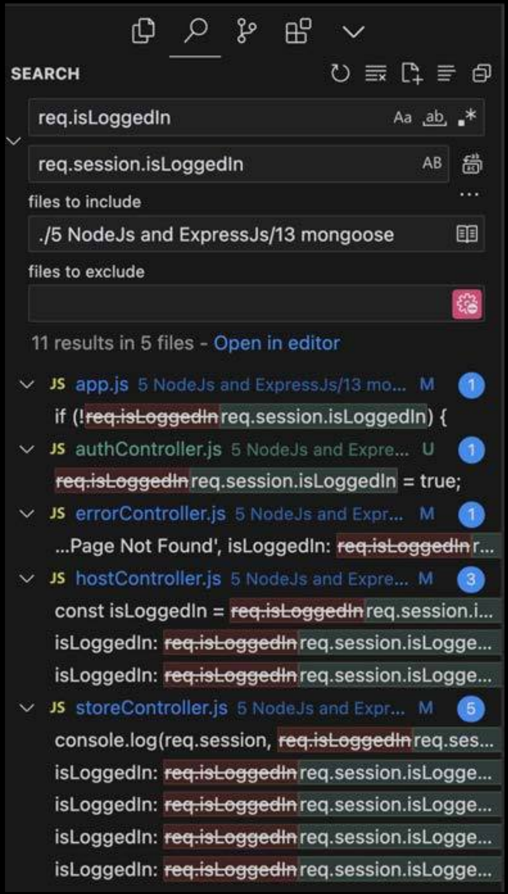
```
app.use(session({
  // Secret key used to sign the session ID cookie and encrypt session data
  secret: 'Complete Coding Secret',
  // Forces session to be saved back to the session store, even if not modified
  resave: false,
  // Forces a session that is "uninitialized" to be saved to the store
  saveUninitialized: true,
  store: store,
}));
```

Practise Milestone

Take your airbnb forward:

- Cleanup cookie code to use session everywhere.
- Remove the cookie middleware.
- Destroy the session on logout.
- Cleanup Logs.
- Understand why leaving the cookie on logout is fine.

Practise Milestone (Solution)




```javascript
exports.postLogout = (req, res, next) => {
  req.session.destroy(() => {
    res.redirect("/login");
  });
};
```

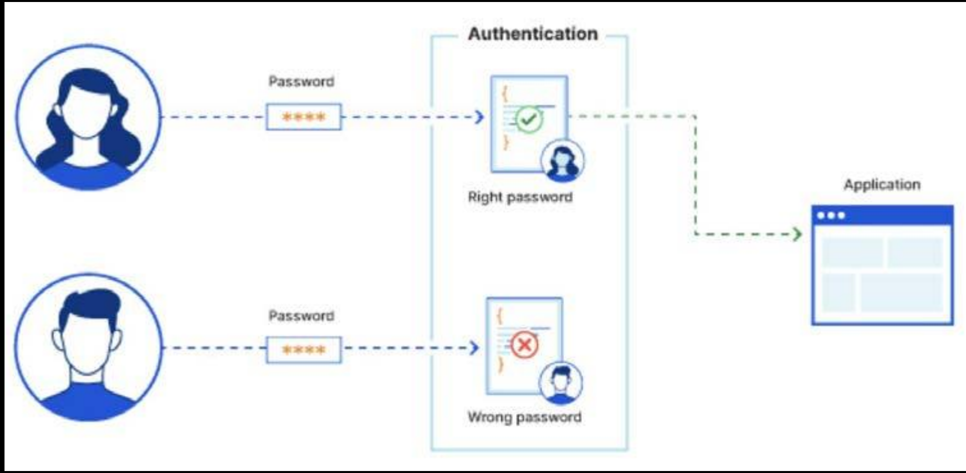
Revision

1. What are Cookies
2. Adding Login Functionality
3. Checking Login State
4. Using a Cookie
5. Define the Logout Feature
6. Problem with Cookies
7. What are Sessions
8. Installing Session Package
9. Creating a Sessions
10. Saving Session in DB

19. Authentication & Authorization

1. What is Authentication
2. What is Authorization
3. Session based Authentication
4. Authentication vs Authorization
5. Signup UI
6. Using Express Validator
7. Adding User Model
8. Encrypting Password
9. Implementing Login
10. Adding User Functions

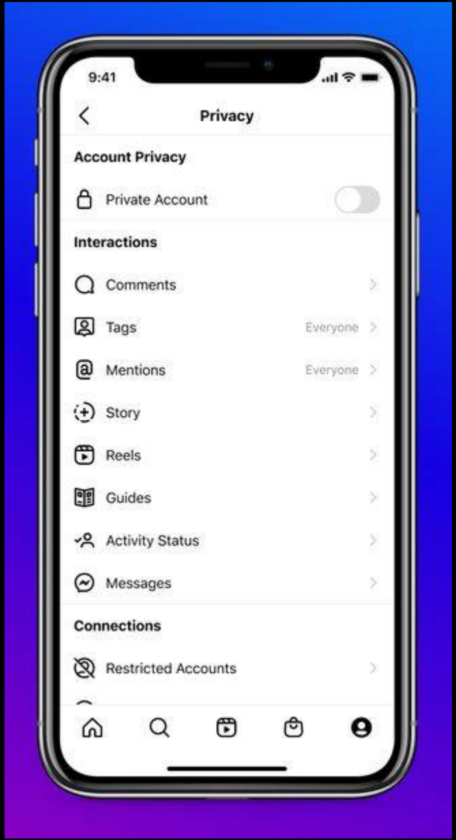
19.1 What is Authentication



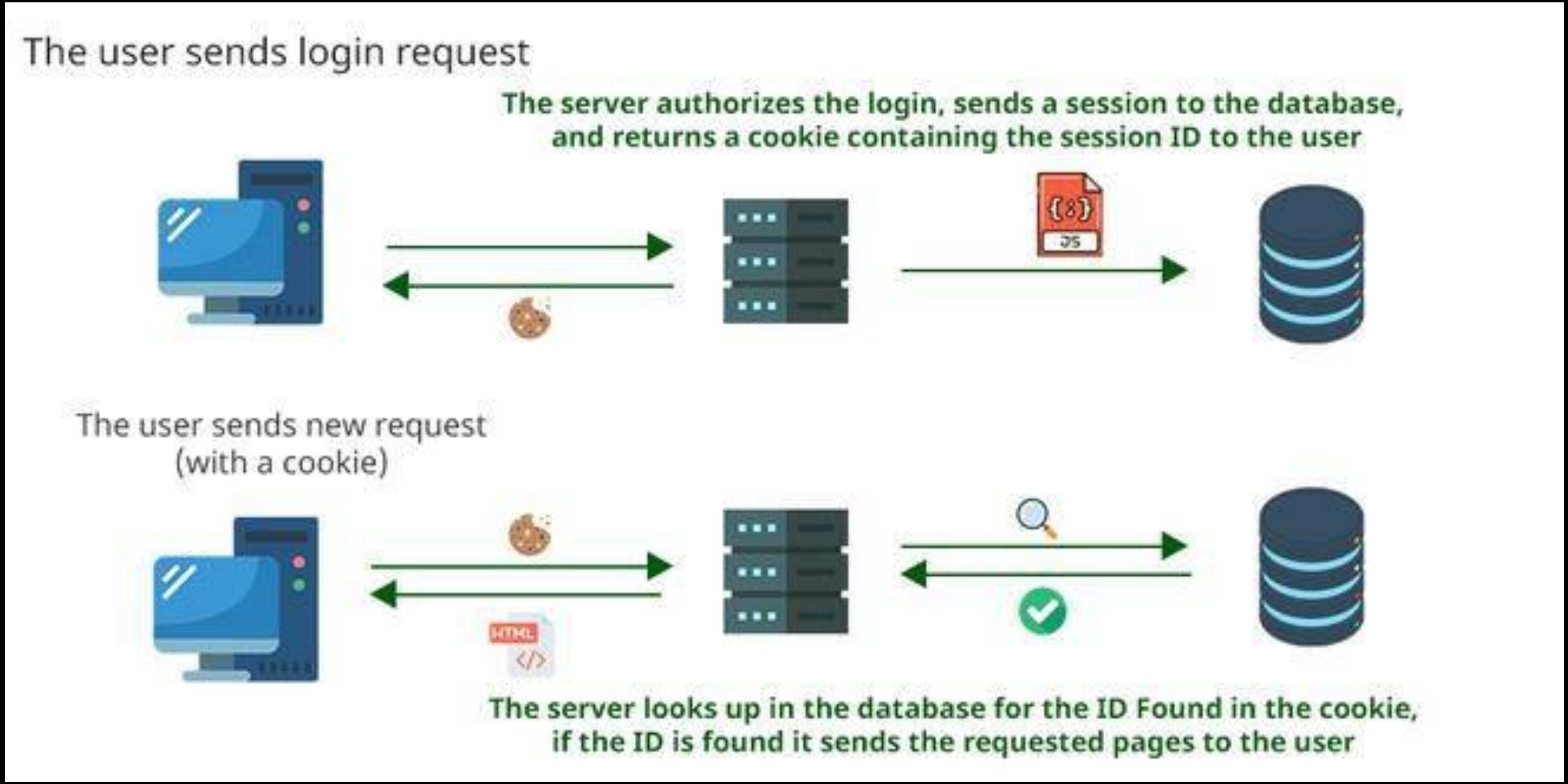
1. Authentication is the process of verifying the identity of a user or system accessing an application.
2. It ensures that only authorized users can access protected resources and features.
3. Authentication is crucial for security, protecting data, and providing personalized experiences in web applications.

19.2 What is Authorization

1. Authorization is the process of determining what actions a user is permitted to perform within an application.
2. It ensures that users can access only the resources and functionalities they have permission for.
3. Authorization enhances security by restricting access to sensitive data and operations, complementing the authentication process.



19.3 Session based Authentication



19.4 Authentication vs Authorization

19.4 Authentication vs Authorization

Aspect	Authentication	Authorization
Definition	Verifies the identity of a user or system	Determines what resources a user can access
Purpose	Ensures users are who they claim to be	Grants or denies permissions to resources and actions
Process	Validates credentials like usernames and passwords	Checks user privileges and access levels
Occurs When	At the start of a session or when accessing secured areas	After authentication, during resource access
User Interaction	Requires user input (e.g., logging in)	Usually transparent unless access is denied
Managed By	Handled by both frontend and backend systems	Mainly enforced by backend servers
Example	User logs into an account with a password	User accesses settings page only if they have admin rights

19.5 Signup UI

1. Define a signup button in navigation bar along with sign-in. It should point to a link /signup
2. Define a auth/signup.ejs file that has the following fields and submits POST request to /signup:
   a. firstName, lastName,
   b. email
   c. password, confirmPassword
   d. userType with possible values ‘guest’, ‘host’
   e. Terms and conditions checkbox.
3. Define routes in authRouter and behaviour in authController.
4. Fix the UI of the app to look pretty.

19.5 Signup UI

1.

```
<% if (!isLoggedIn) { %>
  <a href="/signup" class="bg-green-600 hover:bg-green-700 text-white font-semibold py-2.5 px-6
  rounded-lg transition duration-300 ease-in-out transform hover:scale-105 shadow-md">
    Sign Up
  </a>
  <a href="/login" class="bg-blue-600 hover:bg-blue-700 text-white font-semibold py-2.5 px-6
  rounded-lg transition duration-300 ease-in-out transform hover:scale-105 shadow-md">
    Login
  </a>
<% } else { %>
```

19.5 Signup UI

2.

```html
<form action="/signup" method="POST" class="w-full max-w-md">
  <input type="text" name="firstName" placeholder="First Name" class="w-full px-4 py-2 mb-4 border
  border-gray-300 rounded-lg focus:outline-none focus:border-blue-500"/>
  <input type="text" name="lastName" placeholder="Last Name" class="w-full px-4 py-2 mb-4 border
  border-gray-300 rounded-lg focus:outline-none focus:border-blue-500"/>
  <input type="email" name="email" placeholder="Enter your Email" class="w-full px-4 py-2 mb-4 border
  border-gray-300 rounded-lg focus:outline-none focus:border-blue-500"/>
  <input type="password" name="password" placeholder="Password" class="w-full px-4 py-2 mb-4 border
  border-gray-300 rounded-lg focus:outline-none focus:border-blue-500"/>
  <input type="password" name="confirm_password" placeholder="Confirm Password" class="w-full px-4 py-2
  mb-4 border border-gray-300 rounded-lg focus:outline-none focus:border-blue-500"/>

  <div class="mb-4">
    <p class="text-gray-700 mb-2">User Type:</p>
    <div class="flex space-x-4">
      <label class="flex items-center"><input type="radio" name="userType" value="guest" class="mr-2"/
      ><span class="text-gray-700">Guest</span></label>
      <label class="flex items-center"><input type="radio" name="userType" value="host" class="mr-2"/
      ><span class="text-gray-700">Host</span></label>
    </div>
  </div>

  <div class="mb-4">
    <label class="flex items-center"><input type="checkbox" name="termsAccepted" class="mr-2"/><span
    class="text-gray-700">I accept the terms and conditions</span></label>
  </div>

  <div class="flex justify-center">
    <input type="submit" value="Sign me up" class="bg-green-500 hover:bg-green-600 text-white
    font-semibold py-2 px-4 rounded-lg transition duration-300 ease-in-out transform hover:scale-105
    cursor-pointer">
  </div>
</form>
```

19.5 Signup UI

3, 4.

```
authRouter.post("/logout", authController.postLogout);
authRouter.get("/signup", authController.getSignup);
```

```
exports.getSignup = (req, res, next) => {
  res.render("auth/signup", { pageTitle: "Sign Up", isLoggedIn: false });
};
```

19.6 Using Express Validator

1. Add handling for POST /signup in auth controller and router.
2. Install Express Validator.
3. Use the email and password validations in the post handler.
4. Change the signup.ejs to show the errors. And accept the old values.

Version: 7.2.0

express-validator

Overview

express-validator is a set of express.js middlewares that wraps the extensive collection of validators and sanitizers offered by validator.js.

It allows you to combine them in many ways so that you can validate and sanitize your express requests, and offers tools to determine if the request is valid or not, which data was matched according to your validators, and so on.

19.6 Using Express Validator

1.

```
authRouter.get("/signup", authController.getSignup);
authRouter.post("/signup", authController.postSignup);
```

```
exports.postSignup = (req, res, next) => {
  console.log(req.body);
  res.redirect("/login");
};
```

2.

```
prashantjain@Prashants-MacBook-Pro 13 mongoose % npm install express-validator

added 2 packages, and audited 269 packages in 553ms

48 packages are looking for funding
  run `npm fund` for details

found   vulnerabilities
```

19.6 Using Express Validator

3.

```
exports.postSignup = [
  // First Name validation
  check('firstName')
    .notEmpty()
    .withMessage('First name is required')
    .trim()
    .isLength({ min: 2 })
    .withMessage('First name must be at least 2 characters long')
    .matches(/^[a-zA-Z\s]+$/)
    .withMessage('First name can only contain letters'),

  // Last Name validation
  check('lastName')
    .notEmpty()
    .withMessage('Last name is required')
    .trim()
    .isLength({ min: 2 })
    .withMessage('Last name must be at least 2 characters long')
    .matches(/^[a-zA-Z\s]+$/)
    .withMessage('Last name can only contain letters'),

  // Email validation
  check('email')
    .isEmail()
    .withMessage('Please enter a valid email')
    .normalizeEmail(),
```

```
  // Password validation
  check('password')
    .isLength({ min: 8 })
    .withMessage('Password must be at least 8 characters long')
    .matches(/[a-z]/)
    .withMessage('Password must contain at least one lowercase letter')
    .matches(/[A-Z]/)
    .withMessage('Password must contain at least one uppercase letter')
    .matches(/[!@#$%^&*(),.?":{}|<>]/)
    .withMessage('Password must contain at least one special character')
    .trim(),

  // Confirm password validation
  check('confirm_password')
    .trim()
    .custom((value, { req }) => {
      if (value !== req.body.password) {
        throw new Error('Passwords do not match');
      }
      return true;
    }),

  // User Type validation
  check('userType')
    .notEmpty()
    .withMessage('User type is required')
    .isIn(['guest', 'host'])
    .withMessage('Invalid user type'),

  // Terms Accepted validation
  check('termsAccepted')
    .notEmpty()
    .withMessage('You must accept the terms and conditions')
    .custom((value) => {
      if (value !== 'on') {
        throw new Error('You must accept the terms and conditions');
      }
      return true;
    }),
```

19.6 Using Express Validator

3.
```
// Final handler middleware
(req, res, next) => {
  const { firstName, lastName, email, password, userType } = req.body;
  const errors = validationResult(req);

  if (!errors.isEmpty()) {
    return res.status(422).render('auth/signup', {
      pageTitle: 'Sign Up',
      isLoggedIn: false,
      errorMessages: errors.array().map(error => error.msg),
      oldInput: {
        firstName,
        lastName,
        email,
        password,
        userType
      }
    });
  }

  res.redirect('/login');
}
```

19.6 Using Express Validator

4.

```
<% if (typeof errorMessages !== 'undefined' && errorMessages && errorMessages.length > 0) { %>
  <div class="bg-red-100 border border-red-400 text-red-700 px-4 py-3 rounded relative mb-4"
  role="alert">
    <ul class="list-disc list-inside space-y-1">
      <% errorMessages.forEach(error => { %>
        <li><%= error %></li>
      <% }); %>
    </ul>
  </div>
<% } %>

<input
  type="text"
  name="firstName"
  placeholder="First Name"
  value="<%= typeof oldInput !== 'undefined' ? oldInput.firstName : '' %>"
  class="w-full px-4 py-2 mb-4 border border-gray-300 rounded-lg focus:outline-none
  focus:border-blue-500"
/>
```

19.7 Adding User Model

1. Define a new User Model with following fields:
   a. firstName & lastName (required)
   b. email (required, unique)
   c. Password (required)
   d. userType (possible values ‘guest’, ‘host’)
2. In the POST signup handler, create a new user with the fields from request and redirect to login after saving the user.

19.7 Adding User Model

1.

```
const mongoose = require('mongoose');

const userSchema = new mongoose.Schema({
  firstName: {
    type: String,
    required: true
  },
  lastName: {
    type: String,
    required: true
  },
  email: {
    type: String,
    required: true,
    unique: true
  },
  password: {
    type: String,
    required: true
  },
  userType: {
    type: String,
    required: true,
    enum: ['guest', 'host']
  }
});

module.exports = mongoose.model('User', userSchema);
```

19.7 Adding User Model

2.

```
const user = new User({
  firstName: firstName,
  lastName: lastName,
  email: email,
  password: password,
  userType: userType
});

user.save()
  .then(() => {
    res.redirect('/login');
})
.catch(err => {
    console.log("Error while creating user: ", err);
});
```

19.8 Encrypting Password

1. Understand the problem with plain text passwords.
2. Install a package named: bcryptjs
3. Hash the password before saving password.
4. Understand that even server does not have the password.

19.8 Encrypting Password

2.

```
prashantjain@Prashants-MacBook-Pro 13 mongoose % npm install bcryptjs

added 1 package, and audited 270 packages in 513ms

48 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
```

19.8 Encrypting Password

3.

```
bcrypt.hash(password, 12).then(hashedPassword => {
  const user = new User({
    firstName: firstName,
    lastName: lastName,
    email: email,
    password: hashedPassword,
    userType: userType
  });

  user.save()
    .then(() => {
      res.redirect('/login');
  })
  .catch(err => {
    console.log("Error while creating user: ", err);
  });
});
```

19.9 Implementing Login

1. Read the email and password from the request body and find the user with the email from the Users collection.
2. If the user is not found send an error and re-render the login page, also make changes to the login page to show errors.
3. If the user is found, then use the bcrypt compare function to match the entered password, If password does not match send another error, otherwise create a login session and redirect user to the home.

19.9 Implementing Login

1.

```
exports.postLogin = async (req, res, next) => {
  const { email, password } = req.body;

  try {
    const user = await User.findOne({ email: email });

    if (!user) {
      return res.render('auth/login', {
        pageTitle: 'Login',
        isLoggedIn: false,
        errorMessage: 'Invalid Email'
      });
    }

    res.redirect("/");
  } catch (err) {
    console.log("Error while logging in: ", err);
  }
};
```

19.9 Implementing Login

2.

```
<% if (typeof errorMessage !== 'undefined' && errorMessage) { %>
  <div class="bg-red-100 border border-red-400 text-red-700 px-4 py-3 rounded relative mb-4"
  role="alert">
    <span class="block sm:inline"><%= errorMessage %></span>
  </div>
<% } %>
```

19.9 Implementing Login

3.

```
  const isMatch = await bcrypt.compare(password, user.password);

  if (!isMatch) {
    return res.render('auth/login', {
      pageTitle: 'Login',
      isLoggedIn: false,
      errorMessage: 'Invalid Password'
    });
  }

  req.session.isLoggedIn = true;
  req.session.user = user;
  await req.session.save();

  res.redirect("/");
} catch (err) {
  console.log("Error while logging in: ", err);
}
```

19.10 Adding User Functions

1. Make the navigation bar items display only on the basis of userType by passing the user object to all views.
2. Add a field in User Model names favouriteHomes, which is an array of home ids.
3. Remove the Favourite Model and the Pre hook from the Home Model.
4. Make the favourite user specific and change the following methods:
    a. postAddFavourites
    b. getFavourites
    c. postRemoveFavourite

19.10 Adding User Functions

1.

```
<% if (isLoggedIn) { %>
  <% if (user.userType === 'guest') { %>
    <a href="/homes" class="bg-blue-600 hover:bg-blue-700 text-white
    font-semibold py-2.5 px-6 rounded-lg transition duration-300 ease-in-out
    transform hover:scale-105 shadow-md">
      Homes
    </a>
    <a href="/favourites" class="bg-blue-600 hover:bg-blue-700 text-white
    font-semibold py-2.5 px-6 rounded-lg transition duration-300 ease-in-out
    transform hover:scale-105 shadow-md">
      Favourites
    </a>
  <% } %>
  <% if (user.userType === 'host') { %>
    <a href="/host/host-homes" class="bg-blue-600 hover:bg-blue-700 text-wh
    font-semibold py-2.5 px-6 rounded-lg transition duration-300 ease-in-out
    transform hover:scale-105 shadow-md">
      Host Homes
    </a>
    <a href="/host/add-home" class="bg-blue-600 hover:bg-blue-700 text-whit
    font-semibold py-2.5 px-6 rounded-lg transition duration-300 ease-in-out
    transform hover:scale-105 shadow-md">
      Add Home
    </a>
  <% } %>
```

```
isLoggedIn: req.session.isLoggedIn,
user: req.session.user,
```

19.10 Adding User Functions

2.

```
userType: {
  type: String,
  required: true,
  enum: ['guest', 'host']
},
favouriteHomes: [{
  type: mongoose.Schema.Types.ObjectId,
  ref: 'Home'
}]
```

19.10 Adding User Functions

4.a.

```
exports.postAddFavourites = (req, res, next) => {
  const homeId = req.body.id;
  const userId = req.session.user._id;

  User.findById(userId)
    .then(user => {
      if (!user.favouriteHomes.includes(homeId)) {
        user.favouriteHomes.push(homeId);
        return user.save();
      }
      return user;
    })
    .then(() => {
      res.redirect("/favourites");
    })
    .catch((err) => {
      console.log("Error while adding to favourites", err);
      res.redirect("/favourites");
    });
};
```

19.10 Adding User Functions

4.b.

```
exports.getFavourites = (req, res, next) => {
  const userId = req.session.user._id;
  User.findById(userId)
    .populate('favouriteHomes')
    .then((user) => {
      res.render("store/favourites", {
        homes: user.favouriteHomes,
        pageTitle: "Favourites",
        isLoggedIn: req.session.isLoggedIn,
        user: req.session.user,
      });
    });
};
```

19.10 Adding User Functions

4.c.

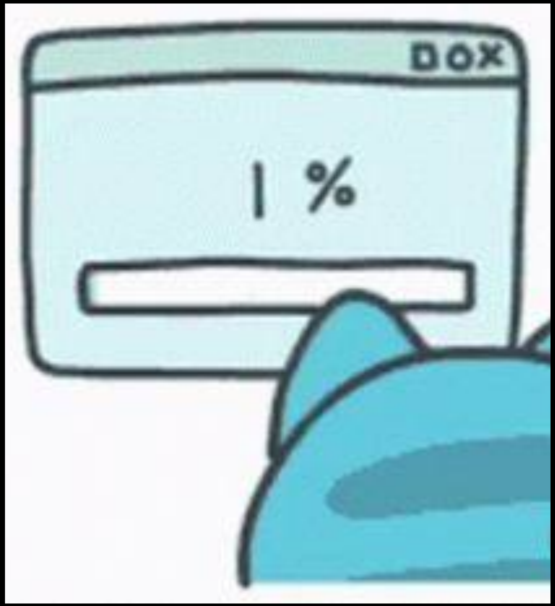
```
exports.postRemoveFavourite = (req, res, next) => {
  const homeId = req.params.homeId;
  const userId = req.session.user._id;

  User.findById(userId)
    .then(user => {
      user.favouriteHomes = user.favouriteHomes.filter(id => id.toString()
      !== homeId);
      return user.save();
    })
    .then(() => {
      res.redirect("/favourites");
    })
    .catch((error) => {
      console.log("Error while remove from favourites ", error);
      res.redirect("/favourites");
    });
};
```

Revision

1. What is Authentication
2. What is Authorization
3. Session based Authentication
4. Authentication vs Authorization
5. Signup UI
6. Using Express Validator
7. Adding User Model
8. Encrypting Password
9. Implementing Login
10. Adding User Functions

20. File Upload & Download

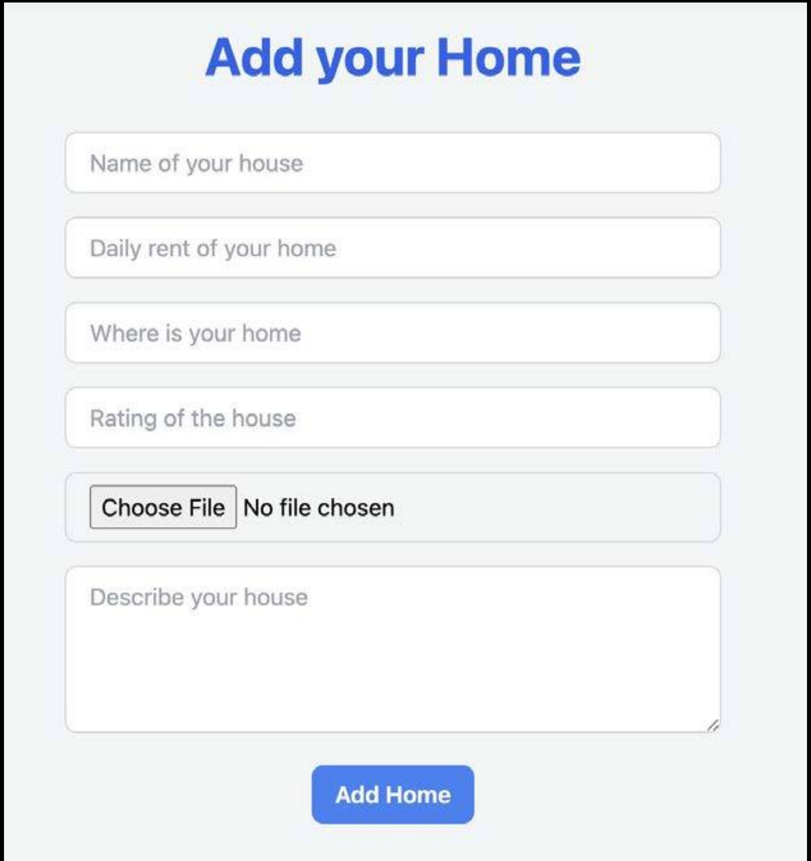
1. Adding a File Picker
2. Creating Multipart Form
3. Handling Multipart Form Data
4. Saving Image Files
5. Custom File Names
6. Restricting Upload File Types
7. Handling Edits
8. Serving Saved Data
9. Serving Files after Auth
10. Deleting Files



20.1 Adding a File Picker

- **Input Element**: Use <input type="file"> to create a file picker.
- **Multiple Files**: Add multiple attribute to allow selecting multiple files.
- **File Types**: Use accept attribute to restrict file types (e.g., accept=".jpg, .png").

```
<input type="file"
  name="photo"
  class="w-full px-4 py-2 mb-4 border border-gray-300
  rounded-lg focus:outline-none focus:border-blue-500"
/>
```



20.2 Creating Multipart Form

```
exports.postAddHome = (req, res, next) => {
  const { houseName, price, location, rating, photoUrl, description } = req.body;
  console.log(req.body);
```

```
Server running at: http://localhost:3001
{
  id: '',
  houseName: 'New House',
  price: '1299',
  location: 'kashmir',
  rating: '3.9',
  photo: 'IMG_4705.HEIC',
  description: 'asdfasdf'
}
```

× Headers Payload Preview Response Initiator Timing Cookies

▼ General

Request URL:	http://localhost:3001/host/edit-home
Request Method:	POST
Status Code:	302 Found
Remote Address:	[::1]:3001
Referrer Policy:	strict-origin-when-cross-origin

▶ Response Headers (8)

▼ Request Headers

Raw

Accept:	text/html,application/xhtml+xml,application/xml;q=0.9,image
Accept-Encoding:	gzip, deflate, br, zstd
Accept-Language:	en-GB,en;q=0.9
Cache-Control:	max-age=0
Connection:	keep-alive
Content-Length:	114
Content-Type:	application/x-www-form-urlencoded
Cookie:	connect.sid=s%3AJHZ1M5GAnP9QuJSwuUTfVrbHSq6HBC
Host:	localhost:3001
Origin:	http://localhost:3001

20.3 Handling Multipart Form Data

npm

Search packages | Search | Sign Up | Sign In

multer DT

1.4.5-lts.1 • Public • Published 2 years ago

Readme | Code Beta | 7 Dependencies | 4,749 Dependents | 46 Versions

Multer

build unknown | npm package 1.4.5-lts.1 | code style standard

Multer is a node.js middleware for handling `multipart/form-data`, which is primarily used for uploading files. It is written on top of busboy for maximum efficiency.

NOTE: Multer will not process any form which is not multipart (`multipart/form-data`).

Translations

This README is also available in other languages:

- Español (Spanish)
- 简体中文 (Chinese)
- 한국어 (Korean)
- Русский язык (Russian)
- Việt Nam (Vietnam)
- Português (Portuguese Brazil)

Installation

```
$ npm install --save multer
```

Install

```
> npm i multer
```

Repository

github.com/expressjs/multer

Homepage

github.com/expressjs/multer#readme

Weekly Downloads

6,313,138

Version	License
1.4.5-lts.1	MIT

Unpacked Size	Total Files
27.6 kB	11

Issues	Pull Requests
195	70

```
prashantjain@Prashants-MacBook-Pro 16 email and advance auth % npm install multer

added 17 packages, and audited 298 packages in 719ms

50 packages are looking for funding
  run `npm fund` for details

1 high severity vulnerability

To address all issues, run:
  npm audit fix

Run `npm audit` for details.
```

20.3 Handling Multipart Form Data

```
<form [illegible] ="/host/<%= editing ? "[illegible]" : "add-home"%>" method="POST" enctype="multipart/form-data"
class="w-full max-w-md">
```

```
const multer = require('multer');
```

```
app.use(express.urlencoded());
app.use(multer().single('photo'));
app.use(express.static(path.join(rootDir, 'public')))
```

```
console.log(req.file);
```

```
{
  fieldname: 'photo',
  originalname: 'IMG_4705.HEIC',
  encoding: '7bit',
  mimetype: 'image/heic',
  buffer: <Buffer 00 00 00 1c 66 74 79 70 68 65 69 63 00 00 00 00 74 6d
00 00 21 68 64 6c 72 00 00 ... 2246669 more bytes>,
  size: 2246719
}
```

20.4 Saving Image Files

```
app.use(multer({ dest: 'uploads/'}).single('photo'));
```

```
{
  fieldname: 'photo',
  originalname: 'house1.png',
  encoding: '7bit',
  mimetype: 'image/png',
  destination: 'uploads/',
  filename: '0c710a9b2903e323fc38e7926650d159',
  path: 'uploads/0c710a9b2903e323fc38e7926650d159',
  size: 387102
}
```

```
> routers
v uploads
    0c710a9b2903e323fc38e7926650d159
v util
```

20.5 Custom File Names

```javascript
// This defines where uploaded files will be stored and how they'll be named
const storage = multer.diskStorage({
  // Set the destination folder for uploaded files
  destination: (req, file, cb) => {
    cb(null, 'uploads/'); // Files will be saved in the 'uploads' directory
  },
  // Set the filename for uploaded files
  filename: (req, file, cb) => {
    cb(null, new Date().toISOString() + '-' + file.originalname);
  }
});
```

```javascript
app.use(multer({ storage }).single('photo'));
```

```
> routers
v uploads
    0c710a9b2903e323fc38e7926650d159
    2024-11-25T11:26:17.582Z-house1.png
v util
```

20.6 Restricting Upload File Types

```
const fileFilter = (req, file, cb) => {
  if ([ 'image/jpeg', 'image/png', 'image/jpg' ].includes(file.mimetype)) {
    cb(null, true);
  } else {
    cb(null, false);
  }
};
```

```
app.use(multer({ storage, fileFilter }).single('photo'));
```

Adding this filter, now multer would not have the file if it does not match the type and req.file will just be undefined.

```
exports.postEditHome = (req, res, next) => {
  const { id, houseName, price, location, rating, photoUrl, description } =
    req.body;

  if (!req.file) {
    return res.status(400).send('No image provided');
  }
```

20.6 Restricting Upload File Types

```
if (!req.file) {
  return res.status(400).send('No image provided');
}

const photoUrl = req.file.path;
```

```
_id: ObjectId('67445aa6e25a19de45a1b82e')
houseName : "New House"
price : 1299
location : "kashmir"
rating : 3.9
description : "asdfasdf"
host : ObjectId('673cacd23e8557da8e22d9b1')
__v : 0
photoUrl : "uploads/2024-11-25T11:35:47.086Z-house1.png"
```

20.7 Handling Edits

```
// business logic outside model
Home.findById(id)
  .then((existingHome) => {
    if (!existingHome) {
      console.log("Home not found for editing");
      return res.redirect("/host/host-homes");
    }
    existingHome.houseName = houseName;
    existingHome.price = price;
    existingHome.location = location;
    existingHome.rating = rating;
    if (req.file) {
      existingHome.photoUrl = req.file.path;
    }
    existingHome.description = description;
    return existingHome.save();
```

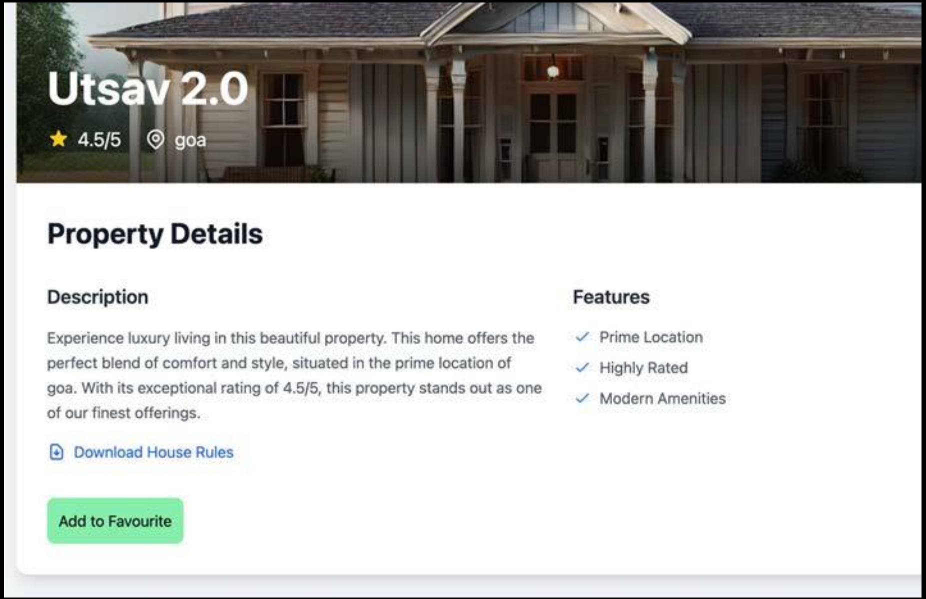
While editing we can make sure that we only overwrite the photoUrl in case a new valid image was uploaded. And don’t force the user to upload images each time they edit.

20.8 Serving Saved Data

```
app.use(express.static(path.join(rootDir, "public")));
app.use('/uploads', express.static(path.join(rootDir, "uploads")));
app.use(bodyParser.urlencoded({ extended: true }));
```

This is done as like in case of public, things inside public will be served like they are in root folder and the url does not have public in it. Here also either remove uploads/ from the image path, or add the qualifier.

20.9 Serving Files after Auth




```html
<div class="mt-4">
  <a href="/rules/<%= home._id %>" class="inline-flex items-center text-blue-600 hover:text-blue-800">
    <svg class="w-5 h-5 mr-2" fill="none" stroke="currentColor" viewBox="0 0 24 24">
      <path stroke-linecap="round" stroke-linejoin="round" stroke-width="2" d="M12 10v6m0 0l-3-3m3 3l3-3m2
      8H7a2 2 0 01-2-2V5a2 2 0 012-2h5.586a1 1 0 01.707.293l5.414 5.414a1 1 0 01.293.707V19a2 2 0 01-2 2z"/>
    </svg>
    Download House Rules
  </a>
</div>
```

20.9 Serving Files after Auth

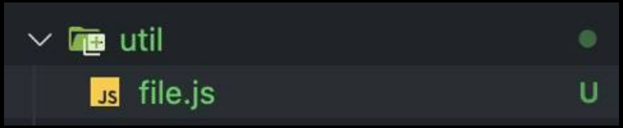
```
storeRouter.get("/rules/:homeId", storeController.getHouseRules);
```

```
const path = require('path');
const rootDir = require('../util/path-util');
```

```
exports.getHouseRules = [(req, res, next) => {
  if (!req.session.isLoggedIn) {
    return res.redirect("/login");
  }
  next();
},

(req, res, next) => {
  const homeId = req.params.homeId;
  // We can make it house specific after have homeId in file name like this: `House Rules-${homeId}.pdf`
  const rulesFileName = `House Rules.pdf`;
  const filePath = path.join(rootDir, 'rules', rulesFileName);
  // res.sendFile(filePath);
  res.download(filePath, 'Rules.pdf');
  }
];
```

20.10 Deleting Files

```javascript
const fs = require('fs');

exports.deleteFile = (filePath) => {
  fs.unlink(filePath, (err) => {
    if (err) throw err;
  });
};
```




```javascript
existingHome.location = location;
existingHome.rating = rating;
if (req.file) {
  deleteFile(existingHome.photoUrl);
  existingHome.photoUrl = req.file.path;
}
```

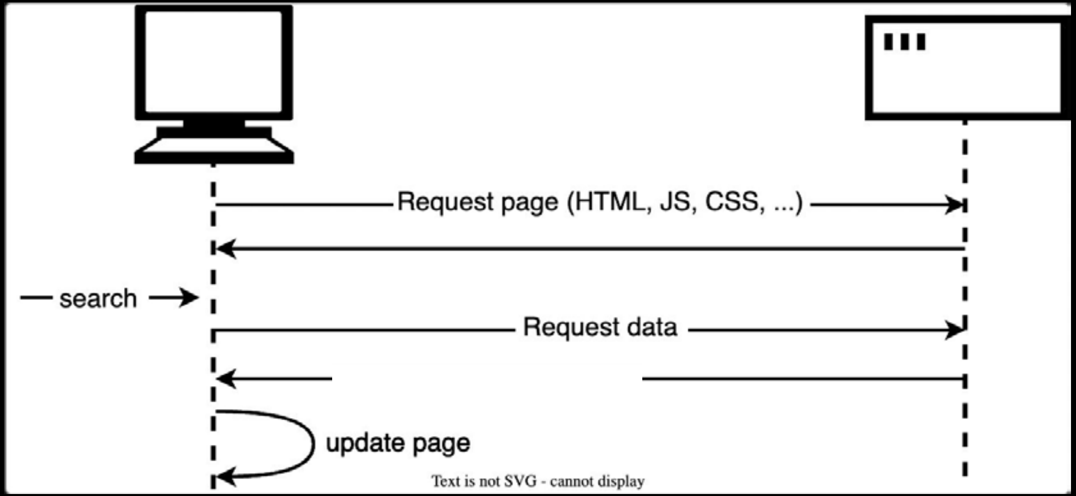
Revision

1. Adding a File Picker
2. Creating Multipart Form
3. Handling Multipart Form Data
4. Saving Image Files
5. Custom File Names
6. Restricting Upload File Types
7. Handling Edits
8. Serving Saved Data
9. Serving Files after Auth
10. Deleting Files

21. REST API / JSON Requests

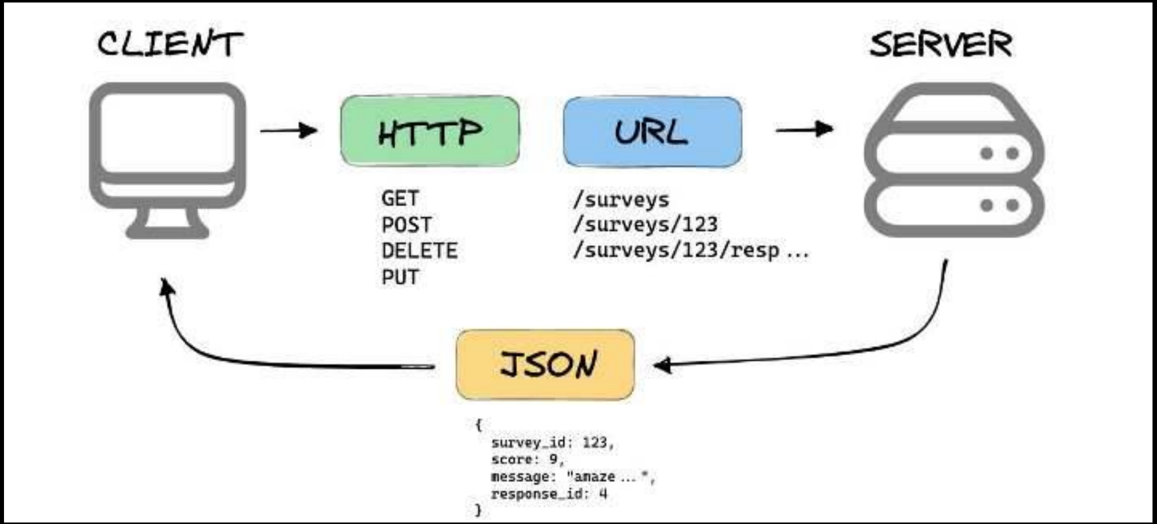
1. What are Async Requests
2. What are REST APIs
3. Decoupling Frontend & Backend
4. Routes & HTTP Methods
5. REST Core Concepts
6. First API Todo App
7. API for Fetch Items
8. API for Deleting Items
9. Improving UI Elements
10. Adding Complete Item functionality
11. Deploying React App

21.1 What are Async Requests



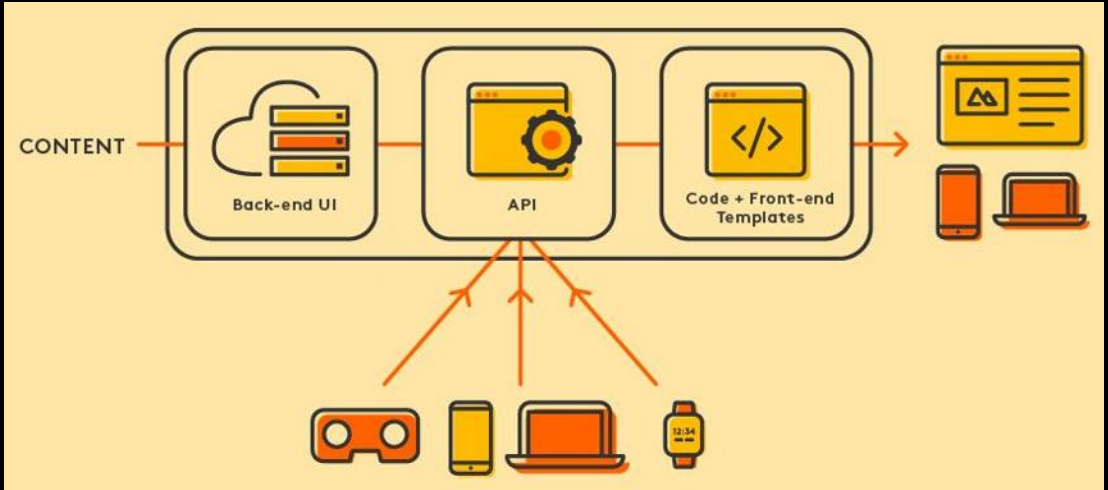
- Async network requests enable web pages to communicate with servers without reloading.
- The client sends JSON requests to the server asynchronously in single-page apps.
- The server processes the request and returns a JSON response.
- The page updates itself dynamically using the received data.

21.2 What are REST APIs



- REST APIs enable communication between clients and servers using HTTP.
- They are mainly identified by a URI.
- They use standard HTTP methods like GET, POST, PUT, and DELETE.
- Data is exchanged in formats like JSON or XML.
- REST APIs are stateless.
- REST APIs allow clients to access and manipulate web resources.

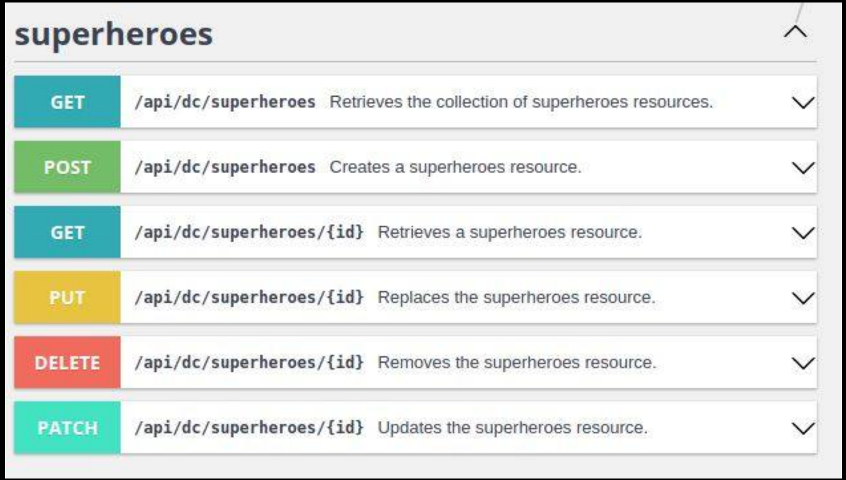
21.3 Decoupling Frontend & Backend



- Separating front-end and back-end allows independent development and scaling.
- REST APIs serve as a communication layer between them.
- Front-end interacts with back-end through standardized RESTful calls.
- Decoupling enhances flexibility and simplifies maintenance.
- REST APIs enable front-end updates without altering back-end code.

21.4 Routes & HTTP Methods

- REST API routes define the endpoints (URLs) where resources can be accessed by clients.
- GET: Retrieves data from the server at the specified route.
- POST: Sends new data to the server to create a resource.
- PUT: Updates or replaces an existing resource at a given route.
- DELETE: Removes a resource from the server at the specified route.
- PATCH: Partially updates an existing resource with new data.



21.5 REST Core Concepts

- Statelessness: Each request contains all necessary information; the server maintains no client session.
- Uniform Interface: Standardized communication using HTTP methods like GET, POST, PUT, DELETE.
- Client-Server Separation: Independent development of front-end and back-end components.
- Cacheability: Responses indicate if they can be cached to improve performance.
- Layered System: Architecture allows for multiple layers between client and server.
- Code on Demand (Optional): Servers can extend client functionality by sending executable code.

21.6 First API Todo App

1. Copy the following from the previous app:
    a. Env files
    b. Package dependencies
    c. app.js
2. Changes in env files
    a. Remove the email data.
    b. Change DB name to todo-app
3. Create empty controllers, models, and routes folders.
4. Remove excess code from app.js.
5. Create the error controller to give out a 404 error message and status.
6. Create a Mongoose model for TodoItem.
7. Create itemsRouter and itemsController for POST /todos request from frontend.
8. Install and setup CORS package in backend.
9. Add express.json() to app.

21.6 First API Todo App

1.

.env.development
.env.example
.env.production
app.js
package.json

2.

backend > .env.development

```
MONGO_DB_USERNAME=root
MONGO_DB_PASSWORD=root
MONGO_DB_DATABASE=todo-app-test
```

backend > .env.production

```
MONGO_DB_USERNAME=root
MONGO_DB_PASSWORD=root
MONGO_DB_DATABASE=todo-app
```

3.

controllers
models
node_modules
routers

21.6 First API Todo App

4.

```
const ENV = process.env.NODE_ENV || 'production'
require('dotenv').config({
  path: `.env.${ENV}`
});

// External Module
const express = require("express");
const mongoose = require("mongoose");
const bodyParser = require("body-parser");

// Local Module
const errorController = require("./controllers/errorController");
const itemsRouter = require("./routers/itemsRouter");

const MONGO_DB_URL =
  `mongodb+srv://${process.env.MONGO_DB_USERNAME}:${process.env.MONGO_DB_PASSWORD}@kgcluster.ie6mb.mongodb.net/${process.
  env.MONGO_DB_DATABASE}`;

const app = express();

app.use(bodyParser.urlencoded({ extended: true }));
app.use(itemsRouter);
app.use(errorController.get404);

const PORT = process.env.PORT || 3000;
mongoose.connect(MONGO_DB_URL).then(() => {
  app.listen(PORT, () => {
    console.log(`Server running at: http://localhost:${PORT}`);
  });
});
```

21.6 First API Todo App

6.

backend > models > TodoItem.js > ...

```javascript
const mongoose = require("mongoose");

const todoItemSchema = new mongoose.Schema({
  task: {
    type: String,
    required: true
  },
  date: {
    type: Date,
    required: true
  },
  completed: {
    type: Boolean,
    default: false
  },
  createdAt: {
    type: Date,
    default: Date.now
  }
});
```

5.

backend > controllers > errorController.js > ...

```javascript
exports.get404 = (req, res, next) => {
  res.status(404).json({ message: 'Page Not Found' });
};
```

21.6 First API Todo App

7.

```
const express = require("express");
const itemsController = require("../controllers/itemsController");
const itemsRouter = express.Router();

itemsRouter.post("/todos", itemsController.postItem);

module.exports = itemsRouter;
```

```
exports.postItem = async (req, res, next) => {
  try {
    console.log(req.body);
    const todoItem = new TodoItem(req.body);
    const item = await todoItem.save();
    res.json(item);
  } catch (err) {
    next(err);
  }
};
```

21.6 First API Todo App

8,9.

```
const bodyParser = require("body-parser");
const cors = require("cors");
```

```
app.use(bodyParser.urlencoded({ extended: true }));
app.use(cors());
app.use(express.json());
app.use(itemsRouter);
app.use(errorController.get404);
```

21.7 API for Fetch Items

```
itemsRouter.get("/todos", itemsController.getItems);
itemsRouter.post("/todos", itemsController.postItem);
```

```
exports.getItems = async (req, res, next) => {
  try {
    const items = await TodoItem.find();
    res.json(items);
  } catch (err) {
    next(err);
  }
};
```

```
export const todoItemToClientModel = (serverItem) => {
  return {
    id: serverItem._id,
    todoText: serverItem.task,
    todoDate: serverItem.date
  }
}
```

```
const formattedDate = new Date(todoDate).toLocaleDateString('en-US', {
  year: 'numeric',
  month: 'long',
  day: 'numeric'
});
```

21.8 API for Deleting Items

```
itemsRouter.get("/todos", itemsController.getItems);
itemsRouter.post("/todos", itemsController.postItem);
itemsRouter.delete("/todos/:id", itemsController.deleteItem);
```

```
exports.deleteItem = async (req, res, next) => {
  try {
    const { id } = req.params;
    await TodoItem.findByIdAndDelete(id);
    res.json({ id, message: "Item deleted" });
  } catch (err) {
    next(err);
  }
};
```

21.9 Improving UI Elements

1. Remove bootstrap
2. Add tailwind to the project.
3. Create the whole app using tailwind classes.

21.10 Adding Complete Item functionality

1. Add a UI element to mark the item complete.
2. Add a component state based on which items will be striked through, default value should come from server.
3. Add a toggleComplete handler in TodoItem that sends a PATCH request to update the completed flag and expects the updated item in return. Also the value of state must be updated based on the final value.
4. Add a route and API in the backend to update the todoItem.

21.10 Adding Complete Item functionality

1,2.

```
<input
  type="checkbox"
  checked={isComplete}
  onChange={toggleComplete}
  className="w-5 h-5 rounded border-gray-300 text-blue-600 focus:ring-blue-500"
/>
```

```
const TodoItem = ({ id, todoText, todoDate, completed }) => {
  const { deleteTodoItem } = useContext(TodoItemsContext);
  const [isComplete, setIsComplete] = useState(completed);
```

```
<div className={`flex flex-col ${isComplete ? 'text-gray-400 line-through' : 'text-gray-700'}`}>
  <span className="font-medium">{todoText}</span>
  <span className="text-sm text-gray-500">{formattedDate}</span>
</div>
```

21.10 Adding Complete Item functionality

3.

```
const toggleComplete = () => {
  fetch(`http://localhost:3000/todos/${id}`, {
    method: 'PATCH',
    headers: { 'Content-Type': 'application/json' },
    body: JSON.stringify({ completed: !isComplete })
  })
  .then(res => res.json())
  .then(data => {
    setIsComplete(data.completed);
  })
  .catch(err => console.log(err));
}
```

21.10 Adding Complete Item functionality

4.

```
itemsRouter.get("/todos", itemsController.getItems);
itemsRouter.post("/todos", itemsController.postItem);
itemsRouter.delete("/todos/:id", itemsController.deleteItem);
itemsRouter.patch("/todos/:id", itemsController.updateItem);
```

```
exports.updateItem = async (req, res, next) => {
  try {
    const { id } = req.params;
    const { completed } = req.body;
    const updatedItem = await TodoItem.findByIdAndUpdate(
      id,
      { completed: completed },
      { new: true }
    );

    res.json(updatedItem);
  } catch (err) {
    next(err);
  }
};
```

Revision

1. What are Async Requests
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